Cochrane Library

Cochrane Database of Systematic Reviews

Antibiotics for acute otitis media in children (Review)

Venekamp RP, Sanders SL, Glasziou PP, Rovers MM

Venekamp RP, Sanders SL, Glasziou PP, Rovers MM.
Antibiotics for acute otitis media in children.
Cochrane Database of Systematic Reviews 2023, Issue 11. Art. No.: CD000219.
DOI: 10.1002/14651858.CD000219.pub5.

www.cochranelibrary.com

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

WILEY

TABLE OF CONTENTS

ABSTRACT 1
PLAIN LANGUAGE SUMMARY 2
SUMMARY OF FINDINGS 4
BACKGROUND 7
OBJECTIVES 7
METHODS 7
Figure 1. 9
Figure 2. 10
RESULTS 11
Figure 3. 12
Figure 4. 17
Figure 5. 18
Figure 6. 19
DISCUSSION 21
AUTHORS' CONCLUSIONS 23
ACKNOWLEDGEMENTS 23
REFERENCES 24
CHARACTERISTICS OF STUDIES 27
DATA AND ANALYSES 55
Analysis 1.1. Comparison 1: Antibiotics versus placebo, Outcome 1: Pain 57
Analysis 1.2. Comparison 1: Antibiotics versus placebo, Outcome 2: Vomiting, diarrhoea or rash 58
Analysis 1.3. Comparison 1: Antibiotics versus placebo, Outcome 3: Abnormal tympanometry 59
Analysis 1.4. Comparison 1: Antibiotics versus placebo, Outcome 4: Tympanic membrane perforation 59
Analysis 1.5. Comparison 1: Antibiotics versus placebo, Outcome 5: Contralateral otitis (in unilateral cases) 60
Analysis 1.6. Comparison 1: Antibiotics versus placebo, Outcome 6: Late AOM recurrences 60
Analysis 2.1. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 1: Pain 61
Analysis 2.2. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 2: Vomiting, diarrhoea or rash 61
Analysis 2.3. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 3: Abnormal tympanometry at 4 weeks 62
Analysis 2.4. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 4: Tympanic membrane perforation 62
Analysis 2.5. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 5: AOM recurrences 62
APPENDICES 62
FEEDBACK 65
WHAT'S NEW 71
HISTORY 71
CONTRIBUTIONS OF AUTHORS 75
DECLARATIONS OF INTEREST 75
SOURCES OF SUPPORT 75
DIFFERENCES BETWEEN PROTOCOL AND REVIEW 75
INDEX TERMS 76

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

[Intervention Review]

Antibiotics for acute otitis media in children

Roderick P Venekamp[1], Sharon L Sanders[2], Paul P Glasziou[2], Maroeska M Rovers[3]

[1]Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht University, Utrecht, Netherlands. [2]Institute for Evidence-Based Healthcare, Bond University, Gold Coast, Australia. [3]Department of Radiology and Nuclear Medicine, Radboud University Nijmegen Medical Centre, Nijmegen, Netherlands

Contact: Roderick P Venekamp, r.p.venekamp@umcutrecht.nl.

Editorial group: Cochrane Acute Respiratory Infections Group.
Publication status and date: Edited (no change to conclusions), published in Issue 12, 2023.

Citation: Venekamp RP, Sanders SL, Glasziou PP, Rovers MM. Antibiotics for acute otitis media in children. *Cochrane Database of Systematic Reviews* 2023, Issue 11. Art. No.: CD000219. DOI: 10.1002/14651858.CD000219.pub5.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ABSTRACT

Background

Acute otitis media (AOM) is one of the most common diseases in childhood for which antibiotics are commonly prescribed; a systematic review reported a pooled prevalence of 85.6% in high-income countries. This is an update of a Cochrane Review first published in the Cochrane Library in 1997 and updated in 1999, 2005, 2009, 2013 and 2015.

Objectives

To assess the effects of antibiotics for children with AOM.

Search methods

We searched CENTRAL, MEDLINE, Embase, Current Contents, CINAHL, LILACS and two trial registers. The date of the search was 14 February 2023.

Selection criteria

We included randomised controlled trials comparing 1) antimicrobial drugs with placebo, and 2) immediate antibiotic treatment with expectant observation (including delayed antibiotic prescribing) in children with AOM.

Data collection and analysis

Two review authors independently screened trials for inclusion and extracted data using the standard methodological procedures recommended by Cochrane. Our primary outcomes were: 1) pain at various time points (24 hours, two to three days, four to seven days, 10 to 14 days), and 2) adverse effects likely to be related to the use of antibiotics. Secondary outcomes were: 1) abnormal tympanometry findings, 2) tympanic membrane perforation, 3) contralateral otitis (in unilateral cases), 4) AOM recurrences, 5) serious complications related to AOM and 6) long-term effects (including the number of parent-reported AOM symptom episodes, antibiotic prescriptions and health care utilisation as assessed at least one year after randomisation). We used the GRADE approach to rate the overall certainty of evidence for each outcome of interest.

Main results

Antibiotics versus placebo

We included 13 trials (3401 children and 3938 AOM episodes) from high-income countries, which we assessed at generally low risk of bias.

Antibiotics do not reduce pain at 24 hours (risk ratio (RR) 0.89, 95% confidence interval (CI) 0.78 to 1.01; 5 trials, 1394 children; high-certainty evidence), or at four to seven days (RR 0.76, 95% CI 0.50 to 1.14; 7 trials, 1264 children), but result in almost a third fewer children having

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

pain at two to three days (RR 0.71, 95% CI 0.58 to 0.88; number needed to treat for an additional beneficial outcome (NNTB) 20; 7 trials, 2320 children; high-certainty evidence), and likely result in two-thirds fewer having pain at 10 to 12 days (RR 0.33, 95% CI 0.17 to 0.66; NNTB 7; 1 trial, 278 children; moderate-certainty evidence). Antibiotics increase the risk of adverse events such as vomiting, diarrhoea or rash (RR 1.38, 95% CI 1.16 to 1.63; number needed to treat for an additional harmful outcome (NNTH) 14; 8 trials, 2107 children; high-certainty evidence).

Antibiotics reduce the risk of children having abnormal tympanometry findings at two to four weeks (RR 0.83, 95% CI 0.72 to 0.96; NNTB 11; 7 trials, 2138 children), slightly reduce the risk of experiencing tympanic membrane perforations (RR 0.43, 95% CI 0.21 to 0.89; NNTB 33; 5 trials, 1075 children) and halve the risk of contralateral otitis episodes (RR 0.49, 95% CI 0.25 to 0.95; NNTB 11; 4 trials, 906 children). However, antibiotics do not reduce the risk of abnormal tympanometry findings at six to eight weeks (RR 0.89, 95% CI 0.70 to 1.13; 3 trials, 953 children) and at three months (RR 0.94, 95% CI 0.66 to 1.34; 3 trials, 809 children) or late AOM recurrences (RR 0.94, 95% CI 0.79 to 1.11; 6 trials, 2200 children). Severe complications were rare, and the evidence suggests that serious complications do not differ between children treated with either antibiotics or placebo.

Immediate antibiotics versus expectant observation

We included six trials (1556 children) from high-income countries.

The evidence suggests that immediate antibiotics may result in a reduction of pain at two to three days (RR 0.53, 95% CI 0.35 to 0.79; NNTB 8; 1 trial, 396 children; low-certainty evidence), but probably do not reduce the risk of pain at three to seven days (RR 0.75, 95% CI 0.50 to 1.12; 4 trials, 959 children; moderate-certainty evidence), and may not reduce the risk of pain at 11 to 14 days (RR 0.91, 95% CI 0.75 to 1.10; 1 trial, 247 children; low-certainty evidence). Immediate antibiotics increase the risk of vomiting, diarrhoea or rash (RR 1.87, 95% CI 1.39 to 2.51; NNTH 10; 3 trials, 946 children; high-certainty evidence).

Immediate antibiotics probably do not reduce the proportion of children with abnormal tympanometry findings at four weeks and evidence suggests that immediate antibiotics may not reduce the risk of tympanic membrane perforation and AOM recurrences. No serious complications occurred in either group.

Authors' conclusions

This review reveals that antibiotics probably have no effect on pain at 24 hours, a slight effect on pain in the days following and only a modest effect on the number of children with tympanic perforations, contralateral otitis episodes and abnormal tympanometry findings at two to four weeks compared with placebo in children with AOM. In high-income countries, most cases of AOM spontaneously remit without complications. The benefits of antibiotics must be weighed against the possible harms: for every 14 children treated with antibiotics, one child experienced an adverse event (such as vomiting, diarrhoea or rash) that would not have occurred if antibiotics were withheld.

For most children with mild disease in high-income countries, an expectant observational approach seems justified. Therefore, clinical management should emphasise advice about adequate analgesia and the limited role for antibiotics.

PLAIN LANGUAGE SUMMARY

Antibiotics for acute middle ear infection (acute otitis media) in children

Key messages

In high-income countries, most children with acute otitis media (AOM) recover spontaneously without complications, and compared with placebo antibiotics have no early effect on pain and only a slight effect on pain in the days following.

Any benefits of antibiotics must be weighed against the possible harms: for every 14 children treated with antibiotics one child experienced an adverse event (such as vomiting, diarrhoea or rash) that would not have occurred if antibiotics were not given.

For most children with mild disease in high-income countries, an expectant observational approach seems justified.

What is AOM?

AOM or acute middle ear infection is one of the most common diseases in early infancy and childhood, causing pain and general symptoms of illness such as fever, irritability and problems feeding and sleeping.

What did we want to find out?

We wanted to find out if antibiotics are effective for relieving signs and symptoms in children with AOM.

What did we do?

We searched for studies that looked at the effectiveness of antibiotics compared with placebo or expectant observation (watchful waiting) in children aged from one month to 15 years with AOM. We compared and summarised their results, and rated our confidence in the evidence, based on factors such as study methods and effect sizes.

What did we find?

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

We found that antibiotics did not decrease the number of children with pain at 24 hours (when 60% of children recovered or improved), only slightly reduced the number of children with pain in the days following, and did not reduce the number of children with late AOM recurrences compared with placebo. However, antibiotics did slightly reduce the number of children with perforations (a rupture) of the eardrum and AOM in the initially unaffected ear compared with placebo. Antibiotics reduced the number of children with abnormal tympanometry (a test to evaluate how well the middle ear is functioning) findings at two to four weeks but not after six to eight weeks and three months.

There was not enough information to know if antibiotics reduced rare complications such as mastoiditis (infection of the bones around the ear). All the studies included in this review were from high-income countries. Data are lacking from populations in which the AOM incidence and risk of progression to mastoiditis is higher.

Antibiotics caused unwanted effects such as diarrhoea, vomiting and rash. It is difficult to balance the small benefits against the small harms of antibiotics in children with AOM. However, for most children with mild disease in high-income countries, an expectant observational approach seems justified.

What are the limitations of the evidence?
From the review of antibiotics against placebo, we are confident that antibiotics have no effect on pain in the first 24 hours and only a slight effect in the days following. We are only moderately confident that antibiotics reduce pain experienced 10 to 12 days after children are initially assessed. Our confidence is moderate because only one study reported this result from a secondary analysis. We are confident that antibiotics modestly reduce the number of children with abnormal tympanometry results after two to four weeks, but that they do not reduce the risk at three months. We are confident that antibiotics increase the risk of unwanted effects.

For the review of immediate antibiotics versus expectant observation, we have little confidence that immediate antibiotics reduce pain two to three days after assessment and are moderately confident that immediate antibiotics probably do not reduce pain at three to seven days. Our confidence is limited because of concerns about how some of the studies were conducted.

How up-to-date is this evidence?
Our evidence is current to 14 February 2023. The findings of this update are consistent with previous versions of the review.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

SUMMARY OF FINDINGS

Summary of findings 1. Antibiotics versus placebo for acute otitis media in children

Antibiotics versus placebo for acute otitis media in children

Patient or population: children with acute otitis media
Settings: primary care and secondary care
Intervention: antibiotics versus placebo

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Control	Antibiotics versus placebo				
Pain - pain at 24 hours	426 per 1000	**379 per 1000** (332 to 431)	**RR 0.89** (0.78 to 1.01)	1394 (5 studies)[a]	⊕⊕⊕⊕ **High**	—
Pain - pain at 2 to 3 days	159 per 1000	**113 per 1000** (92 to 140)	**RR 0.71** (0.58 to 0.88)	2320 (7 studies)	⊕⊕⊕⊕ **High**	—
Pain - pain at 4 to 7 days	248 per 1000	**189 per 1000** (124 to 283)	**RR 0.76** (0.50 to 1.14)	1264 (7 studies)[a]	⊕⊕⊕⊕ **High**	—
Pain - pain at 10 to 12 days	216 per 1000	**71 per 1000** (37 to 142)	**RR 0.33** (0.17 to 0.66)	278 (1 study)	⊕⊕⊕⊖ **Moderate**[b]	—
Vomiting, diarrhoea or rash	196 per 1000	**270 per 1000** (227 to 319)	**RR 1.38** (1.16 to 1.63)	2107 (8 studies)	⊕⊕⊕⊕ **High**	—
Abnormal tympanometry - 2 to 4 weeks	481 per 1000	**399 per 1000** (347 to 462)	**RR 0.83** (0.72 to 0.96)	2138 (7 studies)	⊕⊕⊕⊕ **High**	—
Abnormal tympanometry - 3 months	241 per 1000	**227 per 1000** (159 to 323)	**RR 0.94** (0.66 to 1.34)	809 (3 studies)	⊕⊕⊕⊕ **High**	—

*The basis for the **assumed risk** for 'Study population' was the average risk in the control groups (i.e. total number of participants with events divided by total number of participants included in the meta-analysis). The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).
CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High-certainty: Further research is very unlikely to change our confidence in the estimate of effect.

Cochrane Library
Trusted evidence.
Informed decisions.
Better health.
Cochrane Database of Systematic Reviews

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Moderate-certainty: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.
Low-certainty: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
Very low-certainty: We are very uncertain about the estimate.

[a]The number of studies reported in the summary of findings table for the outcomes 'Pain at 24 hours' and 'Pain at 4 to 7 days' differs slightly from those reported in the Data and Analysis Table 1 - Antibiotics versus placebo (five versus six studies and seven versus eight studies, respectively). This is due to the van Buchem trial. This trial is included as one study in our review (and in the summary of findings table), but we included data from two different comparisons from this 2 x 2 factorial design trial (van Buchem 1981) in our analyses.
[b]We downgraded the evidence for pain at days 10 to 12 from high certainty due to inconsistency. Only one trial reported this outcome and the outcome was not specified a priori.

Summary of findings 2. Immediate antibiotics versus expectant observation for acute otitis media in children

Immediate antibiotics versus expectant observation for acute otitis media in children

Patient or population: children with acute otitis media
Settings: primary care and secondary care
Intervention: immediate antibiotics versus expectant observation

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Control	Antibiotics versus expectant observation				
Pain - Pain at 2 to 3 days	284 per 1000	**150 per 1000** (99 to 224)	**RR 0.53** (0.35 to 0.79)	396 (1 study)	⊕⊕⊖⊖ **Low**[a]	—
Pain - Pain at 3 to 7 days	356 per 1000	**267 per 1000** (178 to 398)	**RR 0.75** (0.50 to 1.12)	959 (4 studies)	⊕⊕⊕⊖ **Moderate**[b]	—
Pain - Pain at 11 to 14 days	669 per 1000	**609 per 1000** (502 to 736)	**RR 0.91** (0.75 to 1.10)	247 (1 study)	⊕⊕⊖⊖ **Low**[c]	—
Vomiting, diarrhoea or rash	114 per 1000	**214 per 1000** (159 to 287)	**RR 1.87** (1.39 to 2.51)	946 (3 studies)	⊕⊕⊕⊕ **High**	—
Abnormal tympanometry at 4 weeks	495 per 1000	**510 per 1000** (386 to 668)	**RR 1.03** (0.78 to 1.35)	207 (1 study)	⊕⊕⊕⊖ **Moderate**[d]	—

*The basis for the **assumed risk** for 'Study population' was the average risk in the control groups (i.e. total number of participants with events divided by total number of participants included in the meta-analysis). The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

Cochrane Library
Trusted evidence.
Informed decisions.
Better health.
Cochrane Database of Systematic Reviews

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence
High-certainty: Further research is very unlikely to change our confidence in the estimate of effect.
Moderate-certainty: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.
Low-certainty: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
Very low-certainty: We are very uncertain about the estimate.

[a]We downgraded the evidence for pain at two to three days from high certainty due study limitations (risk of detection bias) and imprecision (outcome data derived from one trial).
[b]We downgraded the evidence for pain at three to seven days from high certainty due study limitations (risk of detection bias).
[c]We downgraded the evidence for pain at days 11 to 14 from high certainty due study limitations (risk of detection bias) and because of the number of children that were 'lost to follow-up' (13%), which introduced a risk of (attrition) bias.
[d]We downgraded the evidence for abnormal tympanometry findings at four weeks from high certainty as the number of children that were 'lost to follow-up' in the immediate antibiotics group was substantially lower than in the expectant observation group (4% versus 11%), thereby introducing a risk of (attrition) bias.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

BACKGROUND

Description of the condition

Despite reports of a decline during the COVID-19 pandemic (Gisselsson-Solen 2022; Hullegie 2021), acute otitis media (AOM) is one of the most frequent diseases in early infancy and childhood. AOM is defined as the presence of middle ear effusion and a rapid onset of signs or symptoms of middle ear inflammation, such as ear pain, otorrhoea or fever (AAP 2013). Approximately 10% of children have an episode of AOM by three months of age and, by three years of age, approximately 50% to 85% of all children have experienced at least one AOM episode (Teele 1989). The peak incidence is within the first year of life (Monasta 2012).

Description of the intervention

Despite a large number of published clinical trials, there is no consensus regarding the most appropriate therapy for AOM; a recent systematic review reported a pooled prevalence of antibiotic prescribing rates in ambulatory care of 85.6% for this condition in high-income countries (Burvenich 2022). One meta-analysis emphasises that AOM resolves spontaneously in most children (Rosenfeld 1994). However, one semi-randomised trial of 1365 participants conducted in Sweden in 1954 reported a rate of mastoiditis of 17% in the untreated group versus none in the penicillin-treated groups (Rudberg 1954). Over recent years, prescription strategies in which antibiotic treatment for acute respiratory infections such as AOM is delayed and instituted only if symptoms persist or worsen after several days have been advocated (AAP 2013).

How the intervention might work

AOM has a multifactorial pathogenesis. Mucosal swelling of the nasopharynx and Eustachian tube due to a viral upper respiratory tract infection can lead to Eustachian tube dysfunction with impaired clearance and pressure regulation of the middle ear. Prolonged dysfunction may be followed by aspiration of potential viral and bacterial pathogens from the nasopharynx to the middle ear. These pathogens might in turn provoke a host inflammatory response, which leads to the clinical manifestations of AOM such as ear pain, otorrhoea, fever and irritability. *Streptococcus pneumoniae* (*S pneumoniae*) has been the predominant pathogen related to AOM for many years, next to *Moraxella catarrhalis* (*M catarrhalis*) and non-typeable *Haemophilus influenzae* (*H influenzae*). However, recent studies suggest that widespread implementation of pneumococcal conjugate vaccination has changed the frequency of otopathogens related to AOM with non-typeable *H influenzae* and non-vaccine *S pneumoniae* serotypes becoming more prevalent (Casey 2013; Coker 2010). Additionally, viral (co-)infection is known to worsen the clinical and bacteriological outcome of AOM (Arola 1990; Chonmaitree 1992). As bacteria are considered to play a predominant role in the causation of AOM-related symptoms, antibiotic treatment may accelerate clinical recovery and may reduce the number of complications related to AOM.

Why it is important to do this review

Although numerous randomised clinical trials (RCTs) on the effectiveness of antibiotic treatment in children with AOM have been performed over the decades, consensus regarding the most appropriate treatment strategy is lacking. As symptoms consistent with AOM resolve spontaneously in most children, an expectant observational approach might be justified. We therefore performed a systematic review to examine the effects of both immediate antibiotic treatment and an expectant observational approach in children with AOM.

OBJECTIVES

To assess the effects of antibiotics for children with AOM.

We attempted to determine to what extent antibiotic therapy was more effective than placebo and what, if any, advantages it offered to children in terms of symptom relief (pain), avoidance of complications (such as tympanic membrane perforations and severe complications such as mastoiditis) and longer-term hearing problems from middle ear effusion (as measured by tympanometry or audiometry). We also assessed the effect of immediate antibiotics versus expectant observation on AOM. Moreover, we aimed to provide information on subgroups of children with AOM that benefit more from antibiotics than others.

METHODS

Criteria for considering studies for this review

Types of studies

Randomised controlled trials (RCTs) of antimicrobial drugs versus placebo control. We also included RCTs comparing immediate antibiotics versus expectant observation. We did not include quasi-RCTs, cross-over trials or cluster-randomised trials.

Types of participants

Studies including children (aged from one month to 15 years) of either gender without ventilation tubes, suffering from AOM irrespective of the setting from which they were recruited.

Types of interventions

Antimicrobial drugs versus placebo control.

Immediate antibiotic versus expectant observation (also known as 'wait and see' or 'watchful waiting' or 'observation therapy'). This includes expectant observational approaches in which prescriptions may or may not be provided.

Types of outcome measures

We focused our data extraction on patient-relevant outcomes, that is, those symptoms or problems that are important to the patient's sense of well-being. While other endpoints, such as microbiological cure, may enhance medical understanding of the disease process, decisions about treatment should focus on helping the patient. We analysed the outcomes listed below in this review, but these outcomes were not used as a basis for including or excluding studies.

Primary outcomes

1. Proportion of children with pain at various time points (24 hours, two to three days, four to seven days, 10 to 14 days).
2. Adverse effects likely to be related to the use of antibiotics such as vomiting, diarrhoea or rash.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Secondary outcomes

1. Abnormal tympanometry findings at various time points (two to four weeks, six to eight weeks, and three months) as a surrogate measure for hearing problems caused by middle ear fluid. Tympanometry is an acoustic evaluation of middle ear function.
2. Tympanic membrane perforation.
3. Contralateral otitis (in unilateral cases).
4. AOM recurrences.
5. Serious complications related to AOM such as mastoiditis and meningitis.
6. Long-term effects (including the number of parent-reported AOM symptom episodes, antibiotic prescriptions and health care utilisation as assessed at least one year after randomisation).

Search methods for identification of studies

We conducted systematic searches for RCTs and controlled clinical trials. There were no language, publication year or publication status restrictions. We searched up to 14 February 2023.

Electronic searches

For this update we searched the Cochrane Central Register of Controlled Trials (CENTRAL 2023, Issue 2) (February 2023), which contains the Acute Respiratory Infections (ARI) Specialised Register; MEDLINE (OVID; April 2015 to February week 2, 2023); Embase (OVID; April 2015 to 2023 Week 06); Current Contents Connect (Web of Science, 2015 to February 2023); CINAHL (EBSCOHost, April 2015 to February 2023) and LILACS (BIREME, 2015 to February 2023). Our previous update using the same search strategies covered the period 2008 to April 2015. See Appendix 1 for details of earlier searches.

We used the search strategy described in Appendix 2 to search CENTRAL and MEDLINE. We combined the MEDLINE search with the Cochrane Highly Sensitive Search Strategy for identifying randomised trials in MEDLINE: sensitivity- and precision-maximising version (2008 revision); Ovid format (Lefebvre 2021). We adapted the search strategy to search Embase (Appendix 3), Current Contents (Appendix 4), CINAHL (Appendix 5) and LILACS (Appendix 6).

Searching other resources

We checked the World Health Organization International Clinical Trials Registry Platform (WHOICTRP; trialsearch.who.int) and ClinicalTrials.gov (clinicaltrials.gov/) for ongoing trials (14 February 2023) using the terms: antibiotics and Otitis Media and Child*.

To increase the yield of relevant studies, we inspected the reference lists of all identified studies and reviews.

Data collection and analysis

Selection of studies

One review author (RPV) screened titles and abstracts obtained from the database searches. Two review authors (RPV, SLS) reviewed the full text of the potentially relevant titles and abstracts against the inclusion criteria. We resolved any disagreements by discussion.

Data extraction and management

Two review authors (RPV, SLS) extracted data from the included studies. We resolved disagreements by discussion.

Assessment of risk of bias in included studies

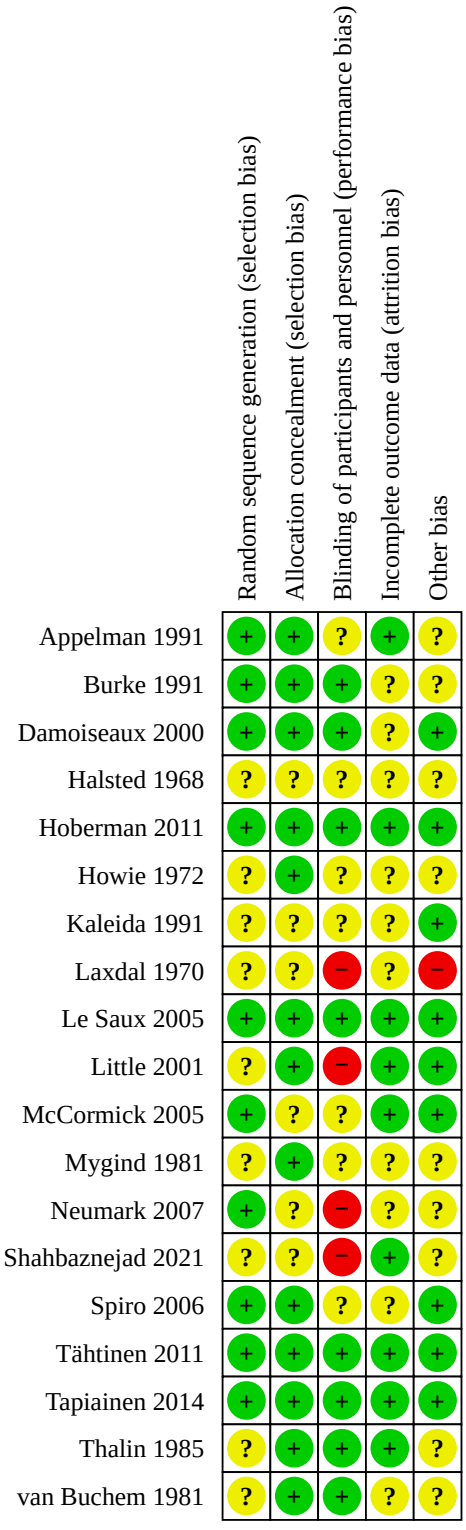
Two review authors (RPV, SLS) independently assessed the methodological quality of the included trials. We resolved any disagreements by discussion. We assessed the methodological quality of the included studies as described in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011):

- sequence generation;
- allocation concealment;
- blinding;
- incomplete outcome data;
- selective outcome reporting; and
- other sources of bias.

We used the Cochrane risk of bias tool to assign a judgement about the adequacy of each domain: 'low', 'high' or 'unclear' risk of bias. Results of the 'Risk of bias' assessment are presented in a risk of bias summary (Figure 1) and a risk of bias graph (Figure 2).

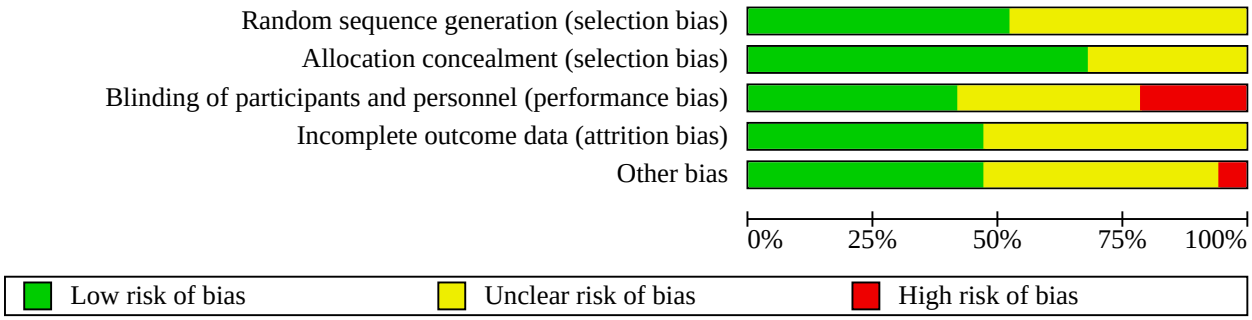
Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Figure 1. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.



Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.



Measures of treatment effect

We expressed dichotomous outcomes as risk ratio (RR) and risk difference (RD) with 95% confidence intervals (CIs). Additionally, we calculated the number needed to treat for an additional beneficial outcome (NNTB) and the number needed to treat for an additional harmful outcome (NNTH) (1/(absolute risk in exposed minus absolute risk in unexposed)).

Unit of analysis issues

We did not identify any studies with non-standard designs, such as cross-over trials and cluster-randomised trials.

Dealing with missing data

We tried to contact the trial authors to provide additional information in case of missing data.

Assessment of heterogeneity

We assessed the level of clinical heterogeneity between the trials by reviewing differences across trials in study population, setting, intervention and outcome measures used. In the absence of substantial clinical heterogeneity, we performed meta-analyses. We used the Chi^2 test (with a significance level set at P < 0.10), and the I^2 statistic (with I^2 values over 50%, suggesting substantial heterogeneity) to assess statistical heterogeneity. For the outcome of pain, we explored the magnitude of baseline risk and heterogeneity using L'Abbé plots (a graph of the proportion of participants with an outcome by the proportion of participants without an outcome) (Song 1999).

Assessment of reporting biases

We assessed reporting bias as outcome reporting bias and publication bias.

We assessed outcome reporting bias by comparing the outcomes reported in the published report against the study protocol or trial registry, whenever this could be obtained. If there was insufficient information to judge the risk of bias we noted this as an 'unclear' risk of bias (Higgins 2011).

If we were able to pool 10 or more studies in a single analysis, we planned to produce a funnel plot to explore possible publication bias. In the absence of bias the plot should approximately resemble a symmetrical (inverted) funnel (Higgins 2021).

Data synthesis

We analysed the data according to the intention-to-treat (ITT) principle, whereby all participants are analysed in the groups to which they were randomly allocated. We performed meta-analysis where we judged clinical heterogeneity to be minimal, to ensure that we would derive clinically meaningful results. We calculated treatment differences by the Mantel-Haenszel method using a random-effects model. We presented results separately for the reviews of antibiotics against placebo and immediate antibiotics versus expectant observation.

Subgroup analysis and investigation of heterogeneity

We did not perform subgroup analyses.

Sensitivity analysis

We did not perform sensitivity analyses.

Summary of findings and assessment of the certainty of the evidence

For each prespecified outcome, we used the GRADE approach to rate the certainty of the body of evidence as it relates to the studies that contribute data to the meta-analyses (GRADEpro GDT). There are four possible ratings: high, moderate, low and very low. The GRADE approach rates evidence from RCTs that do not have serious limitations as high certainty. However, several factors can lead to the downgrading of the evidence. The degree of downgrading is determined by the seriousness of these factors: study limitations, consistency of effect, imprecision, indirectness and publication bias. We used the methods and recommendations described in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2021). We justified all decisions to downgrade the certainty of studies using footnotes, and we made comments to aid the reader's understanding of the review where necessary.

We downgraded the certainty of evidence to moderate, low or very low depending on the presence of each of the following factors:

1. study limitations (risk of bias);
2. indirectness of evidence (directness of evidence);
3. imprecision (precision of results);
4. inconsistency (consistency of results); and
5. publication bias (existence of publication bias).

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

We included Summary of findings 1 for the review of antibiotics against placebo and Summary of findings 2 for the review of immediate antibiotics versus expectant observation. We included our primary outcomes and important secondary outcomes in the summary of findings tables:

1. pain at 24 hours;
2. pain at two to three days;
3. pain at four to seven days;
4. pain at 10 to 12 days;
5. adverse effects likely to be related to the use of antibiotics (vomiting, diarrhoea or rash);
6. abnormal tympanometry findings at two to four weeks; and
7. abnormal tympanometry findings at three months.

RESULTS

Description of studies

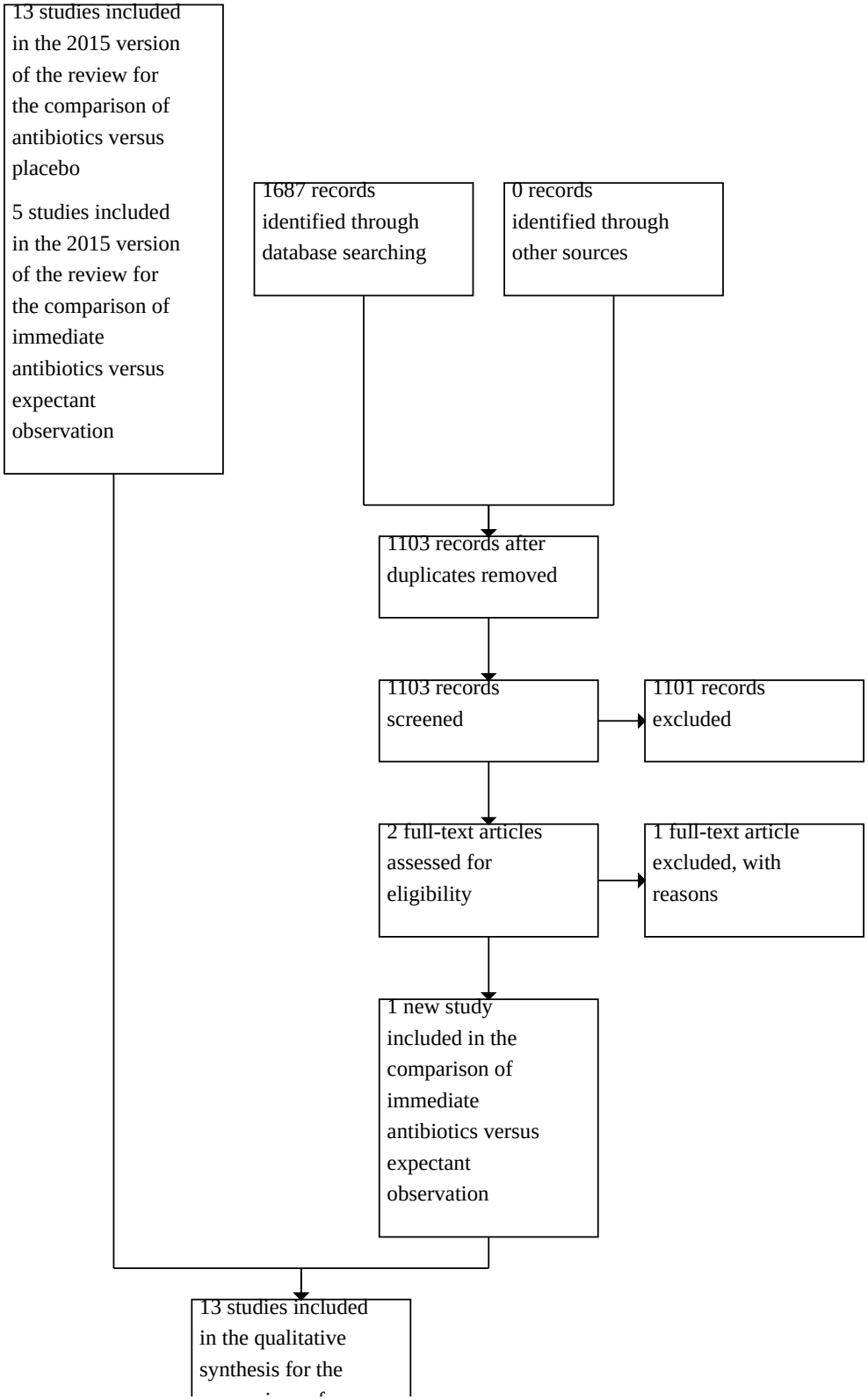
See Characteristics of included studies, Characteristics of excluded studies and Characteristics of ongoing studies tables.

Results of the search

This is an update of a Cochrane Review first published in the Cochrane Library in Issue 1, 1997 (Glasziou 1997) and updated in 1999 (Glasziou 1999), 2005 (Glasziou 2005), 2009 (Sanders 2009), 2013 (Venekamp 2013) and 2015 (Venekamp 2015). In the 2015 update of our review (Venekamp 2015), we identified 13 RCTs for the review of antibiotics against placebo (Appelman 1991; Burke 1991; Damoiseaux 2000; Halsted 1968; Hoberman 2011; Howie 1972; Kaleida 1991; Le Saux 2005; Mygind 1981; Tähtinen 2011; Tapiainen 2014; Thalin 1985; van Buchem 1981), while we judged five RCTs eligible for the review of immediate antibiotics versus expectant observation (Laxdal 1970; Little 2001; McCormick 2005; Neumark 2007; Spiro 2006) (Figure 3). We excluded a total of 11 studies for various reasons (Arguedas 2011; Casey 2012; Chaput 1982; Engelhard 1989; Liu 2011; Ostfeld 1987; Rudberg 1954; Ruohola 2003; Sarrell 2003; Tähtinen 2012; van Buchem 1985).

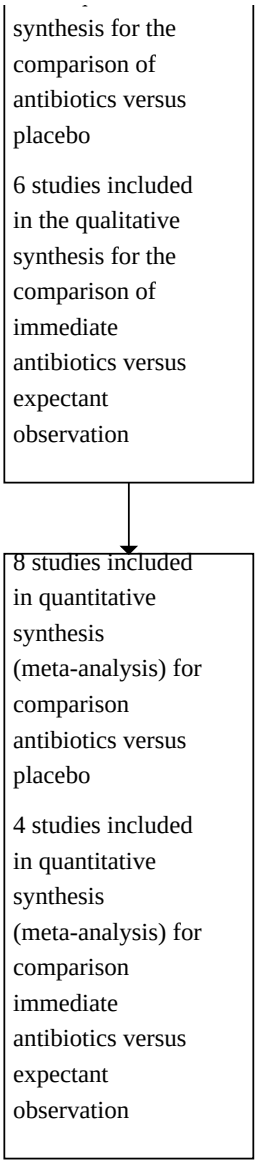
Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Figure 3.



Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Figure 3. (Continued)



With the updated search (April week 3, 2015 to February week 2, 2023), we retrieved a total of 1687 records. Removing duplicates left 1103 records. After screening titles and abstracts, we identified two potentially eligible articles. After reviewing the full text, we excluded one trial since this focussed on a specific subgroup of children with AOM, i.e. those with AOM and ear discharge due to a spontaneous perforation of the ear drum (Hay 2021). This left one new trial eligible for inclusion in the review of immediate antibiotics versus expectant observation (Shahbaznejad 2021) (Figure 3). We did not identify any additional trials after reviewing the reference lists of the full-text papers and relevant systematic reviews. We did not identify any additional ongoing trials apart from the one identified previously, which is still ongoing (the WATCH trial).

Included studies

Methods, participants, interventions and outcomes of the included studies are described in more detail in the table of Characteristics of included studies.

Antibiotics versus placebo

Thirteen trials including 3401 children (3938 AOM episodes) were eligible for the review of antibiotics against placebo (Appelman 1991; Burke 1991; Damoiseaux 2000; Halsted 1968; Hoberman 2011; Howie 1972; Kaleida 1991; Le Saux 2005; Mygind 1981; Tähtinen 2011; Tapiainen 2014; Thalin 1985; van Buchem 1981).

Design

Twelve trials were double-blind, placebo-controlled, parallel-group randomised clinical trials (Appelman 1991; Burke 1991;

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Damoiseaux 2000; Halsted 1968; Hoberman 2011; Howie 1972; Kaleida 1991; Le Saux 2005; Mygind 1981; Tähtinen 2011; Tapiainen 2014; Thalin 1985), while one trial had a 2 x 2 factorial design (van Buchem 1981).

Participants and settings

The sample size of the 13 individual trials ranged from 84 children (Tapiainen 2014) to 536 children (Kaleida 1991). The children were aged between two months and 15 years and 50% to 60% of included children were male. Three trials were performed in primary care (Burke 1991; Damoiseaux 2000; Tähtinen 2011), six in secondary care (Halsted 1968; Hoberman 2011; Howie 1972; Kaleida 1991; Le Saux 2005; Thalin 1985), and four in both primary and secondary care (Appelman 1991; Mygind 1981; Tapiainen 2014; van Buchem 1981). AOM was diagnosed by the presence of acute symptoms and otoscopic signs in nine trials (Appelman 1991; Burke 1991; Damoiseaux 2000; Halsted 1968; Hoberman 2011; Howie 1972; Kaleida 1991; Mygind 1981; van Buchem 1981), and by the presence of middle ear effusion at pneumatic otoscopy and/or tympanometry in three trials (Le Saux 2005; Tähtinen 2011; Tapiainen 2014), while the criteria were not clearly described in one trial (Thalin 1985).

Interventions and comparators

Two trials compared penicillin for seven days with placebo (Mygind 1981; Thalin 1985), four trials compared amoxicillin for seven to 14 days with or without myringotomy (surgical incision made in the tympanic membrane) with placebo (Burke 1991; Damoiseaux 2000; Kaleida 1991; Le Saux 2005), and four trials compared amoxicillin/clavulanate for seven to 10 days with placebo (Appelman 1991; Hoberman 2011; Tähtinen 2011; Tapiainen 2014). In one trial, ampicillin for 10 days was compared with pheneticillin and sulfisoxazole and placebo (Halsted 1968), while another trial compared erythromycin and triple sulphonamide with ampicillin, triple sulphonamide, erythromycin and placebo (Howie 1972). One trial, van Buchem 1981, had a 2 x 2 factorial design resulting in four treatment groups: (1) sham myringotomy plus antibiotics; (2) sham myringotomy plus placebo; (3) myringotomy plus antibiotics; and (4) myringotomy plus placebo. We used all arms of this trial, so this trial was included twice in certain analyses by including both the sham myringotomy plus antibiotic versus the sham myringotomy plus placebo comparison and the myringotomy plus antibiotic versus myringotomy plus placebo comparison data.

Outcomes

Pain

Five trials (1394 children) reported data on pain at 24 hours (Burke 1991; Le Saux 2005; Thalin 1985; Tähtinen 2011; van Buchem 1981), seven (2320 children) on pain at two to three days (Appelman 1991; Halsted 1968; Kaleida 1991; Le Saux 2005; Mygind 1981; Tähtinen 2011; Thalin 1985), seven (1347 children) on pain at four to seven days (Burke 1991; Damoiseaux 2000; Mygind 1981; Tähtinen 2011; Tapiainen 2014; Thalin 1985; van Buchem 1981), and one (278 children) on pain at 10 to 12 days (Hoberman 2011).

Adverse effects likely to be related to the use of antibiotics (vomiting, diarrhoea or rash)

Eight trials (2107 children) reported data on adverse effects likely to be related to the use of antibiotics, such as vomiting, diarrhoea or rash (Burke 1991; Damoiseaux 2000; Hoberman 2011; Le Saux 2005; Mygind 1981; Tähtinen 2011; Tapiainen 2014; Thalin 1985).

Abnormal tympanometry findings as a surrogate measure for hearing problems

Seven trials (2138 children) reported data on abnormal tympanometry findings at two to four weeks (Appelman 1991; Burke 1991; Kaleida 1991; Le Saux 2005; Mygind 1981; Tapiainen 2014; Thalin 1985), three (953 children) on abnormal tympanometry findings at six to eight weeks (Damoiseaux 2000; Kaleida 1991; Tapiainen 2014), and three (809 children) on abnormal tympanometry findings at three months (Burke 1991; Le Saux 2005; Mygind 1981), as a surrogate measure for hearing problems caused by middle ear fluid.

Tympanic membrane perforation

Five trials (1075 children) reported data on tympanic membrane perforation (Burke 1991; Hoberman 2011; Mygind 1981; Tähtinen 2011; Tapiainen 2014).

Progression of symptoms (contralateral otitis or late AOM recurrences)

Four trials (906 children) reported data on contralateral otitis (in unilateral cases) (Burke 1991; Hoberman 2011; Mygind 1981; Thalin 1985), while six trials (2200 children) reported data on late AOM recurrences (Hoberman 2011; Kaleida 1991; Le Saux 2005; Mygind 1981; Thalin 1985; van Buchem 1981).

Serious complications

Ten trials reported on serious complications including mastoiditis or meningitis (Burke 1991; Damoiseaux 2000; Hoberman 2011; Howie 1972; Kaleida 1991; Le Saux 2005; Mygind 1981; Tähtinen 2011; Tapiainen 2014; van Buchem 1981), while information on complications was not explicitly reported in three trials (Appelman 1991; Halsted 1968; Thalin 1985).

Long-term effects

One trial reported data on secondary care referrals at one year after randomisation as assessed by reviewing the children's notes (Burke 1991). Four children in the antibiotic group (4%) and seven in the placebo group (6%) were lost to follow-up.

One trial reported data on the proportion of children with AOM recurrences, secondary care referrals and ENT surgery at approximately 3.5 years after randomisation (Damoiseaux 2000). These long-term outcome data were collected by questionnaires. Questionnaires were returned in 168 of the 240 children (70%) that were originally randomised.

Funding

Seven studies received public funding from national funding bodies or from professional organisations (Damoiseaux 2000; Halsted 1968; Hoberman 2011; Kaleida 1991; Tähtinen 2011; Le Saux 2005; Tapiainen 2014). For two of these, additional treatments provided to both study groups were provided by pharmaceutical companies (Damoiseaux 2000; Le Saux 2005), and for one, the study drugs were provided by the pharmaceutical industry (Halsted 1968). Three studies indicated that part funding was received from industry for conduct of the trial or for the provision of study drugs (Burke 1991; Howie 1972; Mygind 1981). One study was fully funded by the pharmaceutical industry (Appelman 1991). Two studies did not report on funding received (Thalin 1985; van Buchem 1981).

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Conflicts of interest

Authors of nine studies did not report on researcher conflicts of interest (Appelman 1991; Burke 1991; Damoiseaux 2000; Halsted 1968; Howie 1972; Kaleida 1991; Mygind 1981; Thalin 1985; van Buchem 1981). Two studies reported some researchers receiving industry funding for consultancy, honoraria or travel (Hoberman 2011; Tähtinen 2011). Two studies declared that researchers had no conflicts of interest (Le Saux 2005; Tapiainen 2014).

Immediate antibiotics versus expectant observation

Six trials including a total of 1556 children were eligible for the review of immediate antibiotics versus expectant observation (Laxdal 1970; Little 2001; McCormick 2005; Neumark 2007; Shahbaznejad 2021; Spiro 2006).

Design

All trials were open-label, parallel-group randomised clinical trials.

Participants and settings

The sample size of the six individual trials ranged from 142 children (Laxdal 1970) to 407 children (Shahbaznejad 2021). The children were aged 15 years and younger, and 50% to 60% of included children were male. Two trials were performed in primary care (Little 2001; Neumark 2007), and four in secondary/tertiary care (Laxdal 1970; McCormick 2005; Shahbaznejad 2021; Spiro 2006). AOM was diagnosed by the presence of acute symptoms and otoscopic signs in three trials (Laxdal 1970; Little 2001; McCormick 2005), by pneumatic otoscopy or preferably an aural microscope in one trial (Neumark 2007), while diagnostic criteria or modalities used were unclear in two trials (Shahbaznejad 2021; Spiro 2006).

Intervention and comparators

In two of these trials, provision of an immediate antibiotic script was compared with an antibiotic script with instructions not to commence antibiotic treatment unless the child was not better or was worse at 48 hours (Spiro 2006) or 72 hours (Little 2001). In these trials, 24% (36/150) and 38% (50/132) of children in the delayed arms reported using antibiotics at some stage during the illness.

The other four trials compared immediate antibiotics with a watchful waiting approach (Laxdal 1970; McCormick 2005; Neumark 2007; Shahbaznejad 2021). In the Laxdal 1970 trial, children in the control group were closely monitored, especially during the first 48 hours and particularly when severe involvement was evident. In the McCormick 2005 trial, antibiotics were administered to the watchful waiting group if a child returned to the office with a treatment failure or recurrence (four children in the expectant observation group had received antibiotics by day four). In the Neumark 2007 trial, 5% (4/87) of children randomised to the watchful waiting group received antibiotics due to treatment failure, while the proportion of children receiving antibiotics during follow-up in the watchful waiting group was not reported in the remaining trial (Shahbaznejad 2021).

Outcomes

One trial did not report any data on our primary or secondary outcomes (Laxdal 1970), leaving four trials from which relevant data could be extracted (Little 2001; McCormick 2005; Neumark 2007; Spiro 2006).

Pain

Data on pain at two to three days and three to seven days could be derived from one trial (396 included in analysis) (Shahbaznejad 2021) and four trials (959 children) (Little 2001; McCormick 2005; Neumark 2007; Spiro 2006), respectively. The data on pain from the Little 2001 trial have been derived from data from the individual patient data (IPD) meta-analysis (Rovers 2006), while the data on pain from the McCormick 2005 trial have been provided by the author. One trial (247 children) reported data on pain at 11 to 14 days (Spiro 2006).

Adverse effects likely to be related to the use of antibiotics (vomiting, diarrhoea or rash)

Three trials (946 children) reported data on adverse events likely to be related to the use of antibiotics such as vomiting, diarrhoea or rash (Little 2001; Shahbaznejad 2021; Spiro 2006).

Abnormal tympanometry findings as a surrogate measure for hearing problems

One trial (207 children) reported data on abnormal tympanometry findings at two to four weeks (McCormick 2005).

Tympanic membrane perforation

Two trials (575 children) reported data on tympanic membrane perforation (Neumark 2007; Shahbaznejad 2021).

Progression of symptoms (contralateral otitis or late AOM recurrences)

None of the trials reported data on contralateral otitis (in unilateral cases), while two trials (584 children) reported data on late AOM recurrences (McCormick 2005; Shahbaznejad 2021).

Serious complications

Four trials reported on serious complications including mastoiditis or meningitis (McCormick 2005; Neumark 2007; Shahbaznejad 2021; Spiro 2006), while information on complications was not explicitly reported in one trial (Little 2001).

Long-term effects

One trial reported data on further ear pain episodes at three months and one year after randomisation (Little 2001). These long-term outcome data were collected by questionnaires. Questionnaires were returned in 219 of the 315 children (70%) that were originally randomised at one year.

Funding

All six studies received public funding from national funding bodies or professional organisations (Laxdal 1970; Little 2001; McCormick 2005; Neumark 2007; Shahbaznejad 2021; Spiro 2006), including one that received study drugs provided by the pharmaceutical industry (Laxdal 1970).

Conflicts of interest

Three studies declared researchers had no conflicts of interest (Little 2001; McCormick 2005; Shahbaznejad 2021), and three studies did not report on researcher conflicts of interest (Laxdal 1970; Neumark 2007; Spiro 2006).

Excluded studies

We excluded 12 studies after reviewing the full text. Three were non-randomised studies (Ostfeld 1987; Rudberg 1954; van

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Buchem 1985), while three other studies had no comparison of antibiotics with placebo or expectant observation (Casey 2012; Engelhard 1989; Sarrell 2003). Two trials studied the effectiveness of short- versus long-course antibiotics (Arguedas 2011; Chaput 1982), one trial studied a single-dose antibiotic with slow versus immediate-release formulations (Liu 2011), whereas another trial was conducted in children with ventilation tubes (Ruohola 2003). Moreover, we excluded one trial report as this study reported on the effectiveness of immediate versus delayed antibiotic prescription based on a secondary analysis of a placebo-controlled trial (Tähtinen 2012). Finally, we excluded another trial report as this study reported on the effectiveness of immediate oral antibiotics versus immediate topical antibiotics versus delayed oral antibiotics for children with acute otitis media and ear discharge due to a spontaneous perforation of the ear drum (Hay 2021). This is a population that generally has a poorer prognosis with higher rates of ear pain and/or fever at three to seven days and more AOM recurrences and hearing problems at three months than children presenting with AOM without ear discharge (Rovers 2006; Smith 2010).

Risk of bias in included studies

The methodological quality of the included studies was generally high. For further details on the risk of bias in included studies, see the risk of bias summary (Figure 1) and risk of bias graph (Figure 2).

Allocation

Random sequence generation was adequate in seven of the 13 trials (Appelman 1991; Burke 1991; Damoiseaux 2000; Hoberman 2011; Le Saux 2005; Tähtinen 2011; Tapiainen 2014), and in three of the six included trials (McCormick 2005; Neumark 2007; Spiro 2006), respectively.

Concealment of allocation was adequately described in 11 of the 13 included trials comparing antibiotics with placebo (Appelman 1991; Burke 1991; Damoiseaux 2000; Hoberman 2011; Howie 1972; Le Saux 2005; Mygind 1981; Tähtinen 2011; Tapiainen 2014; Thalin 1985; van Buchem 1981), and two out of six trials comparing immediate antibiotics with expectant observation (Little 2001; Spiro 2006).

Blinding

All included trials in the review of antibiotics against placebo stated that they were double-blinded. However, we judged blinding to be adequate in eight of the 13 included trials (Burke 1991; Damoiseaux 2000; Hoberman 2011; Le Saux 2005; Tähtinen 2011; Tapiainen 2014; Thalin 1985; van Buchem 1981). All six trials comparing immediate antibiotics with expectant observation were open-label trials (Laxdal 1970; Little 2001; McCormick 2005; Neumark 2007; Shahbaznejad 2021; Spiro 2006). As a consequence, reporting of the child's symptoms by parents was not blinded in these trials. However, investigators were blinded in two of the six trials (McCormick 2005; Spiro 2006).

Incomplete outcome data

The loss to follow-up was below 5% in eight of the 13 trials comparing antibiotics with placebo (Appelman 1991; Burke 1991; Hoberman 2011; Howie 1972; Le Saux 2005; Tähtinen 2011; Tapiainen 2014; Thalin 1985). Loss to follow-up was high in three trials with a total loss to follow-up of 15% (van Buchem 1981), 7% (Kaleida 1991) and 12% (Damoiseaux 2000), respectively. However, one of these trials included all randomised patients in the primary analysis at day four (Damoiseaux 2000). In two of the 13 trials, the total numbers of loss to follow-up/exclusions are described, but it was unclear from which treatment group children were excluded (Halsted 1968; Mygind 1981). For the review of immediate antibiotics against expectant observation, the loss to follow-up was below 5% in two of the five trials (McCormick 2005; Neumark 2007). The total loss to follow-up in the other trials was 11% (Laxdal 1970), 10% (Little 2001), 6% (Spiro 2006) and 5.3% (Shahbaznejad 2021), respectively.

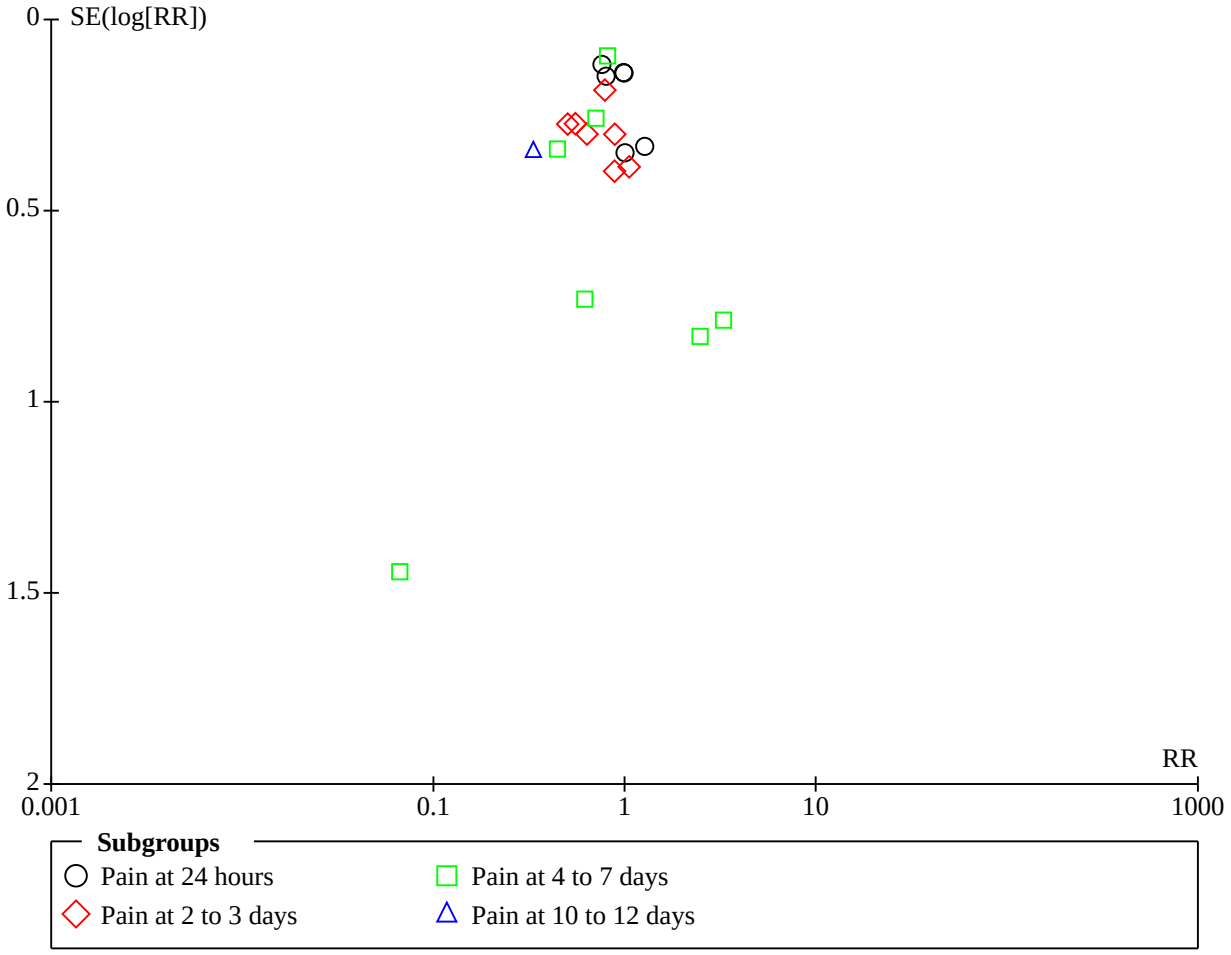
Selective reporting

Three of the 13 included trials comparing antibiotics with placebo had their study protocol registered (Hoberman 2011; Tähtinen 2011; Tapiainen 2014), while in the remaining 10 trials no study protocol or trial registry was found and we judged the risk of outcome reporting bias to be unclear (Appelman 1991; Burke 1991; Damoiseaux 2000; Halsted 1968; Howie 1972; Kaleida 1991; Le Saux 2005; Mygind 1981; Thalin 1985; van Buchem 1981). For the review of immediate antibiotics versus expectant observation, two of the six included trials had their study protocol registered (Shahbaznejad 2021; Spiro 2006), while in the remaining four trials no study protocol or trial registry was found and we judged the risk of outcome reporting bias to be unclear (Laxdal 1970; Little 2001; McCormick 2005; Neumark 2007).

The funnel plot for pain at the various time points did not reveal asymmetry (Figure 4).

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Figure 4. Funnel plot of comparison: 1 Antibiotic versus placebo, outcome: 1.1 Pain.



Other potential sources of bias

Eight of the 13 included trials comparing antibiotics with placebo used intention-to-treat (ITT) analyses, while in the other five this was not clear (Halsted 1968; Howie 1972; Mygind 1981; Thalin 1985; van Buchem 1981). For the review of immediate antibiotics versus expectant observation, three of the six included trials used ITT analyses, while this was not clear in the other three trials (Laxdal 1970; Neumark 2007; Shahbaznejad 2021).

No other potential sources of bias could be detected in the included trials, except for the Laxdal 1970 trial, which we judged as having a high risk of detection bias since children in the control group were subjected to very close scrutiny, especially during the first 48 hours and particularly when severe involvement was evident. However, this trial did not report any data on our primary or secondary outcomes.

Effects of interventions

See: **Summary of findings 1** Antibiotics versus placebo for acute otitis media in children; **Summary of findings 2** Immediate antibiotics versus expectant observation for acute otitis media in children

Antibiotics versus placebo

Primary outcomes

1. Proportion of children with pain at various time points

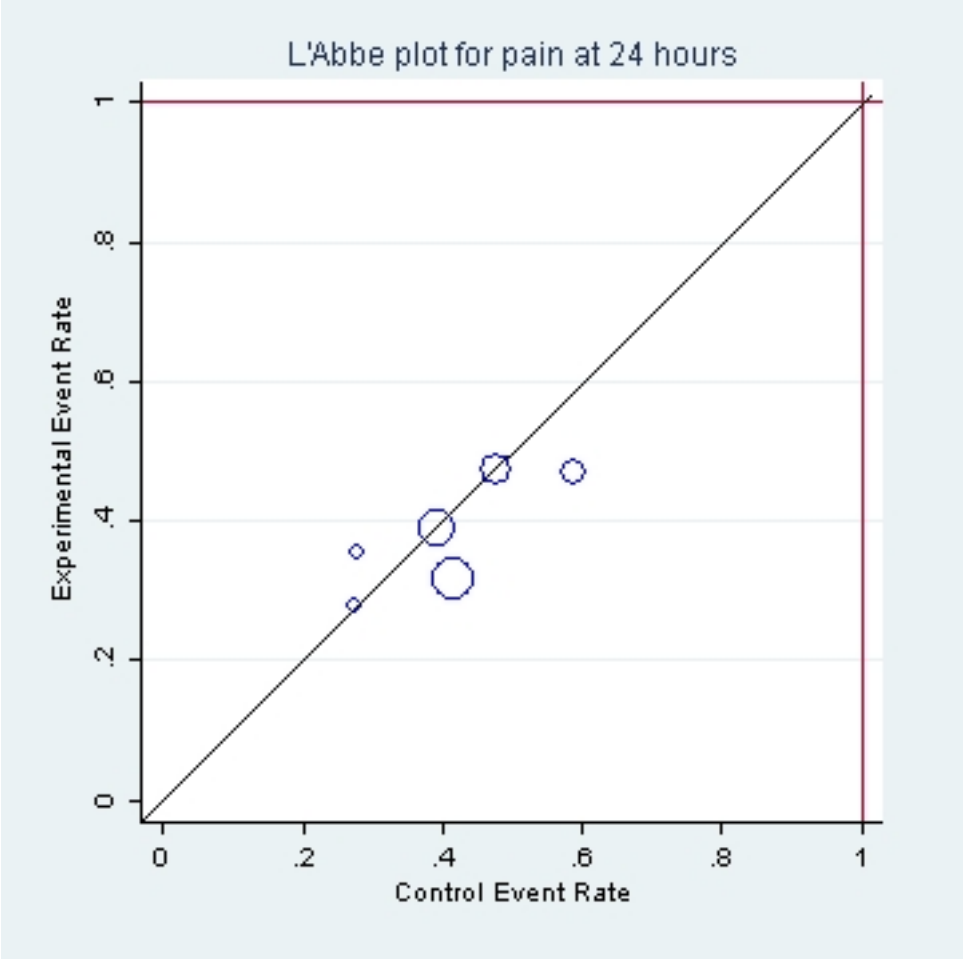
Antibiotics do not reduce pain at 24 hours (risk ratio (RR) 0.89, 95% confidence interval (CI) 0.78 to 1.01; 5 trials, 1394 children) and at four to seven days (RR 0.76, 95% CI 0.50 to 1.14; 7 trials, 1264 children), but result in almost a third fewer children having pain at two to three days (RR 0.71, 95% CI 0.58 to 0.88; 7 trials, 2320 children), and likely result in two-thirds fewer having pain at 10 to 12 days (RR 0.33, 95% CI 0.17 to 0.66; 1 trial, 278 children) (Analysis 1.1).

This means that 5% (95% CI 2% to 7%) fewer children had pain after two to three days (number needed to treat for an additional beneficial outcome (NNTB) 20; 95% CI 14 to 50), and 14% (95% CI 6% to 22%) fewer children had pain after 10 to 12 days (NNTB 7; 95% CI 4 to 16) with antibiotics, respectively.

Plots of the event rate (pain) in the treatment and control groups for each study at 24 hours and two to three days are reported in Figure 5 and Figure 6.

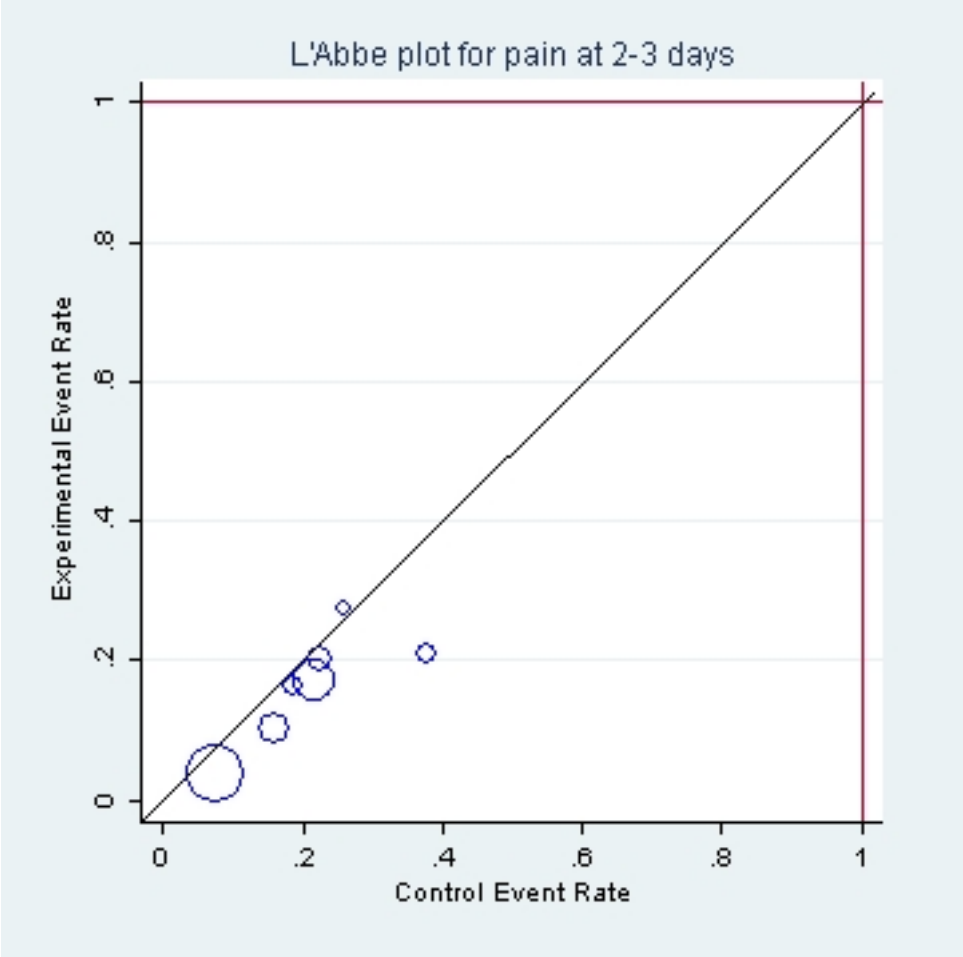
Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Figure 5. L'Abbé plot of the rates of pain at 24 hours for the placebo (control) versus antibiotic (experimental) group.



Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Figure 6. L'Abbé plot of the rates of pain at two to three days for the placebo (control) versus antibiotic (experimental) group.



Certainty of evidence

We judged the evidence on pain at 24 hours, two to three days and four to seven days to be of high certainty, while we judged the evidence on pain at 10 to 12 days to be of moderate certainty. We downgraded the evidence for pain at days 10 to 12 as this outcome was not specified a priori in this trial (secondary analysis).

2. Adverse effects likely to be related to the use of antibiotics

Antibiotics result in a 38% (95% CI 16% to 63%; 8 trials, 2107 children) relative increase in the risk of adverse effects likely to be related to the use of antibiotics (defined as vomiting, diarrhoea or rash) compared with placebo; 27% (283/1044) of children treated with antibiotics versus 20% (208/1063) of children treated with placebo experienced vomiting, diarrhoea or rash (Analysis 1.2). The number needed to treat for an additional harmful outcome (NNTH) was 14 (9 to 26).

Certainty of evidence

We judged the evidence for adverse effects likely to be related to the use of antibiotics (vomiting, diarrhoea or rash) to be of high certainty.

Secondary outcomes

1. Abnormal tympanometry findings at various time points

Antibiotics result in a 17% (95% CI 4% to 28%; 7 trials, 2138 children) relative reduction in the risk of abnormal tympanometry findings at two to four weeks (Analysis 1.3). This means that 9% (95% CI 5% to 13%) fewer children had abnormal tympanometry findings at two to four weeks (NNTB 11, 95% CI 7 to 20).

However, antibiotics do not reduce the risk of abnormal tympanometry findings at six to eight weeks (RR 0.89, 95% CI 0.70 to 1.13; 3 trials, 953 children), and at three months (RR 0.94, 95% CI 0.66 to 1.34; 3 trials, 809 children) (Analysis 1.3). Furthermore, audiometry was done in only two studies and incompletely reported. The two studies that used audiograms were van Buchem 1981 and Kaleida 1991: (i) van Buchem 1981 reported that, "After one month, 31% of the patients showed an air/bone gap of more than 20 dB. After two months, this was still the case with 19% of the patients. Here again, there were no significant differences between the groups"; (ii) Kaleida 1991 stated that "Analysis of hearing acuity in children two years of age and older indicated that elevated hearing thresholds ... bore no apparent relationship ... to mode of treatment (amoxicillin versus placebo)".

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Certainty of evidence

We judged the evidence for abnormal tympanometry findings at the various time points to be of high certainty.

2. Tympanic membrane perforation

Antibiotic treatment is associated with a 57% (95% CI 11% to 79%; 5 trials, 1075 children) relative reduction in the risk of tympanic membrane perforation compared with placebo (Analysis 1.4). However, absolute benefits of antibiotics appeared to be small: 3% (95% CI 1% to 5%) fewer children had a tympanic membrane perforation. Therefore, 33 children (95% CI 20 to 100) needed to be treated to prevent one child experiencing a tympanic membrane perforation.

Certainty of evidence

We judged the evidence for tympanic membrane perforation to be of high certainty.

3. Contralateral otitis

Antibiotics result in a 51% (95% CI 5% to 75%; 4 trials, 906 children) relative reduction in the development of contralateral otitis compared with placebo (Analysis 1.5). This means that 9% (95% CI 5% to 13%) fewer children had contralateral otitis (NNTB 11, 95% CI 7 to 20).

Certainty of evidence

We judged the evidence for contralateral otitis to be of high certainty.

4. Acute otitis media (AOM) recurrences

AOM recurrences were common. Burke 1991 stated "The mean number of recorded recurrences of otitis media or acute red ear was 0.70 (range 0 to 4) in the antibiotic group and 0.63 (range 0 to 7) in the placebo group and this difference was not significant (difference 0.06, 95% CI -0.22 to 0.339)." Six other trials reported the proportions who relapsed; the combined results were consistent with Burke's findings and indicate that antibiotics do not reduce the occurrence of late AOM recurrences compared with placebo (RR 0.94, 95% CI 0.79 to 1.11; 6 trials, 2200 children) (Analysis 1.6).

Certainty of evidence

We judged the evidence for late AOM recurrences to be of high certainty.

5. Serious complications related to AOM

Few serious complications occurred in either the antibiotic treatment group or the control group. In just over 3000 children studied, only one case of mastoiditis occurred in both the antibiotic group (Mygind 1981) and the placebo group (Hoberman 2011). Moreover, one child suffered from meningitis (Damoiseaux 2000), pneumococcal bacteraemia and radiologically confirmed pneumonia (Hoberman 2011) in the placebo group and one child had transient facial paralysis in the antibiotic group (Kaleida 1991). Hence, the applicability of these findings to groups of children in whom serious complications such as mastoiditis is common is uncertain. One of the excluded studies did report high rates of mastoiditis (Rudberg 1954). This was an open, semi-randomised study conducted in Sweden in 1954. Participants were randomised by case-sheet number but a proportion (about 30 of 220) requested, and were granted, entry to the penicillin group. The rate of mastoiditis was 17% in the untreated group versus 1.5% in the sulphonamide-treated group and 0% in the penicillin-treated group. The biases of this study (semi-randomisation and unblinded outcome assessment) are unlikely to explain such a large difference.

Certainty of evidence

We judged the evidence for serious complications to be of low certainty. We downgraded the evidence from high certainty due to serious concerns about imprecision (we considered the sample size to be insufficient to draw any definite conclusions based on these data).

6. Long-term effects

Based on reviewing children's notes, antibiotics do not reduce the number of secondary care referrals at one year after randomisation: 7/110 (6%) in the antibiotic group and 9/111 (8%) in the placebo group (RR 0.78, 95% 0.30 to 2.03).

Based on questionnaires returned by parents approximately 3.5 years after initial randomisation, AOM recurrence was observed in 63% (47/75) of children in the immediate antibiotics group and in 43% (37/86) of the placebo group. Immediate antibiotics therefore likely result in a 46% (95% CI 8% to 97%) relative increase in the risk of AOM recurrences compared to placebo. This means that 20% (95% CI 5% to 35%) more children had AOM recurrences (NNTH 5, 95% CI 2 to 20) when receiving immediate antibiotics. No between-group differences were observed for secondary care referrals. Furthermore, antibiotics were not associated with a statistically significant reduction in the number of ear, nose and throat surgeries (RR 0.68, 95% CI 0.40 to 1.17).

Certainty of evidence

We judged the evidence for long-term effects at one year to be of high certainty, while we judged the 3.5 years data to be of moderate certainty. We mainly downgraded the evidence because of the high proportion of children that were not included in the analysis (30%), which introduced a significant risk of (attrition) bias.

Immediate antibiotics versus expectant observation

Primary outcomes

1. Proportion of children with pain at various time points

The evidence suggests that immediate antibiotics may result in a reduction of pain at two to three days (RR 0.53, 95% CI 0.35 to 0.79; NNTB 8; 1 trial, 396 children), but probably do not reduce the risk of pain at three to seven days (RR 0.75, 95% CI 0.50 to 1.12; 4 trials, 959 children), and may not reduce the risk of pain at 11 to 14 days (RR 0.91, 95% CI 0.75 to 1.10; 1 trial, 247 children) compared with expectant observation (Analysis 2.1).

Certainty of evidence

We judged the evidence on pain at three to seven days to be of moderate certainty, while we judged the evidence on pain at two to three days and 11 to 14 days to be of low certainty. We downgraded the evidence because of study limitations (risk of detection bias) and further downgraded the evidence for pain at two to three days because of imprecision (outcome data derived from one trial), and pain at days 11 to 14 because of the number of children that were 'lost to follow-up' (13%), which introduced a risk of (attrition) bias.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

2. Adverse effects likely to be related to the use of antibiotics

Immediate antibiotics result in an 87% (95% CI 39% to 151%; 3 trials, 946 children) relative increase in the risk of adverse effects likely to be related to the use of antibiotics (defined as vomiting, diarrhoea or rash) compared with expectant observation; 22% (101/456) of children treated with immediate antibiotics versus 11% (56/490) of children treated with expectant observation experienced vomiting, diarrhoea or rash (Analysis 2.2). The NNTH was 10.

Certainty of evidence

We judged the evidence for adverse effects likely to be related to the use of antibiotics (vomiting, diarrhoea or rash) to be of high certainty.

Secondary outcomes

1. Abnormal tympanometry findings at various time points

Immediate antibiotics probably do not reduce the proportion of children with abnormal tympanometry findings at four weeks compared with expectant observation (RR 1.03, 95% CI 0.78 to 1.35; 1 trial, 207 children; Analysis 2.3).

Certainty of evidence

We judged the evidence for abnormal tympanometry findings at four weeks to be of moderate certainty. We downgraded the evidence from high certainty as the number of children that were 'lost to follow-up' in the immediate antibiotics group was substantially lower than in the expectant observation group (4% versus 11%), thereby introducing a risk of (attrition) bias.

2. Tympanic membrane perforation

Only three tympanic membrane perforations (two in the immediate antibiotics group versus one in the expectant observation group) were observed in one trial (396 children) (Shahbaznejad 2021), whereas none were observed in either group in the other trial (179 children) (Neumark 2007). The evidence suggests that immediate antibiotics may not reduce the risk of tympanic membrane perforation (RR 2.21, 95% CI 0.20 to 24.21; 3 trials, 575 children; Analysis 2.4) compared with expectant observation.

Certainty of evidence

We judged the data on tympanic membrane perforation to be of low certainty. We downgraded the evidence because of study limitations (risk of detection bias) and imprecision.

3. Contralateral otitis

None of the trials reported data on contralateral otitis.

4. AOM recurrences

The evidence suggests that immediate antibiotics may not reduce the risk of AOM recurrences (RR 1.04, 95% CI 0.77 to 1.41; 2 trials, 584 children; Analysis 2.5) compared with expectant observation.

Certainty of evidence

We judged the evidence for AOM recurrences to be of low certainty. We downgraded the evidence because of study limitations (risk of detection bias and risk of attrition bias).

5. Serious complications related to AOM

No serious complications occurred in either the immediate antibiotic group or the expectant observation group.

Certainty of evidence

We judged the evidence for serious complications to be of very low certainty since we considered the sample size to be insufficient to draw any robust conclusions based on these data.

6. Long-term effects

Immediate antibiotics may not reduce the risk of parent-reported ear pain episodes at one year (odds ratio (OR) 1.03, 95% CI 0.60 to 1.78; 1 trial, 219 children) compared with expectant observation (delayed antibiotics).

Certainty of evidence

We judged the evidence for long-term effects to be of low certainty. We mainly downgraded because this evidence was derived from a secondary analysis and because of the high proportion of children that were not included in the analysis at one year (30%), which introduced a significant risk of (attrition) bias.

DISCUSSION

Summary of main results

This review reveals that antibiotics have no early effect on pain, a slight effect on pain in the days following and only a modest effect on the number of children with tympanic perforations, contralateral otitis episodes and abnormal tympanometry findings at two to four weeks compared with placebo in children with acute otitis media (AOM). However, in applying these results, there are a number of issues to consider, including the individual potential for serious complications and subgroups of children in whom there may be greater benefits.

Overall completeness and applicability of evidence

Does the effect vary in different clinical groups? The estimated numbers needed to treat for the outcome pain are for the 'average' case and may vary in subgroups. Several studies reported higher rates of failure of placebo treatment amongst children less than two years of age and those with bilateral disease (Appelman 1991; Burke 1991; Damoiseaux 2000; Hoberman 2011; Tähtinen 2011), and another trial has suggested that most benefit is seen in children with high fever or vomiting (Little 2001). Moreover, some studies found that children with bilateral AOM differ with regards to clinical and microbiological (increased presence of (non-typeable) *H influenzae*) characteristics compared with children with unilateral AOM (Barkai 2009; McCormick 2007). However, an individual patient data (IPD) meta-analysis performed in 2006 (Rovers 2006), using data from six high-certainty RCTs, including a total of 1643 children, which were also included in this review as individual trials (Appelman 1991; Burke 1991; Damoiseaux 2000; Le Saux 2005; Little 2001; McCormick 2005), demonstrated that the relative effects of antibiotics were not significantly modified by either age or bilateral disease alone, but the absolute differences were larger in the younger patients (less than two years) with bilateral disease and in those with both AOM and otorrhoea. Whether topical antibiotics are an appropriate alternative to oral antibiotics in children with both AOM and otorrhoea due to a spontaneous tympanic membrane perforation remains to be elucidated (Hay

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

2021). Further analysis of the IPD has shown that age younger than two years is an independent predictor of the development of asymptomatic middle ear effusion (Koopman 2008). This analysis also found that antibiotic therapy has a marginal effect on the development of asymptomatic middle ear effusion in children with AOM.

Does the impact vary by duration and dose of antibiotics? Most trials use seven days of antibiotic treatment. A meta-analysis of a short (less than seven days) versus long (more than seven days) course of antibiotics reported that risk of treatment failure at one month was higher with short courses of antibiotics (odds ratio (OR) 1.34, 95% confidence interval (CI) 1.15 to 1.55) (Kozyrskyj 2010). However, the absolute difference in treatment effect was small (3%) and short courses of antibiotics were associated with a statistically significant reduction in gastrointestinal adverse events compared with longer courses. A recommendation regarding the most appropriate dose of antibiotics is not possible due to a lack of sufficient data.

What are the potential consequences of not using antibiotics? Besides the immediate pain associated with AOM, there are some more serious complications. Though only two cases of mastoiditis were reported in the included trials (one child received antibiotics and one child was assigned to placebo), a semi-randomised trial in Sweden in 1954 reported a rate of 17% in the untreated group versus none in the penicillin-treated groups (Rudberg 1954). In populations or sub-populations where mastoiditis is still judged a frequent problem, such as in some low-income countries, antibiotic treatment would be strongly advised (Berman 1995).

Of note is an article that revealed that doctors commonly over-diagnose AOM (Rothman 2003). What effect might this have on the efficacy of antibiotics (or any treatment)? One effect will be to blunt any treatment effect by dilution (from the cases of non-AOM). The results of two performed trials (Hoberman 2011; Tähtinen 2011), in which AOM had been diagnosed with the use of stringent criteria (including pneumatic otoscopic examination in one trial) (Tähtinen 2011), underline this phenomenon. Nevertheless, physicians in daily practice are likely to use the same diagnostic methods (perhaps even less stringent) as used in the majority of the included trials in this review. As a consequence, the effectiveness of antibiotics reported in this review is likely to be a true reflection of the effectiveness in actual clinical practice. However, if new and more accurate diagnostic procedures are introduced in future daily practice, then the current estimate of effectiveness will have to be reconsidered.

Certainty of the evidence

The methodological quality of the included studies was generally high. We judged the evidence to be of high certainty for most of the outcomes in the review of antibiotics against placebo. We judged the certainty of evidence to be moderate for pain at 10 to 12 days and long-term effects (3.5 years data), and low for serious complications. We downgraded the evidence mainly because of the risk of reporting bias (pain at 10 to 12 days) and attrition bias (long-term effects), and because of sample size considerations (serious complications).

For the review of immediate antibiotics versus expectant observation, we judged the evidence to be of low certainty for most of the outcomes. We downgraded the evidence mainly because of study limitations (risk of detection bias and/or selection bias), imprecision and sample size considerations.

Potential biases in the review process

There was some clinical heterogeneity amongst the included trials. For example, patients were recruited from different settings (general practice, ear, nose and throat, and paediatric clinics). However, the majority of included trials did use a diagnostic method (clinical diagnosis of AOM as inclusion criteria) that resembles daily clinical practice. Besides, duration and dosage of the antibiotic treatment varied to some extent. For the review of antibiotics against placebo, the duration of antibiotic treatment varied from seven to 14 days. However, we do not consider this as a major drawback, since most trials used seven days of antibiotic treatment and current evidence indicates only a small absolute treatment difference (3%) in treatment failure at one month in favour of a long (more than seven days) versus a short (less than seven days) course of antibiotics. Moreover, the primary outcome of this review (proportion of children with pain) is reported within the first seven days of antibiotic treatment. In addition, we assessed funnel plots for potential reporting biases for the primary analysis (Figure 4). No asymmetry could be detected in the included trials.

Agreements and disagreements with other studies or reviews

This review demonstrated that at 24 hours, pain had recovered spontaneously in 60% of children and that the majority had recovered in the following two to 12 days, regardless of whether they had received placebo or antibiotics. However, the IPD meta-analysis, which included six of the trials included in this review, revealed a slower rate of recovery with only 22% of children experiencing spontaneous recovery at 24 hours (Rovers 2006). There are a number of possible explanations for this. First, data from older trials were not included in the IPD meta-analysis and consequently the study population may reflect a higher threshold of doctor visitation; for example, the children may be 'sicker' or presenting to the doctor later in the course of their illness. Variation in the definitions of pain/no pain cut-offs amongst the trials included in the reviews may also explain some of this variation. The IPD meta-analysis indicated that antibiotics had the greatest effect compared with placebo at day three (Rovers 2006).

A previous meta-analysis has examined whether antibiotics were indicated for AOM in children and concluded that the answer is a qualified 'yes' (Rosenfeld 1994). It estimated a NNTB of seven for "primary control" (complete clinical resolution), compared with our NNTB of 20 for symptom relief. The difference may be the consequence of our focus on patient-oriented outcomes, such as pain, rather than clinical signs, such as eardrum appearance. The previous systematic review suggests that where mastoiditis is not a concern, primary care physicians could weigh the benefits against the risks of adverse effects from antibiotics with their patients. This statement is in agreement with the findings of our review as adverse events such as diarrhoea, vomiting or rash were more common in children receiving antibiotics. In the IPD meta-analysis, the most commonly described adverse effect of antibiotic treatment was diarrhoea, ranging from 2% to 14% in controls and from 4% to 21% in those given antibiotics (Rovers 2006). Occurrence of rash ranged from 2% to 6% in the control groups and from 1% to 8% in the antibiotic groups. A recent systematic review and meta-analysis on common harms of amoxicillin revealed that harms were

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

poorly reported in most placebo-controlled trials (Gillies 2014). In this review, diarrhoea was attributed to amoxicillin only in the form of amoxicillin/clavulanate. Amoxicillin did increase the risk of candidiasis compared with placebo, but no association between amoxicillin and rash or vomiting was observed (Gillies 2014). Bacterial resistance to antibiotics is also a consideration, with an association between antibiotic use and resistant bacteria demonstrated for many important pathogens (Arnold 2005).

Several trials evaluated a management strategy for AOM in which an expectant observational approach is used (Laxdal 1970; Little 2001; McCormick 2005; Neumark 2007; Spiro 2006). In one of these trials, pain and malaise at day three were greater amongst those randomised to receive an antibiotic prescription with advice to fill it only if there was no improvement after 72 hours compared with those receiving immediate antibiotics (Little 2001). In a secondary analysis of the trial, no difference was found between delayed and immediate treatment groups in ear function and ear pain at three and 12 months (Little 2006). Another study using a similar prescribing approach and examining clinical outcomes at four to six days found no difference between immediate and delayed antibiotic groups (Spiro 2006). In the third study (McCormick 2005), immediate antibiotic treatment was associated with decreased numbers of treatment failures and improved symptom control at day four and day 12 compared with those allocated to expectant observation with no prescription. Neumark 2007, in a similar comparison, found that immediate antibiotics provided some symptomatic benefit; children who received antibiotics had less pain, used fewer analgesics and consulted less during the first seven days. Meta-analysis of data from these four trials found no difference in pain between immediate antibiotics and expectant observational approaches at three to seven days. Another review (Spurling 2013), which evaluated the effect of delayed versus immediate or no antibiotics for respiratory infections and which included two studies on AOM (Little 2001; Spiro 2006), concluded that immediate antibiotics was the strategy most likely to provide the best clinical outcomes for AOM. One randomised study found that observation therapy with or without a prescription in children with AOM was well accepted by parents (Chao 2008). Antibiotic use was less in those randomised to observation without prescription and no complications were reported.

AUTHORS' CONCLUSIONS

Implications for practice

Antibiotics produce a (small) reduction in the number of children with pain at two to three days (number needed to treat with antibiotics for an additional beneficial outcome (NNTB) 20) and 10 to 12 days (NNTB 7) from initial assessment, and reduce the number of children with tympanic membrane perforations (NNTB 33), contralateral otitis episodes (NNTB 11) and abnormal tympanometry findings at two to four weeks (NNTB 11) compared with placebo. However, in high-income countries, most cases of acute otitis media (AOM) spontaneously remit without complications. The benefits of antibiotics must be weighed against the possible harms: for every 14 children treated with antibiotics one child experienced an adverse event (such as vomiting, diarrhoea or rash) that would not have occurred if antibiotics were withheld. Therefore, management should emphasise advice about adequate analgesia and the limited role for antibiotics. Based on a separate individual participant data meta-analysis performed in 2006, using data from six trials (1643 children), which were also included in this review as individual trials, antibiotics are most useful in children under two years of age with bilateral AOM, or with both AOM and otorrhoea (Rovers 2006). For most other children with mild disease, an expectant observational approach seems justified.

Implications for research

Further research is needed to determine if it is possible to predict which children are more likely to suffer from the complications of AOM and whether an expectant observation approach can be safely applied to children with mild AOM in low-income countries.

ACKNOWLEDGEMENTS

The Methods section of this protocol is based on a standard template developed by Cochrane Airways and adapted by Cochrane Acute Respiratory Infections.

We are grateful to acknowledge the late Chris Del Mar for substantial contributions as an author on previous versions of this review.

We also thank Vittoria Lutje, Information Specialist, for her help in conducting the searches.

The following people conducted the editorial process for this review update:

- Sign-off Editors (final editorial decision): Mark Jones (Bond University, Australia); Mieke van Driel (The University of Queensland, Australia).
- Managing Editor (provided editorial guidance to authors, edited the review, selected peer reviewers, collated peer reviewer comments): Liz Dooley (Bond University, Australia).
- Contact Editor (provided valuable comments during the process of updating this review, and recommended an editorial decision): Allen Cheng (Monash University, Australia).
- Statistical Editor (provided comments): Mark Jones (Bond University, Australia).
- Copy Editor (copy editing and production): Jenny Bellorini, Cochrane Central Production Service.
- Peer reviewers provided comments and recommended an editorial decision:
  - Clinical/content review: Brian Westerberg (University of British Columbia, Canada).
  - Consumer review: Wendy Free and J. Larosa.
  - Methods review: Emma Axon (Cochrane).
  - Search review: Jo Platt, Central Editorial Information Specialist.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

REFERENCES

References to studies included in this review

Appelman 1991 *{published data only}*

* Appelman CL, Claessen JQ, Touw Otten FW, Hordijk GJ, de Melker RA. Co-amoxiclav in recurrent acute otitis media: placebo controlled study. *BMJ* 1991;**303**:1450-2.

Appelman CL, Claessen JQ, Touw Otten FW, Hordijk GJ, de Melker RA. Severity of inflammation of tympanic membrane as predictor of clinical course of recurrent acute otitis media. *BMJ* 1993;**306**:895.

Claessen JQ, Appelman CL, Touw Otten FW, de Melker RA, Hordijk GJ. Persistence of middle ear dysfunction after recurrent acute otitis media. *Clinical Otolaryngology* 1994;**19**:35-40.

Burke 1991 *{published data only}*

Burke P, Bain J, Robinson D, Dunleavey J. Acute red ear in children: controlled trial of non-antibiotic treatment in general practice. *BMJ* 1991;**303**:558-62.

Damoiseaux 2000 *{published data only}*

Bezáková N, Damoiseaux RA, Hoes AW, Schilder AG, Rovers MM. Recurrence up to 3.5 years after antibiotic treatment of acute otitis media in very young Dutch children: survey of trial participants. *BMJ* 2009;**338**:b2525.

* Damoiseaux RAMJ, van Balen FAM, Hoes AW, Verheij TJM, de Melker RA. Primary care based randomised, double blind trial of amoxicillin versus placebo for acute otitis media in children aged under 2 years. *BMJ* 2000;**320**:350-4.

Halsted 1968 *{published data only}*

Halsted C, Lepow ML, Balassanian N, Emmerich J, Wolinsky E. Otitis media: clinical observation, microbiology and evaluation of therapy. *American Journal of Diseases of Children* 1968;**115**(5):542-51.

Hoberman 2011 *{published data only}*

* Hoberman A, Paradise JL, Rockette HE, Shaikh N, Wald ER, Kearney DH, et al. Treatment of acute otitis media in children under 2 years of age. *New England Journal of Medicine* 2011;**364**(2):105-15.

Paradise JL, Hoberman A, Rockette HE, Shaikh N. Treating acute otitis media in young children: what constitutes success? *Pediatric Infectious Disease Journal* 2013;**32**(7):745-7.

Howie 1972 *{published data only}*

Howie VM, Ploussard JH. Efficacy of fixed combination antibiotics versus separate components in otitis media: effectiveness of erythromycin estolate, triple sulfonamide, ampicillin, erythromycin estolate-triple sulfonamide, and placebo in 280 patients with acute otitis media under two and one-half years of age. *Clinical Pediatrics* 1972;**11**(4):205-14.

Kaleida 1991 *{published data only}*

Kaleida PH, Casselhrant ML, Rockette HE, Paradise JL, Bluestone CD, Blatter MM, et al. Amoxicillin or myringotomy or both for acute otitis media: results of a randomized clinical trial. *Paediatrics* 1991;**87**(4):466-74.

Laxdal 1970 *{published data only}*

Laxdal OE, Merida J, Jones RHT. Treatment of acute otitis media: a controlled study of 142 children. *Canadian Medical Association Journal* 1970;**102**(3):263-8.

Le Saux 2005 *{published data only}*

Le Saux N, Gaboury I, Baird M, Klassen TP, MacCormick J, Blanchard C, et al. A randomized, double-blind, placebo-controlled noninferiority trial of amoxicillin for clinically diagnosed acute otitis media in children 6 months to 5 years of age. *Canadian Medical Association Journal* 2005;**172**(3):335-41.

Little 2001 *{published data only}*

* Little P, Gould C, Williamson I, Moore M, Warner G, Dunleavey J. Pragmatic randomised controlled trial of two prescribing strategies for childhood acute otitis media. *BMJ* 2001;**322**:336-42.

Little P, Moore M, Warner G, Dunleavy J, Williamson I. Longer term outcomes from a randomised trial of prescribing strategies in otitis media. *British Journal of General Practitioners* 2006;**56**(524):176-82.

McCormick 2005 *{published data only}*

McCormick DP, Chonmaitree T, Pittman C, Saeed K, Friedman NR, Uchida T, et al. Nonsevere acute otitis media: a clinical trial comparing outcomes of watchful waiting versus immediate antibiotic treatment. *Pediatrics* 2005;**115**(6):1455-65.

Mygind 1981 *{published data only}*

* Mygind N, Meistrup-Larsen K-I, Thomsen J, Thomsen VF, Josefsson K, Sorensen H. Penicillin in acute otitis media: a double-blind placebo-controlled trial. *Clinical Otolaryngology* 1981;**6**:5-13.

Thomsen J, Meistrup-Larsen KI, Sorensen H, Larsen PK, Mygind N. Penicillin and acute otitis: short and long term results. *Annals of Otology, Rhinology and Laryngology, Supplement* 1980;**89**:271-4.

Neumark 2007 *{published data only}*

Neumark T, Molstad S, Rosen C, Persson L, Torngren A, Brudin L, et al. Evaluation of phenoxymethylpenicillin treatment of acute otitis media in children aged 2-16. *Scandinavian Journal of Primary Health Care* 2007;**25**:166-71.

Shahbaznejad 2021 *{published data only}*

Shahbaznejad L, Talaei E, Hosseinzadeh F, Masoumi B, Rezai S, Rezai MS. Comparing watchful waiting approach vs. antibiotic therapy in children with nonsevere acute otitis media: a randomized clinical trial. *International Journal of Pediatrics* 2021;**-**:5515467. [DOI: https://doi.org/10.1155/2021/5515467] [PMID: 34135974]

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Spiro 2006 *{published data only}*

Spiro DM, Tay K, Arnold DH, Dziura J, Baker M, Shapiro ED. Wait-and-see prescription for the treatment of acute otitis media. *JAMA* 2006;**296**(10):1235-41.

Tähtinen 2011 *{published data only}*

Tähtinen PA, Laine MK, Huovinen P, Jalava J, Ruuskanen O, Ruohola A. A placebo-controlled trial of antimicrobial treatment for acute otitis media. *New England Journal of Medicine* 2011;**364**(2):116-26.

Tapiainen 2014 *{published data only}*

Tapiainen T, Kujala T, Renko M, Koivunen P, Kontiokari T, Kristo A, et al. Effect of antimicrobial treatment of acute otitis media on the daily disappearance of middle ear effusion: a placebo-controlled trial. *JAMA Pediatrics* 2014;**168**(7):635-41.

Thalin 1985 *{published data only}*

Thalin A, Densert O, Larsson A, Lyden E, Ripa T. Is penicillin necessary in the treatment of acute otitis media? In: Proceedings of the International Conference on Acute and Secretory Otitis Media 1985, Jerusalem. Amsterdam: Kugler Publications, 1985:441-6.

van Buchem 1981 *{published data only}*

van Buchem FL, Dunk JHM, van't Hof MA. Therapy of acute otitis media: myringotomy, antibiotics or neither? A double-blind study in children. *Lancet* 1981;**2**:883-7.

References to studies excluded from this review

Arguedas 2011 *{published data only}*

Arguedas A, Soley C, Kamicker BJ, Jorgensen DM. Single-dose extended-release azithromycin versus a 10-day regimen of amoxicillin/clavulanate for the treatment of children with acute otitis media. *International Journal of Infectious Diseases* 2011;**15**(4):e240-8.

Casey 2012 *{published data only}*

Casey JR, Block SL, Hedrick J, Almudevar A, Pichichero ME. Comparison of amoxicillin/clavulanic acid high dose with cefdinir in the treatment of acute otitis media. *Drugs* 2012;**72**:1991-7.

Chaput 1982 *{published data only}*

Chaput de Saintonge DM, Levine DF, Temple Savage IT, Burgess GW, Sharp J, Mayhew SR, et al. Trial of three-day and ten-day courses of amoxycillin in otitis media. *British Medical Journal Clinical Research Edition* 1982;**284**(6322):1078-81.

Engelhard 1989 *{published data only}*

Engelhard D, Strauss N, Jorczak-Sarni L, Cohen D, Sacjs TG, Shapiro M. Randomised study of myringotomy, amoxycillin/clavulanate, or both for acute otitis media in infants. *Lancet* 1989;**2**(8655):141-3.

Hay 2021 *{published data only}*

Hay AD, Moore MV, Taylor J, Turner N, Noble S, Cabral C, et al. Immediate oral versus immediate topical versus delayed oral antibiotics for children with acute otitis media with discharge: the REST three-arm non-inferiority electronic platform-supported RCT. *Health Technology Assessment* 2021;**25**(67):1-76.

Liu 2011 *{published data only}*

Liu P, Fang AF, LaBadie RR, Crownover PH, Arguedas AG. Comparison of azithromycin pharmacokinetics following single oral doses of extended-release and immediate-release formulations in children with acute otitis media. *Antimicrobial Agents and Chemotherapy* 2011;**55**(11):5022-6.

Ostfeld 1987 *{published data only}*

Ostfeld E, Segal J, Kaufstein M, Gelernter I. Management of acute otitis media without primary administration of systemic antimicrobial agents. In: Lim DJ, Bluestone CD, Klein JO, Nelson JD , editors(s). Recent advances in otitis media. Proceedings of the Fourth International Symposium. Toronto: BC Decker, 1987:235-9.

Rudberg 1954 *{published data only}*

Rudberg RD. Acute otitis media: comparative therapeutic results of sulphonamide and penicillin administered in various forms. *Acta Oto-Laryngologica* 1954;**113**(Suppl):1-79.

Ruohola 2003 *{published data only}*

Ruohola A, Heikkinen T, Meurman O, Puhakka T, Lindblad N, Ruuskanen O. Antibiotic treatment of acute otorrhea through tympanostomy tube: randomized double-blind placebo-controlled study with daily follow-up. *Pediatrics* 2003;**111**(5):1061-7.

Sarrell 2003 *{published data only}*

Sarrell EM, Cohen HA, Kahan E. Naturopathic treatment for ear pain in children. *Pediatrics* 2003;**111**(5):574-9.

Tähtinen 2012 *{published data only}*

Tähtinen PA, Laine MK, Ruuskanen O, Ruohola A. Delayed versus immediate antimicrobial treatment for acute otitis media. *Pediatric Infectious Disease Journal* 2012;**31**(12):1227-32.

van Buchem 1985 *{published data only}*

van Buchem F, Peeters M, Van' t Hof M. Acute otitis media: a new treatment strategy. *BMJ* 1985;**290**:1033-7.

References to ongoing studies

the WATCH trial *{published data only}*

* Abbott P, Gunasekera H, Leach AJ, Askew D, Walsh R, Kong K, et al. A multi-centre open-label randomised non-inferiority trial comparing watchful waiting to antibiotic treatment for acute otitis media without perforation in low-risk urban Aboriginal and Torres Strait Islander children (the WATCH trial): study protocol for a randomised controlled trial. *Trials* 2016;**17**(1):119. [DOI: https://doi.org/10.1186/s13063-016-1247-y] [PMID: 26941013]

Additional references

AAP 2013

American Academy of Pediatrics and American Academy of Family Physicians. Clinical practice guidelines: diagnosis

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

and management of acute otitis media. *Pediatrics* 2013;**131**(3):e964-99.

Arnold 2005

Arnold SR, Straus SE. Interventions to improve antibiotic prescribing in ambulatory care. *Cochrane Database of Systematic Reviews* 2005, Issue 4. Art. No: CD003539. [DOI: 10.1002/14651858.CD003539]

Arola 1990

Arola M, Ruuskanen O, Ziegler T, Mertsola J, Näntö-Salonen K, Putto-Laurila A, et al. Clinical role of respiratory virus infection in acute otitis media. *Pediatrics* 1990;**86**:848-55.

Barkai 2009

Barkai G, Leibovitz E, Givon-Lavi N, Dagan R. Potential contribution by nontypable Haemophilus influenzae in protracted and recurrent acute otitis media. *Pediatric Infectious Disease Journal* 2009;**28**:466-71.

Berman 1995

Berman S. Otitis media in developing countries. *Pediatrics* 1995;**96**:126-31.

Burvenich 2022

Burvenich R, Dillen H, Trinh NT, Freer J, Wynants L, Heytens S, et al. Antibiotic use in ambulatory care for acutely ill children in high-income countries: a systematic review and meta-analysis. *Archives of Disease in Childhood* 2022;**107**(12):1088-94.

Casey 2013

Casey JR, Kaur R, Friedel VC, Pichichero ME. Acute otitis media otopathogens during 2008 to 2010 in Rochester, New York. *Pediatic Infectious Disease Journal* 2013;**32**(8):805-9.

Chao 2008

Chao JH, Kunkov S, Reyes LB, Lichten S, Crain EF. Comparison of two approaches to observation therapy for acute otitis media in the emergency department. *Pediatrics* 2008;**121**:31352-e6.

Chonmaitree 1992

Chonmaitree T, Owen MJ, Patel JA, Hedgpeth D, Horlick D, Howie VM. Effect of viral respiratory tract infection on outcome of acute otitis media. *Journal of Pediatrics* 1992;**120**:856-62.

Coker 2010

Coker TR, Chan LS, Newberry SJ, Limbos MA, Suttorp MJ, Shekelle PG, et al. Diagnosis, microbial epidemiology, and antibiotic treatment of acute otitis media in children: a systematic review. *JAMA* 2010;**304**(19):2161-9.

Dickersin 1994

Dickersin K, Scherer R, Lefebvre C. Identifying relevant studies for systematic reviews. *BMJ* 1994;**309**(6964):1286-91.

Gillies 2014

Gillies M, Ranakusuma A, Hoffmann T, Thorning S, McGuire T, Glasziou P, et al. Common harms from amoxicillin: a systematic review and meta-analysis of randomized placebo-controlled trials for any indication. Canadian Medical Association Journal 2014 Nov 17 [Epub ahead of print]. [DOI: 10.1503/cmaj.140848]

Gisselsson-Solen 2022

Gisselsson-Solen M, Hermansson A. Trends in upper respiratory tract infections and antibiotic prescriptions during the COVID-19 pandemic - a national observational study. *Acta Otolaryngology* 2022;**142**(7-8):549-52.

GRADEpro GDT [Computer program]

GRADEpro GDT. Hamilton (ON): McMaster University (developed by Evidence Prime), accessed 1 September 2023. Available at gradepro.org.

Higgins 2011

Higgins JP, Green S (editors). Cochrane Handbook for Systematic Reviews of Interventions version 5.1.0 [updated March 2011]. The Cochrane Collaboration, 2011. Available from training.cochrane.org/handbook/archive/v5.1/.

Higgins 2021

Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Cochrane Handbook for Systematic Reviews of Interventions Version 6.2 (updated February 2021). Cochrane, 2021. Available from training.cochrane.org/handbook.

Hullegie 2021

Hullegie S, Schilder AG, Marchisio P, de Sévaux JL, van der Velden AW, van de Pol AC, et al. A strong decline in the incidence of childhood otitis media during the COVID-19 pandemic in the Netherlands. *Frontiers in Cellular and Infection Microbiology* 2021;**11**:768377. [DOI: 10.3389/fcimb.2021.768377]

Koopman 2008

Koopman L, Hoes AW, Glasziou PP, Appelman CL, Burke P, McCormick DP, et al. Antibiotic therapy to prevent the development of asymptomatic middle ear effusion in children with acute otitis media. *Archives of Otolaryngology - Head and Neck Surgery* 2008;**134**(2):128-32.

Kozyrskyj 2010

Kozyrskyj A, Klassen TP, Moffatt M, Harvey K. Short-course antibiotics for acute otitis media. *Cochrane Database of Systematic Reviews* 2010, Issue 9. Art. No: CD001095. [DOI: 10.1002/14651858.CD001095]

Lefebvre 2021

Lefebvre C, Glanville J, Briscoe S, Littlewood A, Marshall C, Metzendorf M-I, et al. Chapter 4: Searching for and selecting studies. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). Cochrane Handbook for Systematic Reviews of Interventions Version 6.2 (updated February 2021). Cochrane, 2021. Available from training.cochrane.org/handbook.

Little 2006

Little P, Moore M, Warner G, Dunleavy J, Williamson I. Longer term outcomes from a randomised trial of prescribing strategies in otitis media. *British Journal of General Practice* 2006;**56**:176-82.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

McCormick 2007

McCormick DP, Chandler SM, Chonmaitree T. Laterality of acute otitis media: different clinical and microbiologic characteristics. *Pediatric Infectious Disease Journal* 2007;**26**:583-8.

Monasta 2012

Monasta L, Ronfani L, Marchetti F, Montico M, Brumatti LV, Bavcar A, et al. Burden of disease caused by otitis media: systematic review and global estimates. *PLOS One* 2012;**7**(4):e36226.

Rosenfeld 1994

Rosenfeld RM, Vertrees JE, Carr J, Cipolle RJ, Uden DL, Giebink GS, et al. Clinical efficacy of antimicrobial drugs for acute otitis media: meta-analysis of 5400 children from thirty-three randomized trials. *Journal of Pediatrics* 1994;**124**:355-67.

Rothman 2003

Rothman R, Owens T, Simel D. Does this child have acute otitis media? *JAMA* 2003;**290**(12):1633-40.

Rovers 2006

Rovers MM, Glasziou P, Appelman CL, Burke P, McCormick DP, Damoiseaux RA, et al. Antibiotics for acute otitis media: a meta-analysis with individual patient data. *Lancet* 2006;**368**:1429-35.

Smith 2010

Smith L, Ewings P, Smith C, Thompson M, Harnden A, Mant D. Ear discharge in children presenting with acute otitis media: observational study from UK general practice. *British Journal of General Practice* 2010;**60**(571):101-5.

Song 1999

Song F. Exploring heterogeneity in meta-analysis: is the L'Abbé plot useful? *Journal of Clinical Epidemiology* 1999;**52**(8):725-30.

Spurling 2013

Spurling GKP, Del Mar CB, Dooley L, Foxlee R, Farley R. Delayed antibiotics for respiratory infections. *Cochrane Database of Systematic Reviews* 2013, Issue 4. Art. No: CD004417. [DOI: 10.1002/14651858.CD004417.pub4]

Teele 1989

Teele DW, Klein JO, Rosner B. Epidemiology of otitis media during the first seven years of life in children in greater Boston: a prospective, cohort study. *Journal of Infectious Diseases* 1989;**160**(1):83-94.

REFERENCES TO OTHER PUBLISHED VERSIONS OF THIS REVIEW

Glasziou 1995

Glasziou PP, Hayem M, Del Mar CB. A meta-analysis of treatments for acute otitis media: antibiotic vs placebo, short vs long antibiotic course, and myringotomy. *Cochrane Database of Systematic Reviews* 1995, Issue 1. Art. No: CD000219. [DOI: 10.1002/14651858.CD000219]

Glasziou 1997

Glasziou PP, Hayem M, Del Mar CB. Treatments for acute otitis media in children: antibiotic versus placebo. *Cochrane Database of Systematic Reviews* 1997, Issue 1. Art. No: CD000219. [DOI: 10.1002/14651858.CD000219]

Glasziou 1999

Glasziou PP, Hayem M, Del Mar CB. Antibiotics versus placebo for acute otitis media in children. *Cochrane Database of Systematic Reviews* 1999, Issue 3. Art. No: CD000219. [DOI: 10.1002/14651858.CD000219.pub2]

Glasziou 2005

Glasziou PP, Del Mar CB, Sanders SL, Hayem M. Antibiotics for acute otitis media in children. *Cochrane Database of Systematic Reviews* 2005, Issue 1. Art. No: CD000219. [DOI: 10.1002/14651858.CD000219.pub2]

Sanders 2009

Sanders SL, Glasziou PP, Del Mar CB, Rovers MM. Antibiotics for acute otitis media in children. *Cochrane Database of Systematic Reviews* 2009, Issue 2. Art. No: CD000219. [DOI: 10.1002/14651858.CD000219.pub2]

Venekamp 2013

Venekamp RP, Sanders S, Glasziou PP, Del Mar CB, Rovers MM. Antibiotics for acute otitis media in children. *Cochrane Database of Systematic Reviews* 2013, Issue 1. Art. No: CD000219. [DOI: 10.1002/14651858.CD000219.pub3]

Venekamp 2015

Venekamp RP, Sanders S, Glasziou PP, Del Mar CB, Rovers MM. Antibiotics for acute otitis media in children. Cochrane Database of Systematic Reviews. *Cochrane Database of Systematic Reviews* 2015, Issue 6. Art. No: CD000219. [DOI: 10.1002/14651858.CD000219.pub4]

* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Appelman 1991

Study characteristics	
Methods	**Randomised -** yes, computer-generated random numbers
	Concealment of allocation - adequate

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Appelman 1991 *(Continued)*

	Double-blind - yes, blinding procedure not described **Intention-to-treat (ITT) -** unclear **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 126 children (N = 121 children included in analysis) **Age -** between 6 months and 12 years **Setting -** general practice and secondary care in the Netherlands; confirmation of diagnosis and randomisation were done by otorhinolaryngologists **Inclusion criteria -** recurrence of acute otitis media (AOM) characterised by a (sub)acute onset, otalgia and otoscopic signs of middle ear infection within 4 weeks to 12 months of the previous attack **Exclusion criteria -** antibiotic treatment < 4 weeks prior to randomisation, previous participation in this study, contraindication for penicillin, serious concurrent disease that necessitated antibiotic treatment **Baseline characteristics -** balanced
Interventions	**Tx -** amoxicillin/clavulanate (weight tailored dose) for 7 days; N = 70 (N = 67 included in analysis) **C -** matching placebo for 7 days; N = 56 (N = 54 included in analysis) **Use of additional medication -** each child was given analgesics (paracetamol) as long as earache was present and decongestant nose drops for 1 week
Outcomes	**Primary outcome -** treatment failure (i.e. presence of otalgia or fever > 38 °C or both at 3 days) Assessment by (blinded) general practitioner at 3 days on the presence or absence of fever (> 38 °C) and otalgia and 14 days on the presence or otorrhoea Assessment by otorhinolaryngologist at 1 month by otoscopy, tympanometry and in children > 3 years of age an audiogram
Notes	Funding- "We thank Smith Kline Beecham Farma for supplying antibiotic and placebo and for an endowment for the expenses of participating doctors; Merck for providing nose drops and financing the data documents; and Nicholas-Mepros for supplying paracetamol and an endowment. Merck Sharp and Dohme gave financial support to this study." No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers
Allocation concealment (selection bias)	Low risk	Treatment allocated by otolaryngologist (independent of trial personnel); treatment code placed in sealed envelopes
Blinding of participants and personnel (performance bias)	Unclear risk	Identical taste and appearance to amoxicillin/clavulanate and placebo not described
Incomplete outcome data (attrition bias)	Low risk	Loss to follow-up - treatment: N = 3 (4%) and placebo: N = 2 (4%) due to loss of their registration forms

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Appelman 1991 *(Continued)*

Other bias	Unclear risk	ITT analysis - unclear, baseline characteristics - balanced

Burke 1991

Study characteristics	
Methods	**Randomised -** yes, computer-generated random numbers **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** yes **Loss to follow-up -** not described **Design -** parallel
Participants	**N -** 232 children **Age -** between 3 and 10 years **Setting -** general practice; 48 general practitioners in 17 general practices in Southampton, Bristol and Portsmouth (UK) **Inclusion criteria -** acute earache and at least 1 abnormal eardrum **Exclusion criteria -** antibiotic treatment or acute otitis media (AOM) < 2 weeks prior to randomisation, strong indication for antibiotic treatment according to general practitioner, contraindication for amoxicillin, serious chronic conditions **Baseline characteristics -** slight imbalance in gender (boys treated with antibiotics versus boys treated with placebo = 52% versus 42%) and figure 1 appears to demonstrate that fewer children were crying at baseline (0 hours) in the amoxicillin arm compared with the placebo arm, suggesting a failure of randomisation
Interventions	**Tx -** amoxicillin 250 mg 3 times daily for 7 days; N = 114 (N = 114 included in analysis for short-term outcome) **C -** matching placebo 3 times daily for 7 days; N = 118 (N = 118 included in analysis for short-term outcome) **Use of additional medication -** analgesics (paracetamol 120 mg/5 mL) for pain as needed
Outcomes	**Main outcomes** were divided into short-term, middle-term and long-term: **Short-term -** (a) duration of symptoms; (b) use of analgesics (assessed by weighing bottles); (c) clinical signs at 1 week; (d) incidence of complications; (e) treatment failure (i.e. second-line antibiotics were required) **Middle-term -** (a) tympanometry findings at 1 and 3 months **Long-term -** (b) number of AOM episodes in 12 months; (b) number of specialist referrals Home visits by researcher at day 1, days 4 to 6 and general practitioner visit at day 7 Symptom diary kept by parents for 21 days
Notes	It is not clear whether the "discharging ears" in Table 1 should be included as perforations. We have now included the number of perforations as summarised in Table 2 in our analysis.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Burke 1991 *(Continued)*

Funding - "We thank the Medical Research Council, The scientic foundation board of the Royal College of General Practitioners, Beecham Research, and Reckitt and Colman for financial support."

No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random numbers
Allocation concealment (selection bias)	Low risk	Randomisation was carried out independently of the investigators; randomisation code was kept sealed and was unknown to any of the participants in the study
Blinding of participants and personnel (performance bias)	Low risk	Each bottle was identified only by number
Incomplete outcome data (attrition bias)	Unclear risk	Loss to follow-up - not described; all randomised patients included in short-outcome analysis
Other bias	Unclear risk	ITT analysis - yes; baseline characteristics - imbalance for gender and crying

Damoiseaux 2000

Study characteristics

Methods	**Randomised -** yes, computerised 2 block randomisation **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** yes **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 240 children (N = 212 children included in analysis) **Age -** between 6 months and 2 years **Setting -** general practice; 53 general practitioners (GPs) in the Netherlands **Inclusion criteria -** acute otitis media (AOM) defined as infection of the middle ear of acute onset and a characteristic eardrum picture (injection along the handle of the malleus and the annulus of the tympanic membrane or a diffusely red or bulging eardrum) or acute otorrhoea. In addition, 1 or more symptoms of acute infection (fever, recent earache, general malaise, recent irritability) **Exclusion criteria -** antibiotic treatment < 4 weeks prior to randomisation, contraindication for amoxicillin, comprised immunity, craniofacial abnormalities, Down's syndrome or being entered in this study before **Baseline characteristics -** slight imbalance in the prevalence of recurrent AOM, regular attendance at a daycare centre and parental smoking; logistic regression was used to adjust for these imbalances

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Damoiseaux 2000 *(Continued)*

Interventions	**Tx -** amoxicillin suspension 40 mg/kg/day 3 times daily for 10 days; N = 117 (N = 107 included in analysis for short-term outcome) **C -** matching placebo suspension for 10 days; N = 123 (N = 105 included in analysis for short-term outcome) **Use of additional medication -** all children received decongestant nose drops for 7 days; analgesics (paracetamol, children < 1 year: 120 mg suppository, > 1 year: 240 mg suppository) were allowed
Outcomes	**Primary outcome -** persistent symptoms at day 4: assessed by the doctor and defined as persistent earache, fever > 38 °C, crying or being irritable. Additionally, prescription of another antibiotic because of clinical deterioration before the first follow-up visit was to be considered a persistent symptom. **Secondary outcomes -** (a) clinical treatment failure at day 11 (i.e. persistent fever, earache, crying, being irritable or no improvement of tympanic membrane (including perforation); (b) duration of fever, pain or crying; (c) mean number of doses analgesics given; (d) adverse effects mentioned in diaries; (e) percentage of children with middle ear effusion at 6 weeks (i.e. combined otoscopy and tympanometry) Follow-up visits at the GP's clinic were scheduled at day 4 and 11; home visit at 6 weeks by the researcher collecting data of symptoms, referrals and both otoscopy and tympanometry was performed Parents were instructed to keep a symptom diary for 10 days
Notes	Funding - Netherlands Organisation for Scientific Research (grant no 904-58-074). Nasivin nose drops for this study were donated by E Merck Nederland BV. No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computerised 2-block randomisation
Allocation concealment (selection bias)	Low risk	Randomisation was carried out independently of the investigators; randomisation code was kept in the pharmacy of the University Medical Centre Utrecht
Blinding of participants and personnel (performance bias)	Low risk	Placebo suspension with same taste and appearance as amoxicillin
Incomplete outcome data (attrition bias)	Unclear risk	Loss to follow-up/exclusion from analysis (received other antibiotics or had grommets inserted) - treatment: N = 10 (9%) and placebo: N = 18 (15%). However, for primary analysis of symptoms at day 4, all randomised patients were included
Other bias	Low risk	ITT analysis - yes, baseline characteristics - slight imbalance, logistic regression was used to adjust for imbalances in prognostic factors

Halsted 1968

Study characteristics

Methods	**Randomised -** yes, pre-determined code, which was unknown to physician; method of random sequence generation unclear

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Halsted 1968 *(Continued)*

	Concealment of allocation - adequate **Double-blind -** yes **Intention-to-treat (ITT) -** unclear **Loss to follow-up -** described, unclear from which treatment group patients were excluded **Design -** parallel
Participants	**N -** 106 children (N = 89 children included in analysis; N = 12 children were excluded because they did not adhere to the double-blind protocol; N = 5 children lost to follow-up or excluded because of persistent fever, development of complications requiring antibiotic treatment or if group A streptococci were cultured from the middle ear) **Age -** between 2 months and 5.5 years **Setting -** secondary care: paediatric department of Cleveland (USA) **Inclusion criteria -** AOM based on otoscopic findings; most of the cases had bulging membrane with loss of normal light reflex and landmarks, in a few the eardrum was only diffusely red **Exclusion criteria -** antibiotic treatment < 10 days prior to randomisation, associated bacterial infection requiring antibiotic treatment, rupture of tympanic membrane, contraindication for study drugs **Baseline characteristics -** not described
Interventions	**Tx 1 -** ampicillin 100 mg/kg/day 4 daily for 10 days; N = ? (N = 30 included in analysis) **Tx 2 -** pheneticillin 30 mg/kg/day 4 daily and sulfisoxazole 150 mg/kg/day 4 daily for 10 days; N = ? (N = 32 included in analysis) **C -** placebo for 10 days; N = ? (N = 27 included in analysis) **Use of additional medication -** phenylephrine nose drops and aspirin for children over 6 months were prescribed as necessary; no other medications were employed
Outcomes	**Primary outcome -** early improvement defined as defervescence and decrease of symptoms at 24 to 72 hours **Secondary outcomes -** (a) late improvement defined as resolution of symptoms and normal tympanic membrane at 14 to 18 days, (b) bacteriological cultures
Notes	Funding - "This word was aided in part by Public Health Service research grant 5 RO1 A1 05770 and training grant 5 TO1 A1 00180 of the National Institute of Allergy and Infectious Deseases. Ampicillin (Polycillin), phenethicillin (Syncillin), and placebo were supplied by Bristol Laboratories, Syracuse, NY. Sulfisoxazole (Gantrisin) was supplied by Hoffman-La Roche, Incl, Nutley, NJ." No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Pre-determined code, which was unknown to physician; method of random sequence generation unclear
Allocation concealment (selection bias)	Unclear risk	Method not described

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Halsted 1968 *(Continued)*

Blinding of participants and personnel (performance bias)	Unclear risk	Identical taste and appearance to antibiotics and placebo not described
Incomplete outcome data (attrition bias)	Unclear risk	Reasons described; unclear from which treatment group patients were excluded
Other bias	Unclear risk	ITT analysis - unclear, baseline characteristics - not described

Hoberman 2011

Study characteristics	
Methods	**Randomised -** yes, stratified block randomisation with computer-generated randomisation lists **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** yes **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 291 (N = 291 included in analysis) **Age -** between 6 months and 2 years **Setting -** secondary care; children's hospital of Pittsburgh and a private paediatric clinic in Kittanning (USA) **Inclusion criteria -** children needed to have received at least 2 doses of pneumococcal conjugate vaccine and to have acute otitis media (AOM) as defined on the basis of 3 criteria: (a) the onset (i.e. within the preceding 48 hours) of symptoms that parents rated with a score of at least 3 on the acute otitis media - severity of symptoms (AOM-SOS) scale (on which scores range from 0 to 14, with higher scores indicating greater severity of symptoms), (b) the presence of middle ear effusion and (c) moderate or marked bulging of the tympanic membrane or slight bulging accompanied by either otalgia or marked erythema of the membrane All the study clinicians were otoscopists who had successfully completed an otoscopic validation programme **Exclusion criteria -** antibiotic treatment < 96 hours prior to randomisation, concomitant acute illness (e.g. pneumonia) or a chronic illness (e.g. cystic fibrosis), contraindication to amoxicillin, presence of otalgia for more than 48 hours, perforation of the tympanic membrane **Baseline characteristics -** balanced
Interventions	**Tx -** amoxicillin-clavulanate 90 mg-6.4 mg/kg daily in 2 doses for 10 days; N = 144 (N = 139 were assessed at day 4 to 5) **C -** matching placebo in 2 doses for 10 days; N = 147 (N = 142 were assessed at day 4 to 5) **Use of additional medication -** acetaminophen (paracetamol) as needed for symptom relief At each visit children were categorised as having met the criteria for either clinical success or clinical failure

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Hoberman 2011 *(Continued)*

	Children who met the criteria for clinical failure were treated with a standardised 10-day regimen of orally administered amoxicillin (90 mg/kg daily) and cefixime (8 mg/kg daily)
Outcomes	**Primary outcomes -** (a) time to resolution of symptoms (i.e. time to the first recording of an AOM-SOS score of 0 or 1 and the time to the second of 2 successive recordings of that score; (b) symptom burden over time (i.e. mean AOM-SOS score over time each day for the first 7 days of follow-up and groups' weighted mean scores for that period) **Secondary outcomes -** (a) clinical failure at day 4 to 5; (b) clinical failure at day 10 to 12; (c) use of acetaminophen (paracetamol); (d) occurrence of adverse events; (e) nasopharyngeal colonisation rates; (f) use of healthcare resources; (g) relapses Clinical failure was defined at or before the day 4 to 5 visit as either a lack of substantial improvement in symptoms, a worsening of signs on otoscopic examination, or both, and at the day 10 to 12 visit as the failure to achieve complete or nearly complete resolution of symptoms and otoscopic signs, without regard to the persistence of resolution of middle ear effusion. Once a child had met the criteria for clinical failure, he or she remained in that category for the analysis. Daily symptoms were assessed with the use of a structured interview of 1 of the child's parents until the first follow-up visit; visits were scheduled at day 4 or 5, day 10 to 12 (end of treatment) and at day 21 to 25 Patients were asked to complete a diary twice a day for 3 days and once a day thereafter
Notes	This study did not report pain data that could be used for the review comparing antibiotics with placebo Funding - supported by a grant (3U01AI066007-02S1) from the National Institute of Allergy and Infectious Diseases Dr. Hoberman reports receiving honoraria and travel-expense reimbursement from GlaxoSmithKline, and Dr. Wald, grant support from Merck and GlaxoSmithKline.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Stratified block randomisation with computer-generated randomisation lists
Allocation concealment (selection bias)	Low risk	A pharmacist (independent of the trial team) provided masked study medication bottles with amoxicillin/clavulanate or placebo
Blinding of participants and personnel (performance bias)	Low risk	Placebo with same taste and appearance as amoxicillin-clavulanate
Incomplete outcome data (attrition bias)	Low risk	Children not assessed at day 4 to 5 - treatment: N = 5 (3%) and placebo: N = 5 (3%). All randomised patients included in analysis
Other bias	Low risk	ITT analysis - yes, baseline characteristics - balanced

Howie 1972

Study characteristics

Methods	**Randomised -** yes, method of randomisation not described

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Howie 1972 *(Continued)*

	Concealment of allocation - adequate
	Double-blind - yes
	Intention-to-treat (ITT) - unclear
	Loss to follow-up - not described
	Design - parallel
Participants	**N -** 280 children
	Age - 2.5 years or younger
	Setting - secondary care: general paediatric practice in Huntsville (USA)
	Inclusion criteria - acute otitis media (AOM) as clinically diagnosed by the participating paediatricians
	Exclusion criteria - if researchers felt that parents would not accept diagnostic aspiration, when the condition of the patient required immediate antibiotic treatment
	Baseline characteristics - not described
Interventions	**Tx 1 -** erythromycin estolate 125 mg/5 mL - triple sulphonamide suspension 0.5 g/5 mL; N = 80
	Tx 2 - ampicillin 250 mg/5 mL; N = 36
	Tx 3 - triple sulphonamide suspension 0.5 g/5 mL; N = 23
	Tx 4 - erythromycin estolate 125 mg/5 mL; N = 25
	C 1 - placebo - equal parts acetaminophen (paracetamol) and chlorpheniramine maleate syrup; N = 33
	C 2 - placebo - 4 parts Kaopectate and 1 part acetaminophen (paracetamol, Tylenol) plus food colouring; N = 83
	Use of additional medication - all children received decongestant nose drops for 7 days; analgesics (paracetamol, children < 1 year: 120 mg suppository, > 1 year: 240 mg suppository) were allowed
Outcomes	**Primary outcomes -** (a) presence or absence of exudate while on medication; (b) bacteriological findings of the exudate when present; no patient-relevant outcomes were described
	At baseline and before treatment was started, the middle ear exudate was aspirated. The decision whether to collect exudate on the first repeat visit was made with no knowledge of the drug regimen to which the patient had been assigned.
Notes	Funding - "The study was supported in part by a grant from Eli Lilly Company."
	No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of randomisation not described
Allocation concealment (selection bias)	Low risk	Randomisation was performed by a collaborating pharmacist
Blinding of participants and personnel (performance bias)	Unclear risk	Identical taste and appearance to amoxicillin/clavulanate and placebo not described

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Howie 1972 *(Continued)*

Incomplete outcome data (attrition bias)	Unclear risk	Loss to follow-up - not described
Other bias	Unclear risk	ITT analysis - unclear, baseline characteristics - not described

Kaleida 1991

Study characteristics	
Methods	**Randomised -** yes, stratified randomisation, method of randomisation not described **Concealment of allocation -** unclear, method not described **Double-blind -** yes, blinding procedure not described **Intention-to-treat (ITT) -** yes **Loss to follow-up -** not described **Design -** parallel
Participants	**N -** 536 children (representing 1049 non-severe acute otitis media (AOM) episodes; 980 non-severe AOM episodes included for primary analysis) **Age -** between 7 months and 12 years **Setting -** secondary care: children's hospital and a private paediatric practice in Pittsburgh (USA) **Inclusion criteria -** AOM based on presence of middle ear effusion, as determined otoscopically, in association with specified symptoms of acute middle ear infection (fever, otalgia or irritability), or signs of acute infection (erythema or white opacification, or both, accompanied by fullness or bulging and impaired mobility), or both **Exclusion criteria -** children who recently received antibiotics, who had potential complicating or confounding conditions (e.g. eardrum perforation, asthma or chronic sinusitis) **Baseline characteristics -** balanced
Interventions	Children were enrolled for a 1-year period. At entry, each child was assigned randomly to a treatment regimen that specified consistent treatments for episodes of non-severe and severe AOM based on severity of otalgia and the presence of fever (> 39 °C orally or > 39.5 °C rectally within the 24-hour period before presentation) **Non-severe AOM episodes** were treated with: Tx - amoxicillin 40 mg/kg/day 3 times daily for 14 days; N = 522 (N = 488 included in primary analysis) C - placebo for 14 days; N = 527 (N = 492 included in primary analysis) **Severe AOM episodes** in children aged < 2 years were treated with: Tx 1 - amoxicillin 40 mg/kg/day 3 times daily for 14 days Tx 2 - amoxicillin 40 mg/kg/day 3 times daily for 14 days and myringotomy **Severe AOM episodes** in children aged ≥ 2 years were treated with: Tx 1 - amoxicillin 40 mg/kg/day 3 times daily for 14 days Tx 2 - amoxicillin 40 mg/kg/day 3 times daily for 14 days and myringotomy

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Kaleida 1991 *(Continued)*

	Tx 3 - placebo and myringotomy
Outcomes	**Primary outcome -** initial treatment failure: in non-severe episodes this was the case when either otalgia, fever or both were present more than 24 hours after treatment was initiated and when 48 hours or more after initial treatment was initiated the child's temperature reached 38 °C orally or 38.5 °C rectally or an otalgia score of ≥ 6 was present **Secondary outcomes -** (a) recurrent AOM defined as the development of AOM 15 days or more after the initiation of treatment for a preceding episode, (b) new episodes of otitis media with effusion defined by otoscopy and tympanometry findings After initial visits, children were followed up by telephone to identify those with persistent symptoms and children younger than 2 years of age were re-examined within 48 to 72 hours Follow-up visits were scheduled routinely after 2 and 6 weeks after initial treatment and monthly thereafter
Notes	We included only the non-severe AOM episodes in this review (N = 1049 of which 980 were included for primary analysis); children experiencing non-severe AOM episodes were randomly allocated to either antibiotics or placebo Funding - "This work was supported by National Institutes of Health grant NIDCD CD 00014." No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of randomisation not described
Allocation concealment (selection bias)	Unclear risk	Method not described
Blinding of participants and personnel (performance bias)	Unclear risk	Identical taste and appearance to amoxicillin and placebo not described
Incomplete outcome data (attrition bias)	Unclear risk	Follow-up/exclusion of non-severe episodes for short-term outcome - treatment: N = 34 (7%) and placebo: N = 35 (7%). Reasons not described
Other bias	Low risk	ITT analysis - yes, baseline characteristics - balanced

Laxdal 1970

Study characteristics

Methods	**Randomised -** yes, method of randomisation not described **Concealment of allocation -** unclear; method not described **Double-blind -** no; open-label study, investigators not blinded **Intention-to-treat (ITT) -** unclear **Loss to follow-up -** not described

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Laxdal 1970 *(Continued)*

	Design - parallel
Participants	**N -** 142 children **Age -** between 0 and 15 years **Setting -** secondary care (private paediatric clinic) in Saskatoon (Canada) **Inclusion criteria -** at least 1 eardrum had to show redness and loss of landmarks **Exclusion criteria -** predominant respiratory symptoms, if allergy appeared to be a significant factor or if rupture of the eardrum had occurred **Baseline characteristics -** not described
Interventions	**Tx 1 -** penicillin G 250 mg/m^2/day 4 times daily (approximately 33 mg/kg/day) for at least 7 days; N = 45 **Tx 2 -** ampicillin 250 mg/m^2/day 4 times daily (approximately 33 mg/kg/day) for at least 7 days; N = 49 **C -** symptomatic therapy (Auralgan ear drops, acetylsalicylic acid, decongestant nose drops); N = 48 **Use of additional medication -** children in treatment groups also received symptomatic therapy as required
Outcomes	**Primary outcomes -** (a) treatment failure (i.e. either deterioration or no improvement observed at day 7), (b) relapses Results were evaluated at 7 days, except in cases where the ear inflammation was severe and the child appeared sufficiently ill (toxic) to warrant further examination 24 to 48 hours after treatment initiation Children in the control group were subjected to very close scrutiny, especially during the first 48 hours and particularly when severe involvement was evident (high risk of detection bias)
Notes	Open-label trial comparing immediate antibiotics (penicillin G and ampicillin) versus expectant observation It was unclear whether otalgia played an important role in the definition of treatment failure Data on relapses: N = 126 included in analysis, no crude numbers for separate treatment groups provided Funding - "Funds for the support of this project were received from National health Grant No. 607-7-61." "We are grateful to Dr Leighton Smith of Ayerst Laboratories for supplying us with ampicillin (Penbritin). Dr. J.R. MacDougal of Frank W Horner & Company kindly made supplies of Penicillin G (P-50) available to our patients." No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of randomisation not described
Allocation concealment (selection bias)	Unclear risk	Not described
Blinding of participants and personnel (performance bias)	High risk	Open-label trial, outcome assessment not blinded

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Laxdal 1970 *(Continued)*

Incomplete outcome data (attrition bias)	Unclear risk	Loss to follow-up - not described for short-term outcome. Loss to follow-up for long-term outcome (acute otitis media (AOM) relapses) - N = 16 (11%); no crude numbers for separate treatment groups provided
Other bias	High risk	ITT analysis - unclear, baseline characteristics - not described; high risk of detection bias due to different follow-up strategies between treatment groups

Le Saux 2005

Study characteristics	
Methods	**Randomised -** yes, computer-generated randomisation sequence **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** yes **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 531 children (N = 512 children included in analysis; N = 19 were excluded post hoc due to inappropriate randomisation (N = 4) or alternative clinical diagnosis (N = 15)) **Age -** between 6 months and 5 years **Setting -** secondary care: emergency department in Ottawa (Canada) **Inclusion criteria -** new onset (< 4 days) of symptoms referable to the upper respiratory tract and either ear pain or fever (> 38 °C). In addition, all patients had to have evidence of middle ear effusion, defined by ≥ 2 of the following signs: opacity, impaired mobility on the basis of pneumatic otoscopy and redness or bulging (or both) of the tympanic membrane **Exclusion criteria -** antibiotic treatment < 2 weeks prior to randomisation, contraindication to amoxicillin or penicillin or sensitivity to ibuprofen or aspirin, presence of otorrhoea, co-morbid disease such as sinusitis or pneumonia, prior middle ear surgery, placement of a ventilation tube, history of recurrent acute otitis media (more than 4 episodes in 12 months), compromised immunity, craniofacial abnormalities, or any chronic or genetic disorder **Baseline characteristics -** balanced
Interventions	**Tx -** amoxicillin suspension (60 mg/kg) 3 times daily for 10 days; N = 258 (N = 253 included in analysis for day 3) **C -** matching placebo for 10 days; N = 254 (N = 246 included in analysis for day 3) **Use of additional medication -** parents were given a 5-day supply of antipyretic and analgesic medication in the form of ibuprofen suspension as required for pain or fever and a 48-hour supply of codeine elixir to be given as required for pain and fever
Outcomes	**Primary outcome -** clinical resolution of symptoms, defined as absence of receipt of an antimicrobial (other than amoxicillin in the treatment group) at any time during the 14-day period. The initiation of antimicrobial therapy was based on persistence or worsening of symptoms, fever or irritability associated with otoscopic signs of unresolving AOM, or development of symptoms indicative for mastoiditis or invasive disease **Secondary outcomes -** (a) presence of symptoms (i.e. fever, pain, irritability, vomiting, activity level) on days 1, 2 and 3; (b) number of analgesic doses, codeine doses on days 1, 2 and 3; (c) occurrence of

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Le Saux 2005 *(Continued)*

	any rash or diarrhoea in the 14 days after randomisation; (d) presence of middle ear effusion assessed by tympanometry at 1 and 3 months after diagnosis The parents were contacted on days 1, 2 and 3 after randomisation and once between day 10 and day 14 for administration of a standard questionnaire. If the parents or research assistant felt that the symptoms were not improving or were worsening, a medical reassessment was advised and the child was seen by a physician in the emergency department or clinic or by the paediatrician The child was clinically assessed at 1 month and 3 months after randomisation to determine the number of subsequent episodes of acute otitis media (AOM) and to undergo tympanometry
Notes	Physician Services Incorporated provided support for study expenses. Whitehall Robbins provided ibuprofen (Children's Advil) free of charge and funded the tympanometers used in the study but was not involved in the design of the study or the analysis of the results. No author conflicts declared.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation sequence stratified by study centre and age using random-permuted blocks of sizes 4 and 6
Allocation concealment (selection bias)	Low risk	Randomisation sequence was kept under secure conditions and was accessible only by the trial pharmacist
Blinding of participants and personnel (performance bias)	Low risk	Placebo was similar to amoxicillin with regard to appearance and taste and was dispensed in identical opaque bottles, which were numbered sequentially
Incomplete outcome data (attrition bias)	Low risk	Loss to follow-up at day 3 - treatment: N = 5 (2%) and placebo: N = 8 (3%)
Other bias	Low risk	ITT analysis - yes, baseline characteristics - balanced

Little 2001

Study characteristics

Methods	**Randomised -** yes, method of randomisation not described **Concealment of allocation -** adequate **Double-blind -** no; open-label study, investigators not blinded **Intention-to-treat (ITT) -** yes **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 315 children (N = 285 children included in analysis) **Age -** between 6 months and 10 years **Setting -** general practice; 42 general practitioners in 3 health authorities in south-west England

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Little 2001 *(Continued)*

	Inclusion criteria - acute otalgia and otoscopic evidence of acute inflammation of the eardrum (dullness or cloudiness with erythema, bulging or perforation). When children were too young for otalgia to be specifically documented from their history (under 3 years old) then otoscopic evidence alone was a sufficient entry criterion **Exclusion criteria -** otoscopic appearances consistent with crying or a fever alone (pink drum alone), appearances and history more suggestive of otitis media with effusion and chronic suppurative otitis media, serious chronic disease (such as cystic fibrosis, valvular heart disease), use of antibiotics < 2 weeks prior to randomisation, previous complications (septic complications, hearing impairment) and if the child was unwell to be left to wait and see (e.g. high fever, floppy, drowsy, not responding to antipyretics) **Baseline characteristics -** balanced
Interventions	**Tx -** immediate treatment with antibiotics: amoxicillin syrup 125 mg/5 mL 3 times daily for 7 days (children who were allergic to amoxicillin received erythromycin 125 mg/5 mL 4 times daily; N = 151 (N = 135 included in analysis) **C -** similar antibiotics were prescribed but parents were asked to wait for 72 hours before considering using the prescription. Parents were instructed that if their child still had substantial otalgia or fever after 72 hours, had discharge for > 10 days or was not starting to get better then they should collect the antibiotic prescription that was left at the practice; N = 164 (N = 150 included in analysis) **Use of additional medication -** for both groups doctors emphasised the importance of paracetamol in full doses for relief of pain and fever
Outcomes	**Primary outcomes -** (a) duration of symptoms (i.e. earache, ear discharge, night disturbance, crying); (b) daily pain score; (c) episodes of distress; (d) spoons of paracetamol used; (e) use of antibiotics Doctors were asked to provide information on days of illness, physical signs and antibiotic prescribing; parents were asked to complete a daily symptom diary
Notes	Open-label trial comparing immediate versus delayed antibiotic prescription (prescription provided but advised to fill only if symptoms did not improve or worsened) Funding - NHS Research and Development (South West and South East Regions) No author conflicts declared

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of randomisation not described
Allocation concealment (selection bias)	Low risk	Sealed, numbered, opaque envelopes
Blinding of participants and personnel (performance bias)	High risk	Open-label trial, outcome assessment not blinded
Incomplete outcome data (attrition bias)	Low risk	Loss to follow-up/exclusion from analysis (intervention ineffective, did not use antibiotics or did not delay) - treatment: N = 16 (12%) and placebo: N = 14 (9%); comparison of the baseline information for the 3 types of responders (those who provided diaries, those who gave information by telephone and those from whom no diary information could be collected) revealed no evidence of significant bias between treatment groups or between patients by age or severity of symptoms

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Little 2001 *(Continued)*

Other bias	Low risk	ITT analysis - yes, baseline characteristics - balanced

McCormick 2005

Study characteristics	
Methods	**Randomised -** yes, computer-generated randomisation sequence **Concealment of allocation -** unclear; method not described **Double-blind -** no, open-label trial, investigators blinded, parents not blinded **Intention-to-treat (ITT) -** yes **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 223 children (N = 218 children included in analysis at day 12) **Age -** between 6 months and 12 years **Setting -** secondary care: paediatric clinic of University of Texas Medical Branch (USA) **Inclusion criteria -** children were required to have (a) symptoms of ear infection; (b) otoscopic evidence of acute otitis media (AOM), including middle ear effusion; (c) non-severe AOM **Exclusion criteria -** co-morbidity requiring antibiotic treatment, anatomic defect of ear or nasopharynx, allergy to study medication, immunologic deficiency, major medical condition and/or indwelling ventilation tube or draining otitis in the affected ear(s) **Baseline characteristics -** balanced
Interventions	**Tx -** immediate treatment with antibiotics: oral amoxicillin 90 mg/kg/day twice daily for 10 days; N = 112 (N = 110 included in analysis at day 12) **C -** expectant observation: no immediate antibiotics; N = 111 (N = 108 included in analysis at day 12) Children in the control group with AOM failure or recurrence received oral amoxicillin 90 mg/kg/day; children in Tx group with AOM failure or recurrence received amoxicillin-clavulanate (90 mg/kg/day of amoxicillin component) **Use of additional medication -** all parents received saline nose drops and/or cerumen-removal drops (if needed), ibuprofen and over-the-counter decongestant/antihistamine to be given as needed
Outcomes	**Primary outcomes -** (a) parent satisfaction with AOM care; (b) resolution of AOM symptoms after treatment, including number of doses of symptom medication given; (c) AOM failure (days 0 to 12) or recurrence (days 13 to 30) defined as attending to the paediatrician clinic with acute ear symptoms, an abnormal tympanic membrane, or an AOM severity score higher than that at enrolment; (d) nasopharyngeal carriage of *Streptococcus pneumoniae* strains resistant to antibiotics **Secondary outcomes -** (a) minor adverse events caused by medication (e.g. allergy, diarrhoea and candidal infection); (b) serious AOM-related adverse events (e.g. invasive pneumococcal disease, mastoiditis, bacteraemia, meningitis, perforation of the tympanic membrane, hospitalisation and emergency ear surgery; (c) parent-child quality of life measures related to AOM Parents were instructed to complete a symptom diary from day 1 to 10 and a satisfaction questionnaire on day 12 and day 30; routine follow-up appointments for data collection were scheduled for day 12 and day 30. Patient-initiated visits were scheduled on request by the parents for children who seemed to not be responding to treatment

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

McCormick 2005 *(Continued)*

Notes	Investigator-blinded trial comparing immediate antibiotic prescribing versus expectant observation (no prescription provided) Funding - "Studies were conducted with assistance from the General Clinical Research Center at the University of Texas Medical Branch, funded by National Center for Research Resources, National Institutes of Health, US Public Health Service grant M01 RR 00073 and Agency for Healthcare Research and Quality grant R01 HS10613-02." The authors declare no conflicts of interest.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation sequence
Allocation concealment (selection bias)	Unclear risk	Method not described
Blinding of participants and personnel (performance bias)	Unclear risk	Investigator-blinded study, parents not blinded
Incomplete outcome data (attrition bias)	Low risk	Loss to follow-up at day 12 - treatment: N = 2 (2%) and expectant observation: N = 3 (3%)
Other bias	Low risk	ITT analysis - yes, baseline characteristics - balanced

Mygind 1981

Study characteristics

Methods	**Randomised -** yes, method of randomisation not described **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** unclear **Loss to follow-up -** reasons described, unclear from which treatment group patients were excluded **Design -** parallel
Participants	**N -** 158 children (N = 149 included in analysis) **Age -** between 1 and 10 years **Setting -** general practice and secondary care: confirmation of diagnosis and trial recruitment were done by otorhinolaryngologists in Copenhagen (Denmark) **Inclusion criteria -** earache for 1 to 24 hours. The diagnosis was made if the child cried because of pain and if the tympanic membrane appeared to be red and inflamed. **Exclusion criteria -** antibiotic treatment < 4 weeks prior to randomisation, other treatment apart from acetylsalicylic acid already commenced, secretion in the external ear, suspected chronic otitis media, treatment for secretory otitis media within last 12 months, concurrent disease (e.g. pneumonia or severe tonsillitis), suspected penicillin allergy

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Mygind 1981 *(Continued)*

	Baseline characteristics - balanced
Interventions	**Tx -** penicillin 50 mg/mL 4 times daily; children aged 1 to 2 years: 10 mL daily, children between 3 and 5 years: 20 mL daily, children between 6 and 10 years: 30 mL daily for 7 days; N = ? (N = 72 included in analysis) **C -** placebo for 7 days; N = ? (N = 77 included in analysis) **Use of additional medication -** acetylsalicylic acid tablets (maximum of 50 mg/kg/day for 3 days) were supplied as the only supplementary treatment permitted
Outcomes	**Main outcomes:** (a) mean symptoms (i.e. pain, fever) scores; (b) number of analgesic tables used; (c) contralateral otitis; (d) spontaneous perforation of tympanic membrane; (e) mean number of days of otorrhoea; (f) tympanometry results at 1 week, 4 weeks and 3 months Initial visits were performed at home: otoscopy and bacterial culture from nasopharynx were performed Score cards were given to parents Follow-up visits at hospital at day 2 to 3, day 7, week 4 and week 12. If supplementary treatment was required at day 2 to 3, then myringotomy was performed. If supplementary treatment was required at day 7, then amoxicillin was given
Notes	Funding - "The penicillin and placebo granulate were supplied in coded bottles by Astra Pharmaceuticals, Sodertalje, Sweden." No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of randomisation not described
Allocation concealment (selection bias)	Low risk	Randomisation performed by pharmaceutical company. Penicillin and placebo were supplied in coded bottles to study personnel
Blinding of participants and personnel (performance bias)	Unclear risk	Identical taste and appearance to amoxicillin and placebo not described
Incomplete outcome data (attrition bias)	Unclear risk	Patients not included in analysis - N = 9 (6%). Reasons described, unclear from which treatment group patients were excluded
Other bias	Unclear risk	ITT analysis - unclear, baseline characteristics - balanced

Neumark 2007

Study characteristics

Methods	**Randomised -** yes, Internet-based random number generator **Concealment of allocation -** unclear; method not described **Double-blind -** no, open-label trial **Intention-to-treat (ITT) -** unclear

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Neumark 2007 *(Continued)*

	Loss to follow-up - reasons described, unclear from which treatment group patients were excluded **Design -** parallel
Participants	**N -** 186 children (N = 179 patients were included in analysis; 7 patients were excluded due to non-compliance with protocol) **Age -** between 2 and 16 years **Setting -** general practice: 32 healthcare centres and 72 general practitioners in Sweden **Inclusion criteria -** acute otitis media (AOM) was based on direct inspection of the eardrum by pneumatic otoscope or preferably an aural microscope. Findings had to include a bulging, red eardrum displaying reduced mobility. **Exclusion criteria -** perforation of the eardrum, chronic ear conditions or impaired hearing, previous adverse reactions to penicillin, concurrent disease that should be treated with antibiotics, recurrent AOM (3 or more AOM episodes during the past 6 months), children with immunosuppressive conditions, genetic disorders and mental disease or retardation **Baseline characteristics -** balanced
Interventions	**Tx -** immediate treatment with antibiotics: phenoxymethylpenicillin 25 mg/kg twice daily for 5 days; N = 92 **C -** expectant observation: no immediate antibiotics; N = 87 The guardians received written information about how to act if the condition did not improve or got worse within 3 days after randomisation **Use of additional medication -** symptomatic treatment with paracetamol or non-steroidal anti-inflammatory drugs (NSAIDs), drugs reducing the swelling of the nasal mucosa (e.g. decongestant nose drops) and nasal steroids were allowed
Outcomes	**Primary outcomes -** (a) pain at day 0, 1, 2 and 3 to 7; (b) use of analgesics at day 0, 1, 2, 3, 4 to 7; (c) fever > 38 °C at day 0, 1, 2 and 3 to 7; (d) subjective recovery at day 14 and 3 months; (e) perforations at 3 months; (f) serous otitis media at 3 months All participants were asked to complete a symptom diary for 7 days; a nurse telephoned all participants after approximately 14 days to supplement the information in the diary and to register all acute contacts that had occurred during the first week of treatment; the final follow-up was performed after 3 months to register perforations and serous otitis media
Notes	Open-label trial comparing immediate antibiotic prescribing versus expectant observation (no prescription provided but advice on what to do if symptoms did not improve or worsened) The study was financially supported by FORSS Medical Research Council of Southeast Sweden, Department of Research and Development, Kalmar County Council, The Swedish Society of Medicine, and STRAMA, The Swedish Strategic Programme for the Rational Use of Antimicrobial Agents and Surveillance of Resistance. No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Internet-based random number generator
Allocation concealment (selection bias)	Unclear risk	Method not described

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Neumark 2007 *(Continued)*

Blinding of participants and personnel (performance bias)	High risk	Open-label trial, outcome assessment not blinded
Incomplete outcome data (attrition bias)	Unclear risk	Patients not included in analysis - N = 7 (4%). Reasons described, unclear from which treatment group patients were excluded
Other bias	Unclear risk	ITT analysis - unclear, baseline characteristics - balanced

Shahbaznejad 2021

Study characteristics	
Methods	**Randomised -** yes **Concealment of allocation -** unclear **Double-blind -** no **Intention-to-treat (ITT) -** unclear **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 407 children (N = 375 children included in analysis) **Age -** between 6 months and 6 years **Setting -** paediatric infectious disease clinic in Buali tertiary hospital in Sari, north of Iran **Inclusion criteria -** AOM diagnosis (acute onset of fever, erythema of tympanic membrane and middle ear effusion), and onset of the symptoms within 48 hours prior to visiting the physician **Exclusion criteria -** severe AOM (fever > 39°C, moderate to severe irritability and otalgia), otorrhoea, concurrent conjunctivitis, underlying diseases such as immunodeficiency and history of penicillin allergy **Baseline characteristics -** balanced
Interventions	**Tx -** amoxicillin (80 mg/kg/day divided in 2 doses) for 7 to 10 days; N = 194 (N = 188 included in primary analysis) **C -** watchful waiting; N = 213 (N = 208 included in primary analysis) **Use of additional medication -** acetaminophen or ibuprofen was prescribed for participants in both groups in case of pain and fever
Outcomes	**Primary outcome -** AOM recovery, defined as elimination of fever, irritability and ear pain after 72 hours **Secondary outcomes -** fever, ear pain, irritability, poor feeding, seizure, vomiting, diarrhoea, coryza, cough, pharyngitis, nasal congestion after 72 hours; severe complications related to AOM; AOM recurrences after 1 and 3 months; middle ear effusion after 1 and 3 months (defined as otorrhoea, loss of tympanic membrane mobility, air-fluid level or bubbles behind tympanic membrane and tympanic oedema); side effects such as diarrhoea, vomiting and rash
Notes	"The deputy of research and technology of Mazandaran University of Medical Sciences provided the financial support for the study."

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Shahbaznejad 2021 *(Continued)*

No author conflicts declared.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Quote: "The patients were randomly divide into two groups, …, by simple randomization method, and physicians were given a table of precoded numbers and patients enrolled the study in order of table numbers. Neither the participants nor the evaluators were aware of the randomization process or group allocation." How the randomisation sequence was generated is not described. Also, trial registration says "entering in blocks of four and randomised division", which is inconsistent with the publication.
Allocation concealment (selection bias)	Unclear risk	Description of method of concealment of allocation unclear
Blinding of participants and personnel (performance bias)	High risk	Open-label study, outcome assessment not blinded
Incomplete outcome data (attrition bias)	Low risk	Loss to follow-up rates low (5.3%) and comparable across groups
Other bias	Unclear risk	ITT unclear, no formal sample size calculations, no details provided on the consumption of analgesics and/or antibiotics during follow-up across groups

Spiro 2006

Study characteristics

Methods	**Randomised -** yes, computer-assisted randomisation **Concealment of allocation -** adequate **Double-blind -** no, open-label study, investigators blinded **Intention-to-treat (ITT) -** yes **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 283 children (N = 265 children included in analysis at days 4 to 6) **Age -** between 6 months and 12 years **Setting -** secondary care: paediatric emergency department of Yale-New Haven Hospital in New Haven (USA) **Inclusion criteria -** the diagnosis of acute otitis media (AOM) was made at the discretion of the clinician according to the diagnostic criteria in the evidence-based guideline published in *Pediatrics* 2004 **Exclusion criteria -** presence of additional intercurrent bacterial infection such as pneumonia, if the patient appeared to be "toxic" as determined by the clinician, hospitalisation, immunocompromised children, antibiotic treatment < 1 week prior to randomisation, children who had either myringotomy

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Spiro 2006 *(Continued)*

	or a perforated tympanic membrane, uncertain access to medical care (e.g. no telephone access), primary language of parents was neither English nor Spanish, previous enrolment in the study **Baseline characteristics -** balanced
Interventions	**Tx -** immediate treatment with antibiotics; N = 145 (N = 133 included in analysis at days 4 to 6) **C -** participants randomised to delayed prescription were given written and verbal instructions "not to fill the antibiotic prescription unless your child either is not better or is worse 48 hours (2 days) after today's visit"; N = 138 (N = 132 included in analysis at days 4 to 6) **Use of additional medication -** all participants received complimentary bottles of ibuprofen suspension (100 mg/5 mL) and analgesic ear drops
Outcomes	**Primary outcome -** proportion of each group that filled the prescription for an antibiotic. This was defined by whether the parent filled the prescription within 3 days of enrolment and was determined by the response to this question at the interview at day 4 to 6. **Secondary outcomes -** (a) clinical course of the illness; (b) adverse effects of medications; (c) days of school or work missed; (d) unscheduled medical visits; (e) comfort of parents with management of AOM without antibiotics for future episodes 2 trained research assistants blinded to group assignment conducted standardised, structured telephone interviews with the parents at day 4 to 6, day 11 to 14, day 30 and day 40 after enrolment
Notes	Investigator-blinded study comparing immediate versus delayed antibiotic prescribing (prescription provided and advised to fill only if symptoms worsen or do not improve) Funding - this work was supported in part by grant AI01703 from the National Institutes of Health, grant M01-RR00125 from the General Clinical ResearchCenter, Yale University School of Medicine, and the Friends of Yale-New Haven Children's Hospital, who provided material support for this research. The funding organisations played no role in the design, conduct, analysis, or interpretation of the research or in any aspect of preparation or approval of the manuscript. No information about author conflicts is provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-assisted randomisation
Allocation concealment (selection bias)	Low risk	Sealed, opaque envelopes
Blinding of participants and personnel (performance bias)	Unclear risk	Investigator-blinded study, parents not blinded
Incomplete outcome data (attrition bias)	Unclear risk	Loss to follow-up at day 4 to 6 treatment: N = 12 (8%) and expectant observation: N = 6 (4%)
Other bias	Low risk	ITT analysis - yes, baseline characteristics - balanced

Tapiainen 2014

Study characteristics

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Tapiainen 2014 *(Continued)*

Methods	**Randomised -** yes, block randomisation, computerised randomisation list **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** yes **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 84 children (N = 84 children included in analysis at 8 weeks) **Age -** between 6 months and 15 years **Setting -** primary and secondary care: children in day care centres attending an AOM prevention trial at the Department of Pediatrics, Oulu University Hospital and children visiting the City of Oulu Health Care Center and Mehiläinen Pediatric Private Practice, Oulu (Finland) **Inclusion criteria -** acute symptoms of respiratory infection and/or ear-related symptoms and signs of tympanic membrane inflammation together with middle ear effusion at pneumatic otoscopy performed by a study physician **Exclusion criteria -** ventilation tubes (grommets), AOM complication, amoxicillin allergy, Down's syndrome, congenital craniofacial abnormality and immunodeficiency **Baseline characteristics -** balanced
Interventions	**Tx -** amoxicillin-clavulanate for 7 days (amoxicillin 40 mg/kg/day divided into 2 daily doses); N = 42 (N = 42 included in analysis) **C -** matching placebo in 2 doses for 7 days; N = 42 (N = 42 included in analysis) **Use of additional medication -** not described
Outcomes	**Primary outcome -** time to middle ear effusion disappearance defined as a normal tympanogram finding (A curve) from both ears on 2 consecutive measurement days (either at home or at the study clinic) **Secondary outcomes -** (a) time to improved tympanogram findings (i.e. A or C curve) from both ears; (b) time to normal pneumatic otoscopy or otomicroscopy findings from both ears; (c) proportions of children with persistent middle ear effusion on days 7, 14 and 60; (d) disappearance of pain; (e) disappearance of fever; (f) use of pain medication; (g) possible adverse effects of antimicrobial treatment Children were examined by the study physician with pneumatic otoscopy or otomicroscopy and tympanometry at study entry, after 3 and 7 days, and then weekly until both ears were healthy according to pneumatic otoscopy or otomicroscopy Families were trained to perform tympanometry using a handheld tympanometer for daily follow-up at home
Notes	Funding - "This work was financially supported by the Pediatric Research Foundation, Finland, and Medical Research Foundation, Finland." The authors declare no conflicts of interest.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Block randomisation, computerised randomisation list

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Tapiainen 2014 *(Continued)*

Allocation concealment (selection bias)	Low risk	Randomisation list was kept in the pharmacy, which delivered the study drugs to the families according to the consecutive study number
Blinding of participants and personnel (performance bias)	Low risk	Bottles containing amoxicillin-clavulanate or placebo were indistinguishable, dosing was similar in both groups and placebo mixture was flavoured and sweetened to resemble the taste of amoxicillin-clavulanate
Incomplete outcome data (attrition bias)	Low risk	All children were included in the analysis
Other bias	Low risk	ITT analysis - yes, baseline characteristics - balanced

Thalin 1985

Study characteristics	
Methods	**Randomised -** yes, block randomisation, method of random sequence generation not described **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** unclear **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 293 children (N = 293 children included in analysis) **Age -** between 2 and 15 years **Setting -** secondary care: department of otorhinolaryngology in Halmstad (Sweden) **Inclusion criteria -** purulent acute otitis media (AOM) (no further criteria described) **Exclusion criteria -** antibiotic treatment or AOM episode < 4 weeks prior to randomisation, suspected penicillin allergy, presence of ventilation tubes, sensorineural hearing loss, existence of concomitant infection for which antibiotic treatment was required and chronic diseases **Baseline characteristics -** not described
Interventions	**Tx -** phenoxymethyl penicillin 50 mg/kg/day twice daily for 7 days; N = 159 (N = 159 included in analysis) **C -** matching placebo in 2 doses for 7 days; N = 158 (N = 158 included in analysis) **Use of additional medication -** all children were given nose drops containing oxymetazoline chloride and, if needed, analgesics (paracetamol)
Outcomes	**Primary outcome -** treatment failure (defined as remaining non-negligible symptoms such as pain and fever, insufficient resolution of infectious signs during treatment period of 7 days, or both **Secondary outcomes -** (a) resolution of symptoms over time; (b) AOM relapses; (c) tympanometry, audiometry, or both, results at 4 weeks The children were examined at day 0, days 3 to 4, days 8 to 10 and at 4 weeks

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Thalin 1985 *(Continued)*

	Parents were instructed to record symptoms (i.e. temperature, otalgia, discharge from ear and consumption of supplied symptomatic drugs)
Notes	There is no information on study funding provided. There is no information about author conflicts provided.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Block randomisation, method of random sequence generation not described
Allocation concealment (selection bias)	Low risk	Randomisation list was kept by the clinical pharmacologist of the hospital and not disclosed to the investigators until the clinical trial was completed
Blinding of participants and personnel (performance bias)	Low risk	Placebo with same taste and appearance as penicillin
Incomplete outcome data (attrition bias)	Low risk	No children lost to follow-up for primary analysis
Other bias	Unclear risk	ITT analysis - unclear; baseline characteristics - not described

Tähtinen 2011

Study characteristics

Methods	**Randomised -** yes, computerised random number generator with block length of 10 **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** yes **Loss to follow-up -** described **Design -** parallel
Participants	**N -** 322 children (N = 319 children were included in analysis) **Age -** between 6 months and 3 years **Setting -** general practice: healthcare centre of Turku (Finland) **Inclusion criteria -** acute otitis media (AOM) based on 3 criteria: (a) middle ear fluid had to be detected by means of pneumatic otoscopic examination that showed at least 2 of the following tympanic membrane findings: bulging position, decreased or absent mobility, abnormal colour or opacity not due to scarring, or air fluid interfaces; (b) at least 1 of the following acute inflammatory signs in the tympanic membrane had to be present: distinct erythematous patches or streaks or increased vascularity over full, bulging, or yellow tympanic membrane; (c) presence of acute symptoms such as fever, otalgia or respiratory symptoms **Exclusion criteria -** ongoing antibiotic treatment; AOM with spontaneous perforation of the tympanic membrane; systemic or nasal steroid therapy within 3 preceding days; antihistamine, oseltamivir or a combination therapy within 3 preceding days; contraindication to penicillin or amoxicillin; presence

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Tähtinen 2011 *(Continued)*

	of ventilation tube; severe infection requiring antibiotic treatment; documented Epstein-Barr virus infection within 7 preceding days; Down's syndrome or other condition affecting middle ear diseases; known immunodeficiency **Baseline characteristics -** balanced
Interventions	**Tx -** amoxicillin-clavulanate 40 mg-5.7 mg/kg daily in 2 doses for 7 days; N = 162 (N = 161 included in analysis) **C -** matching placebo in 2 doses for 7 days; N = 160 (N = 158 included in analysis) **Use of additional medication -** the use of analgesics and antipyretic agents was encouraged and the use of analgesic ear drops and decongestant nose drops or sprays was allowed
Outcomes	**Primary outcome -** time to treatment failure (i.e. a composite endpoint consisting of 6 independent components: (a) no improvement in overall condition at day 2, (b) worsening of the child's overall condition at any time, (c) no improvement in otoscopic signs at day 7, (d) perforation of tympanic membrane at any time, (e) severe infection (e.g. mastoiditis or pneumonia) necessitating systemic open-label antimicrobial treatment at any time, (f) any other reason for stopping the study drug at any time **Secondary outcomes -** assessed by study physician - (a) time to the initiation of rescue treatment; (b) time to development of contralateral AOM - diary symptom assessment; (c) resolution of symptoms; (d) use of analgesics Parents were given a diary to record symptoms, doses of study drugs and any other medications, and adverse events First visit after enrolment (= day 0) was scheduled at day 2. End-of-treatment visit was scheduled at day 7.
Notes	Supported by the Fellowship Award of the European Society for Paediatric Infectious Diseases (to Dr. Ruohola) and by grants from the Foundation for Paediatric Research; Research Funds from Specified Government Transfers; the Jenny and Antti Wi- huri Foundation; the Paulo Foundation; the Maud Kuistila Me- morial Foundation; the Emil Aaltonen Foundation; the Finnish Cultural Foundation, Varsinais-Suomi Regional Fund; the Turku University Hospital Research Foundation; and the Finnish–Norwegian Medical Foundation. Author OR declares receipt of consultancy fees from Abbott and Novartis for work outside the submitted work.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computerised random number generator with block length of 10
Allocation concealment (selection bias)	Low risk	Concealment of allocation by the pharmacist (independent of trial team) by labelling the identical opaque study drug containers with allocation numbers; allocation list was kept at the paediatric infectious disease ward behind locked doors
Blinding of participants and personnel (performance bias)	Low risk	Placebo with same taste and appearance as amoxicillin-clavulanate
Incomplete outcome data (attrition bias)	Low risk	Loss to follow-up - treatment: N = 1 (1%) and placebo: N = 2 (1%)
Other bias	Low risk	ITT analysis - yes, baseline characteristics - balanced

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

van Buchem 1981

Study characteristics	
Methods	**Randomised -** yes, method of randomisation not described **Concealment of allocation -** adequate **Double-blind -** yes **Intention-to-treat (ITT) -** unclear **Loss to follow-up -** reasons not described, unclear from which treatment group patients were excluded **Design -** 2 x 2 factorial design
Participants	**N -** 202 children (N = 171 children included in analysis; N = 31 were excluded from the study) **Age -** between 2 and 12 years **Setting -** both general practice and secondary care: 12 general practitioners in or near Tilburg (the Netherlands) recruited patients and referred them to 1 of the 3 otorhinolaryngologists, which excluded those cases where there was disagreement with the diagnosis **Inclusion criteria -** acute otitis media (AOM) was based on history and clinical picture (i.e. diffuse redness, bulging of the eardrum, or both) **Exclusion criteria -** antibiotic treatment < 2 weeks prior to randomisation, chronic otitis or otitis media serosa, contraindication for antibiotic treatment **Baseline characteristics -** balanced
Interventions	**Tx -** sham myringotomy and amoxicillin 250 mg 3 times daily for 7 days; N = 47 **C -** sham myringotomy and matching placebo for 7 days; N = 40 **Use of additional medication -** all participants were allowed to use decongestant nose drops and analgesic suppositories (i.e. children aged 2 to 7 years: acetylsalicylic acid 50 mg, phenacetin 50 mg, phenobarbitone 15 mg, codeine phosphate 2.5 mg, caffeine 1.25 mg; children aged 8 to 12 years: acetylsalicylic acid 100 mg, phenacetin 100 mg, phenobarbitone 30 mg, codeine phosphate 5 mg, caffeine 2.5 mg
Outcomes	**Main outcomes -** (a) parent report of pain at day 0, 1 and 7; (b) otoscopic findings at day 0, 1 and 7; (c) discharge from ear at day 1, 7 and 14; (d) mean temperature at day 0, 1 and 7; (e) AOM relapses at 6 months; (f) audiogram findings after 4 and 8 weeks
Notes	This trial had a 2 x 2 factorial design resulting in 4 treatment groups: (1) sham myringotomy plus antibiotics; (2) sham myringotomy plus placebo; (3) myringotomy plus antibiotics; and (4) myringotomy plus placebo. We used all arms of this trial, so this trial was included twice in certain analyses by including both the sham myringotomy plus antibiotic versus the sham myringotomy plus placebo comparison and the myringotomy plus antibiotic versus myringotomy plus placebo comparison data. No information on study funding is provided. No information about author conflicts is provided.

Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

van Buchem 1981 *(Continued)*

Random sequence generation (selection bias)	Unclear risk	Method of randomisation not described
Allocation concealment (selection bias)	Low risk	Randomisation performed by otorhinolaryngologists; general practitioner and parent/child were outcome assessors and remained blinded
Blinding of participants and personnel (performance bias)	Low risk	Sham myringotomy and placebo was similar to amoxicillin with regard to appearance and taste
Incomplete outcome data (attrition bias)	Unclear risk	Loss to follow-up/exclusions - N = 31 (15%). Reasons not described
Other bias	Unclear risk	ITT analysis - unclear, baseline characteristics - balanced

AOM: acute otitis media
AOM-SOS: otitis media - severity of symptoms
C: control
ITT: intention-to-treat
MEE: middle ear effusion
Tx: treatment

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Arguedas 2011	No comparison of antibiotic to placebo or expectant observation: trial comparing single-dose, extended-release azithromycin versus a 10-day regimen of amoxicillin/clavulanate
Casey 2012	No comparison of antibiotic to placebo or expectant observation: trial comparing high-dose amoxicillin/clavulanate versus cefdinir
Chaput 1982	Short versus long course of therapy
Engelhard 1989	No comparison of antibiotic to placebo; the 3 arms were Augmentin, myringotomy or both
Hay 2021	Study population - children with acute otitis media and ear discharge due to a spontaneous perforation of the ear drum (specific population with poorer prognosis than children with acute otitis media without ear discharge)
Liu 2011	No comparison of antibiotic to placebo or expectant observation: trial comparing single oral doses of azithromycin in extended-release versus immediate-release formulations
Ostfeld 1987	Non-randomised study
Rudberg 1954	Non-randomised study: assigned "randomly" based on case number but then allowed to change groups
Ruohola 2003	Conducted in children with ventilation tubes
Sarrell 2003	No comparison of antibiotic to placebo. Method of randomisation not provided and groups appear to be unbalanced at baseline
Tähtinen 2012	Secondary analysis of placebo-controlled trial. This study included the total group of children allocated to immediate antimicrobial treatment (N = 161) and a subgroup of children from the placebo group that received delayed antibiotics (N = 53). As a consequence, comparability of prognosis

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Study	Reason for exclusion
	achieved through randomisation is violated, producing groups of children that are incomparable, which may lead to biased effect estimates
van Buchem 1985	Non-randomised study

N: number

Characteristics of ongoing studies *[ordered by study ID]*

the WATCH trial

Study name	the WATCH trial
Methods	Non-inferiority, multi-centre randomised controlled trial
Participants	Urban Aboriginal and Torres Strait Islander children aged 2 to 16 years with AOM without perforation and who are considered at low risk for complications
Interventions	Immediate prescription of antibiotic therapy (choice of antibiotic recommended to be based on Australian prescribing guidelines) versus watchful waiting (no immediate provision of antibiotic therapy at the time of enrolment)
Outcomes	**Primary outcome -** proportion of children with clinical resolution of AOM, defined as all of the following: no pain, fever not higher than 38 °C, no bulging eardrum and no complications of OM (no perforation or mastoiditis) on day 7 **Secondary outcomes -** (a) proportion of children with resolution of signs of AOM on day 7; (b) proportion of children with MEE, perforation and CSOM at week 7; (c) proportion of children with new antibiotic prescriptions for an index case of AOM; (d) parent/carer-reported time to resolution of AOM symptoms assessed by parent/carer symptom report at days 3, 7 and 14, including the AOM Faces Scale; (e) usage of analgesia for AOM symptom relief; (f) parent/carer satisfaction with AOM treatment
Starting date	The trial is registered with Australian New Zealand Clinical Trials Registry (ACTRN12613001068752) Date of registration: 24 September 2013
Contact information	Penelope Abbott: School of Medicine, Western Sydney University, Sydney, NSW, Australia Email: p.abbott@westernsydney.edu.au
Notes	—

AOM: acute otitis media
CSOM: chronic suppurative otitis media
MEE: middle ear effusion
OM: otitis media

DATA AND ANALYSES

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

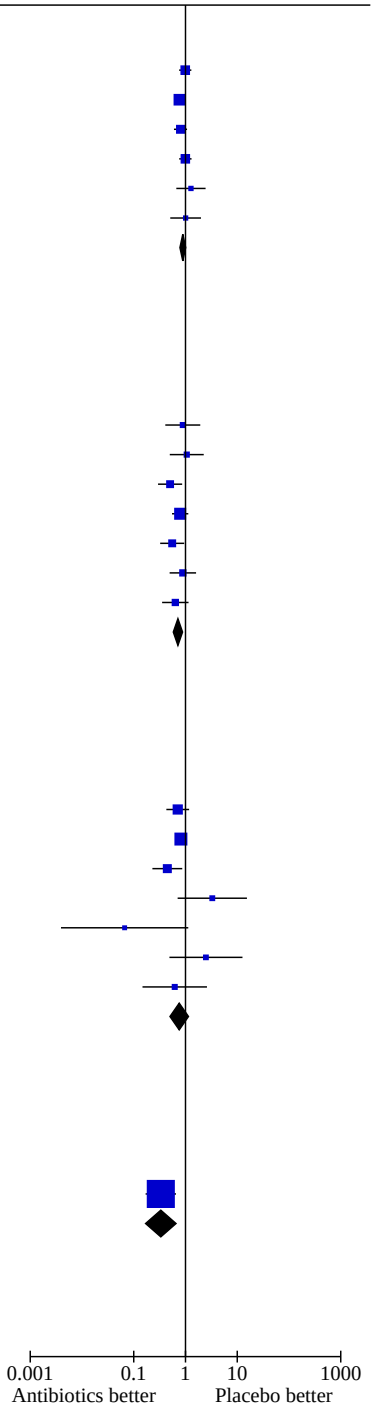
Comparison 1. Antibiotics versus placebo

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Pain	12		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
1.1.1 Pain at 24 hours	5	1394	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.78, 1.01]
1.1.2 Pain at 2 to 3 days	7	2320	Risk Ratio (M-H, Random, 95% CI)	0.71 [0.58, 0.88]
1.1.3 Pain at 4 to 7 days	7	1264	Risk Ratio (M-H, Random, 95% CI)	0.76 [0.50, 1.14]
1.1.4 Pain at 10 to 12 days	1	278	Risk Ratio (M-H, Random, 95% CI)	0.33 [0.17, 0.66]
1.2 Vomiting, diarrhoea or rash	8	2107	Risk Ratio (M-H, Random, 95% CI)	1.38 [1.16, 1.63]
1.3 Abnormal tympanometry	8		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
1.3.1 2 to 4 weeks	7	2138	Risk Ratio (M-H, Random, 95% CI)	0.83 [0.72, 0.96]
1.3.2 6 to 8 weeks	3	953	Risk Ratio (M-H, Random, 95% CI)	0.89 [0.70, 1.13]
1.3.3 3 months	3	809	Risk Ratio (M-H, Random, 95% CI)	0.94 [0.66, 1.34]
1.4 Tympanic membrane perforation	5	1075	Risk Ratio (M-H, Random, 95% CI)	0.43 [0.21, 0.89]
1.5 Contralateral otitis (in unilateral cases)	4	906	Risk Ratio (M-H, Random, 95% CI)	0.49 [0.25, 0.95]
1.6 Late AOM recurrences	6	2200	Risk Ratio (M-H, Random, 95% CI)	0.94 [0.79, 1.11]

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Analysis 1.1. Comparison 1: Antibiotics versus placebo, Outcome 1: Pain

Study or Subgroup	Antibiotics Events	Total	Placebo Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
1.1.1 Pain at 24 hours						
Burke 1991	53	112	56	117	21.8%	0.99 [0.75 , 1.30]
Le Saux 2005	82	258	106	254	30.4%	0.76 [0.60 , 0.96]
Tähtinen 2011	40	85	47	80	19.1%	0.80 [0.60 , 1.07]
Thalin 1985	62	159	62	158	21.4%	0.99 [0.76 , 1.31]
van Buchem 1981	17	48	10	36	3.8%	1.27 [0.67 , 2.44]
van Buchem 1981	13	47	11	40	3.5%	1.01 [0.51 , 1.99]
Subtotal (95% CI)		**709**		**685**	**100.0%**	**0.89 [0.78 , 1.01]**
Total events:	267		292			
Heterogeneity: Tau² = 0.00; Chi² = 4.75, df = 5 (P = 0.45); I² = 0%						
Test for overall effect: Z = 1.84 (P = 0.07)						
1.1.2 Pain at 2 to 3 days						
Appelman 1991	11	67	10	54	6.9%	0.89 [0.41 , 1.93]
Halsted 1968	17	62	7	27	7.4%	1.06 [0.50 , 2.25]
Kaleida 1991	19	488	38	492	14.6%	0.50 [0.29 , 0.86]
Le Saux 2005	43	253	53	246	32.1%	0.79 [0.55 , 1.13]
Mygind 1981	15	72	29	77	14.7%	0.55 [0.32 , 0.94]
Tähtinen 2011	17	85	18	80	12.1%	0.89 [0.49 , 1.60]
Thalin 1985	16	159	25	158	12.2%	0.64 [0.35 , 1.14]
Subtotal (95% CI)		**1186**		**1134**	**100.0%**	**0.71 [0.58 , 0.88]**
Total events:	138		180			
Heterogeneity: Tau² = 0.00; Chi² = 4.82, df = 6 (P = 0.57); I² = 0%						
Test for overall effect: Z = 3.22 (P = 0.001)						
1.1.3 Pain at 4 to 7 days						
Burke 1991	20	111	29	114	24.4%	0.71 [0.43 , 1.18]
Damoiseaux 2000	69	117	89	123	36.0%	0.82 [0.68 , 0.98]
Mygind 1981	10	72	24	77	19.3%	0.45 [0.23 , 0.87]
Tähtinen 2011	7	85	2	80	6.0%	3.29 [0.71 , 15.39]
Tapiainen 2014	0	42	7	42	2.0%	0.07 [0.00 , 1.13]
Thalin 1985	5	159	2	158	5.5%	2.48 [0.49 , 12.62]
van Buchem 1981	3	46	4	38	6.8%	0.62 [0.15 , 2.60]
Subtotal (95% CI)		**632**		**632**	**100.0%**	**0.76 [0.50 , 1.14]**
Total events:	114		157			
Heterogeneity: Tau² = 0.11; Chi² = 11.63, df = 6 (P = 0.07); I² = 48%						
Test for overall effect: Z = 1.33 (P = 0.18)						
1.1.4 Pain at 10 to 12 days						
Hoberman 2011	10	139	30	139	100.0%	0.33 [0.17 , 0.66]
Subtotal (95% CI)		**139**		**139**	**100.0%**	**0.33 [0.17 , 0.66]**
Total events:	10		30			
Heterogeneity: Not applicable						
Test for overall effect: Z = 3.19 (P = 0.001)						



Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

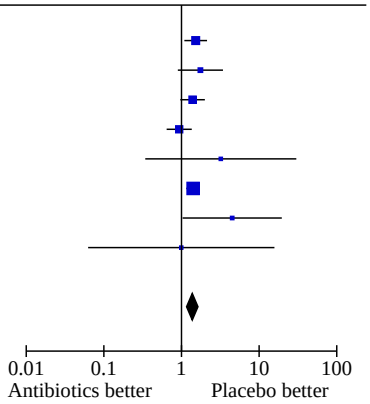
Analysis 1.2. Comparison 1: Antibiotics versus placebo, Outcome 2: Vomiting, diarrhoea or rash

Study or Subgroup	Antibiotics Events	Antibiotics Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI
Burke 1991	53	114	36	118	19.5%	1.52 [1.09 , 2.13]
Damoiseaux 2000	20	117	12	123	5.9%	1.75 [0.90 , 3.42]
Hoberman 2011	49	144	36	147	17.2%	1.39 [0.97 , 2.00]
Le Saux 2005	43	235	47	240	16.6%	0.93 [0.64 , 1.36]
Mygind 1981	3	72	1	77	0.6%	3.21 [0.34 , 30.14]
Tähtinen 2011	105	161	73	158	38.6%	1.41 [1.15 , 1.73]
Tapiainen 2014	9	42	2	42	1.3%	4.50 [1.03 , 19.60]
Thalin 1985	1	159	1	158	0.4%	0.99 [0.06 , 15.75]
Total (95% CI)		**1044**		**1063**	**100.0%**	**1.38 [1.16 , 1.63]**
Total events:	283		208			

Heterogeneity: Tau² = 0.01; Chi² = 8.16, df = 7 (P = 0.32); I² = 14%

Test for overall effect: Z = 3.72 (P = 0.0002)

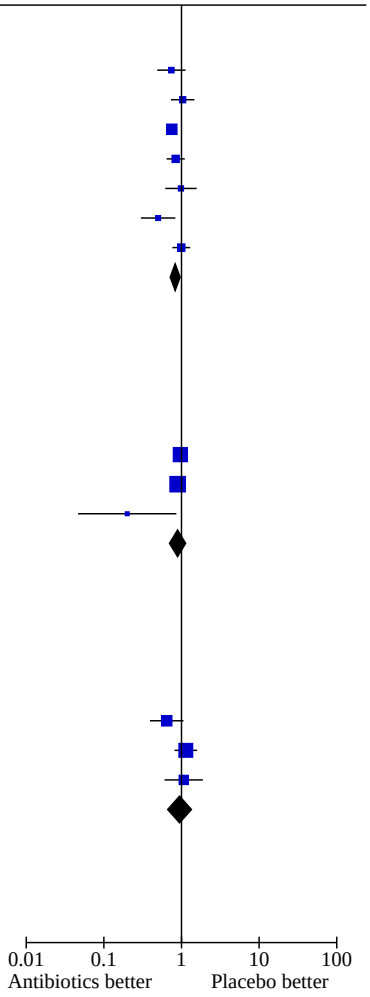
Test for subgroup differences: Not applicable



Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

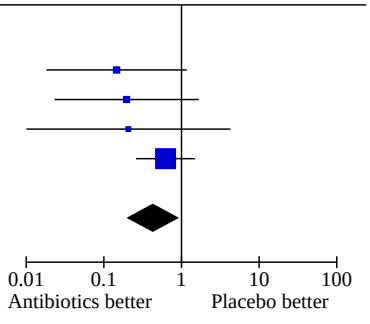
Analysis 1.3. Comparison 1: Antibiotics versus placebo, Outcome 3: Abnormal tympanometry

Study or Subgroup	Antibiotics Events	Antibiotics Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI
1.3.1 2 to 4 weeks						
Appelman 1991	21	51	25	45	9.3%	0.74 [0.49 , 1.13]
Burke 1991	41	112	41	116	12.2%	1.04 [0.73 , 1.46]
Kaleida 1991	188	401	255	408	30.0%	0.75 [0.66 , 0.85]
Le Saux 2005	68	233	77	222	16.8%	0.84 [0.64 , 1.10]
Mygind 1981	23	72	25	77	7.8%	0.98 [0.62 , 1.57]
Tapiainen 2014	13	42	26	42	6.8%	0.50 [0.30 , 0.83]
Thalin 1985	65	159	65	158	17.2%	0.99 [0.76 , 1.29]
Subtotal (95% CI)		**1070**		**1068**	**100.0%**	**0.83 [0.72 , 0.96]**
Total events:	419		514			
Heterogeneity: Tau² = 0.01; Chi² = 10.01, df = 6 (P = 0.12); I² = 40%						
Test for overall effect: Z = 2.52 (P = 0.01)						
1.3.2 6 to 8 weeks						
Damoiseaux 2000	69	107	70	105	46.0%	0.97 [0.80 , 1.18]
Kaleida 1991	151	329	169	328	51.4%	0.89 [0.76 , 1.04]
Tapiainen 2014	2	42	10	42	2.6%	0.20 [0.05 , 0.86]
Subtotal (95% CI)		**478**		**475**	**100.0%**	**0.89 [0.70 , 1.13]**
Total events:	222		249			
Heterogeneity: Tau² = 0.02; Chi² = 4.81, df = 2 (P = 0.09); I² = 58%						
Test for overall effect: Z = 0.95 (P = 0.34)						
1.3.3 3 months						
Burke 1991	20	111	31	111	30.1%	0.65 [0.39 , 1.06]
Le Saux 2005	58	228	47	210	44.6%	1.14 [0.81 , 1.59]
Mygind 1981	18	72	18	77	25.3%	1.07 [0.61 , 1.89]
Subtotal (95% CI)		**411**		**398**	**100.0%**	**0.94 [0.66 , 1.34]**
Total events:	96		96			
Heterogeneity: Tau² = 0.04; Chi² = 3.56, df = 2 (P = 0.17); I² = 44%						
Test for overall effect: Z = 0.32 (P = 0.75)						



Analysis 1.4. Comparison 1: Antibiotics versus placebo, Outcome 4: Tympanic membrane perforation

Study or Subgroup	Antibiotics Events	Antibiotics Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI
Tapiainen 2014	0	42	0	42		Not estimable
Hoberman 2011	1	144	7	147	12.4%	0.15 [0.02 , 1.17]
Tähtinen 2011	1	161	5	158	11.8%	0.20 [0.02 , 1.66]
Burke 1991	0	114	2	118	5.9%	0.21 [0.01 , 4.26]
Mygind 1981	7	72	12	77	70.0%	0.62 [0.26 , 1.50]
Total (95% CI)		**533**		**542**	**100.0%**	**0.43 [0.21 , 0.89]**
Total events:	9		26			
Heterogeneity: Tau² = 0.00; Chi² = 2.59, df = 3 (P = 0.46); I² = 0%						
Test for overall effect: Z = 2.28 (P = 0.02)						
Test for subgroup differences: Not applicable						

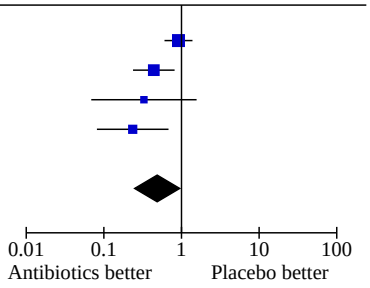


Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Analysis 1.5. Comparison 1: Antibiotics versus placebo, Outcome 5: Contralateral otitis (in unilateral cases)

Study or Subgroup	Antibiotics Events	Antibiotics Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI
Burke 1991	29	98	33	102	36.1%	0.91 [0.60 , 1.38]
Hoberman 2011	13	161	29	158	30.9%	0.44 [0.24 , 0.81]
Mygind 1981	2	64	6	63	12.7%	0.33 [0.07 , 1.56]
Thalin 1985	4	130	17	130	20.3%	0.24 [0.08 , 0.68]
Total (95% CI)		**453**		**453**	**100.0%**	**0.49 [0.25 , 0.95]**
Total events:	48		85			

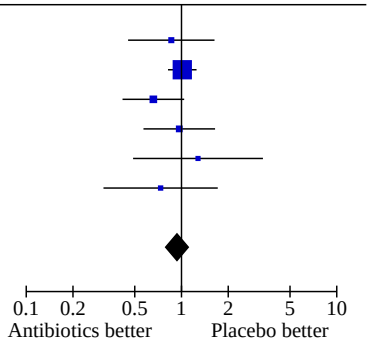
Heterogeneity: Tau² = 0.28; Chi² = 8.79, df = 3 (P = 0.03); I² = 66%
Test for overall effect: Z = 2.12 (P = 0.03)
Test for subgroup differences: Not applicable



Analysis 1.6. Comparison 1: Antibiotics versus placebo, Outcome 6: Late AOM recurrences

Study or Subgroup	Antibiotics Events	Antibiotics Total	Placebo Events	Placebo Total	Weight	Risk Ratio M-H, Random, 95% CI
Hoberman 2011	19	119	13	70	6.8%	0.86 [0.45 , 1.63]
Kaleida 1991	125	448	123	446	62.8%	1.01 [0.82 , 1.25]
Le Saux 2005	27	248	39	236	13.4%	0.66 [0.42 , 1.04]
Mygind 1981	19	72	21	77	10.0%	0.97 [0.57 , 1.65]
Thalin 1985	9	159	7	158	3.0%	1.28 [0.49 , 3.35]
van Buchem 1981	9	92	10	75	3.9%	0.73 [0.31 , 1.71]
Total (95% CI)		**1138**		**1062**	**100.0%**	**0.94 [0.79 , 1.11]**
Total events:	208		213			

Heterogeneity: Tau² = 0.00; Chi² = 3.60, df = 5 (P = 0.61); I² = 0%
Test for overall effect: Z = 0.78 (P = 0.43)
Test for subgroup differences: Not applicable



Comparison 2. Immediate antibiotics versus expectant observation

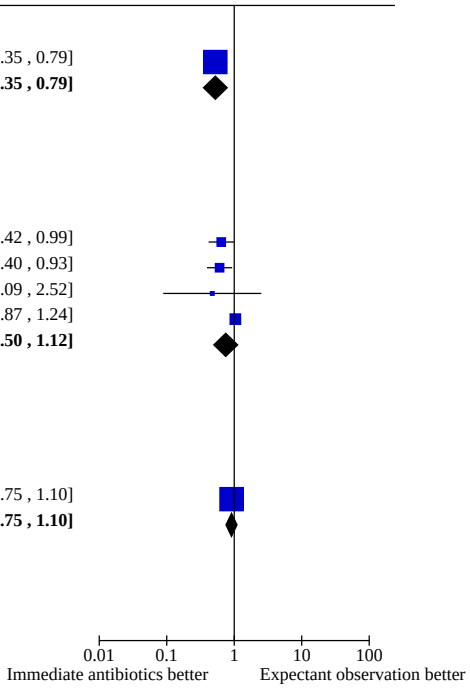
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Pain	5		Risk Ratio (M-H, Random, 95% CI)	Subtotals only
2.1.1 Pain at 2 to 3 days	1	396	Risk Ratio (M-H, Random, 95% CI)	0.53 [0.35, 0.79]
2.1.2 Pain at 3 to 7 days	4	959	Risk Ratio (M-H, Random, 95% CI)	0.75 [0.50, 1.12]
2.1.3 Pain at 11 to 14 days	1	247	Risk Ratio (M-H, Random, 95% CI)	0.91 [0.75, 1.10]
2.2 Vomiting, diarrhoea or rash	3	946	Risk Ratio (M-H, Random, 95% CI)	1.87 [1.39, 2.51]
2.3 Abnormal tympanometry at 4 weeks	1	207	Risk Ratio (M-H, Random, 95% CI)	1.03 [0.78, 1.35]
2.4 Tympanic membrane perforation	2	575	Risk Ratio (M-H, Random, 95% CI)	2.21 [0.20, 24.21]

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.5 AOM recurrences	2	584	Risk Ratio (M-H, Random, 95% CI)	1.04 [0.77, 1.41]

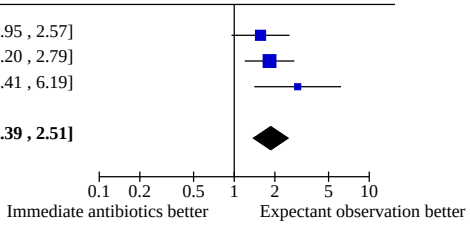
Analysis 2.1. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 1: Pain

Study or Subgroup	Antibiotics Events	Antibiotics Total	Expectant observation Events	Expectant observation Total	Weight	Risk Ratio M-H, Random, 95% CI
2.1.1 Pain at 2 to 3 days						
Shahbaznejad 2021	28	188	59	208	100.0%	0.53 [0.35 , 0.79]
Subtotal (95% CI)		**188**		**208**	**100.0%**	**0.53 [0.35 , 0.79]**
Total events:	28		59			
Heterogeneity: Not applicable						
Test for overall effect: Z = 3.12 (P = 0.002)						
2.1.2 Pain at 3 to 7 days						
Little 2001	26	151	44	164	28.2%	0.64 [0.42 , 0.99]
McCormick 2005	24	102	38	98	28.3%	0.61 [0.40 , 0.93]
Neumark 2007	2	92	4	87	5.2%	0.47 [0.09 , 2.52]
Spiro 2006	89	133	85	132	38.4%	1.04 [0.87 , 1.24]
Subtotal (95% CI)		**478**		**481**	**100.0%**	**0.75 [0.50 , 1.12]**
Total events:	141		171			
Heterogeneity: Tau² = 0.10; Chi² = 10.28, df = 3 (P = 0.02); I² = 71%						
Test for overall effect: Z = 1.40 (P = 0.16)						
2.1.3 Pain at 11 to 14 days						
Spiro 2006	75	123	83	124	100.0%	0.91 [0.75 , 1.10]
Subtotal (95% CI)		**123**		**124**	**100.0%**	**0.91 [0.75 , 1.10]**
Total events:	75		83			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.97 (P = 0.33)						
Test for subgroup differences: Chi² = 0.00, df = 2 (P < 0.00001), I² = 0%						



Analysis 2.2. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 2: Vomiting, diarrhoea or rash

Study or Subgroup	Antibiotics Events	Antibiotics Total	Expectant observation Events	Expectant observation Total	Weight	Risk Ratio M-H, Random, 95% CI
Little 2001	31	135	22	150	35.6%	1.57 [0.95 , 2.57]
Spiro 2006	46	133	25	132	48.5%	1.83 [1.20 , 2.79]
Shahbaznejad 2021	24	188	9	208	15.9%	2.95 [1.41 , 6.19]
Total (95% CI)		**456**		**490**	**100.0%**	**1.87 [1.39 , 2.51]**
Total events:	101		56			
Heterogeneity: Tau² = 0.00; Chi² = 1.98, df = 2 (P = 0.37); I² = 0%						
Test for overall effect: Z = 4.14 (P < 0.0001)						
Test for subgroup differences: Not applicable						

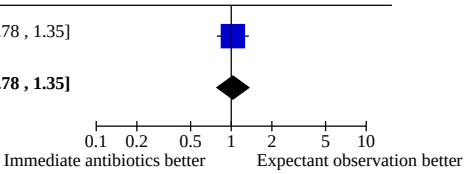


Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Analysis 2.3. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 3: Abnormal tympanometry at 4 weeks

Study or Subgroup	Antibiotics Events	Antibiotics Total	Expectant observation Events	Expectant observation Total	Weight	Risk Ratio M-H, Random, 95% CI
McCormick 2005	55	108	49	99	100.0%	1.03 [0.78 , 1.35]
Total (95% CI)		**108**		**99**	**100.0%**	**1.03 [0.78 , 1.35]**
Total events:	55		49			

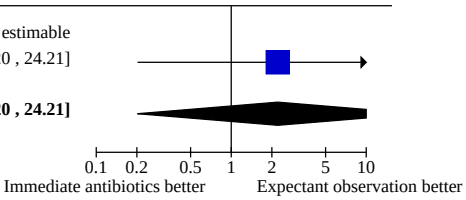
Heterogeneity: Not applicable
Test for overall effect: Z = 0.21 (P = 0.84)
Test for subgroup differences: Not applicable



Analysis 2.4. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 4: Tympanic membrane perforation

Study or Subgroup	Antibiotics Events	Antibiotics Total	Expectant observation Events	Expectant observation Total	Weight	Risk Ratio M-H, Random, 95% CI
Neumark 2007	0	92	0	87		Not estimable
Shahbaznejad 2021	2	188	1	208	100.0%	2.21 [0.20 , 24.21]
Total (95% CI)		**280**		**295**	**100.0%**	**2.21 [0.20 , 24.21]**
Total events:	2		1			

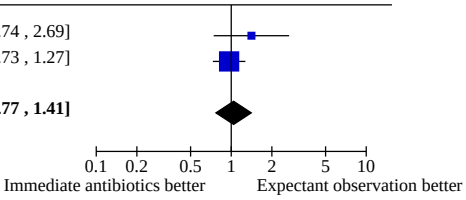
Heterogeneity: Not applicable
Test for overall effect: Z = 0.65 (P = 0.52)
Test for subgroup differences: Not applicable



Analysis 2.5. Comparison 2: Immediate antibiotics versus expectant observation, Outcome 5: AOM recurrences

Study or Subgroup	Antibiotics Events	Antibiotics Total	Expectant observation Events	Expectant observation Total	Weight	Risk Ratio M-H, Random, 95% CI
McCormick 2005	20	109	13	100	20.1%	1.41 [0.74 , 2.69]
Shahbaznejad 2021	61	178	70	197	79.9%	0.96 [0.73 , 1.27]
Total (95% CI)		**287**		**297**	**100.0%**	**1.04 [0.77 , 1.41]**
Total events:	81		83			

Heterogeneity: Tau² = 0.01; Chi² = 1.15, df = 1 (P = 0.28); I² = 13%
Test for overall effect: Z = 0.26 (P = 0.79)
Test for subgroup differences: Not applicable



APPENDICES

Appendix 1. Previous search

Several electronic databases were used to compile relevant published RCTs of antibiotic treatment for AOM in children. The Cochrane Controlled Trials Register, MEDLINE and Current Contents were searched from 1966 to January 2000 by an expert librarian in conjunction with one researcher, using combinations of "OTITIS MEDIA" and a search strategy described by (Dickersin 1994) for optimally identifying controlled trials. In addition, titles in Index Medicus were checked from 1958 to 1965. The references of all relevant retrieved trials were checked to identify other articles.

The search was updated in March 2003, and again in July 2008. We searched the Cochrane Central Register of Controlled Trials (CENTRAL) (*The Cochrane Library* 2008, Issue 2), which contains the ARI Group's Specialized Register; MEDLINE (1966 to June week 4 2008); OLDMEDLINE (1958 to 1965); EMBASE (January 1990 to July 2008); and Current Contents (1966 to July 2008). The bibliographies of relevant articles were checked. A forward search of relevant articles was conducted in Web of Science®.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

The following search strategy was run on MEDLINE (Ovid) combined with terms from Phase 1 and 2 of the Cochrane highly sensitive search strategy for identifying reports of RCTs (Lefebvre 2021). Modified terms were used to search the other databases:

MEDLINE (Ovid)

#1 exp Otitis Media/
#2 exp Otitis Media with Effusion/
#3 exp Otitis Media, Suppurative/
#4 glue ear.mp.
#5 otitis media.mp.
#6 OME.mp.
#7 AOM.mp.
#8 #1 or #2 or #3 or #4 or #5 or #6 or #7
#9 exp Anti-Bacterial Agents/
#10 exp Drug Therapy/
#11 exp Anti-Infective Agents/
#12 antibiotic$.mp.
#13 #9 or #10 or #11 or #12
#14 #8 and #13

There were no language or publication restrictions.

Appendix 2. MEDLINE (Ovid) search strategy

1 exp Otitis Media/
2 otitis media.tw.
3 glue ear*.tw.
4 (middle ear adj5 (infect* or inflam*)).tw.
5 (ome or aom).tw.
6 or/1-5
7 exp Anti-Bacterial Agents/
8 Drug Therapy/
9 Anti-Infective Agents/
10 antibiotic*.tw.
11 antibacterial*.tw.
12 exp Ampicillin/
13 exp Cephalosporins/
14 exp Macrolides/
15 exp Penicillins/
16 (ampicillin* or cephalosporin* or macrolide* or penicillin* or amoxicillin* or amoxycillin* or cefdinir or cefpodoxime or cefuroxime or azithromycin or clarithromycin or erythromycin*).tw,nm.
17 or/7-16
18 6 and 17
19 (Controlled Clinical Trial or Randomized Controlled Trial).pt.
20 (placebo or randomly or trial or groups).ti,ab.
21 drug therapy.fs.
22 randomized.ab.
23 19 or 20 or 21 or 22
24 exp animals/ not humans/
25 23 not 24

Appendix 3. Embase.com search strategy

18 #14 AND #17
17 #15 OR #16
16 random*:ab,ti OR placebo*:ab,ti OR factorial*:ab,ti OR crossover*:ab,ti OR 'cross-over':ab,ti OR 'cross over':ab,ti OR volunteer*:ab,ti OR assign*:ab,ti OR allocat*:ab,ti
15 'randomized controlled trial'/exp OR 'single blind procedure'/exp OR 'double blind procedure'/exp OR 'crossover procedure'/exp
14 #4 AND #13
13 #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12
12 ampicillin*:ab,ti OR cephalosporin*:ab,ti OR macrolide*:ab,ti OR penicillin*:ab,ti OR amoxycillin*:ab,ti OR amoxicillin*:ab,ti OR cefdinir*:ab,ti OR cefpodoxime*:ab,ti OR cefuroxime*:ab,ti OR
azithromycin*:ab,ti OR clarithromycin*:ab,ti OR erythromycin*:ab,ti
11 'penicillin g'/exp

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

10 'macrolide'/exp
9 'cephalosporin derivative'/exp
8 'ampicillin'/exp
7 antibiotic*:ab,ti OR antibacterial*:ab,ti
6 'drug therapy'/de OR 'antiinfective agent'/de
5 'antibiotic agent'/exp
4 #1 OR #2 OR #3
3 ('middle ear' NEAR/5 (inflam* OR infect*)):ab,ti
2 'otitis media':ab,ti OR 'glue ear':ab,ti OR 'glue ears':ab,ti OR ome:ab,ti OR aom:ab,ti
1 'otitis media'/exp

Appendix 4. Current Contents search strategy

# 3	578
# 2	528,401
# 1	2,624

Appendix 5. CINAHL search strategy

S30 S19 and S29
S29 S20 or S21 or S22 or S23 or S24 or S25 or S26 or S27 or S28
S28 (MH "Quantitative Studies")
S27 TI placebo* or AB placebo*
S26 (MH "Placebos")
S25 TI random* or AB random*
S24 (MH "Random Assignment")
S23 TI (singl* blind* or doubl* blind* or tripl* blind* or trebl* blind* or singl* mask* or doubl* mask* or tripl* mask* or trebl* mask*) or AB (singl* blind* or doubl* blind* or tripl* blind* or trebl* blind* or singl* mask* or doubl* mask* or tripl* mask* or trebl* mask*)
S22 TI clinic* N1 trial* or AB clinic* N1 trial*
S21 PT clinical trial
S20 (MH "Clinical Trials+")
S19 S7 and S18
S18 S8 or S9 or S10 or S11 or S12 or S13 or S14 or S15 or S16 or S17
S17 TI (ampicillin* or cephalosporin* or macrolide* or amoxicillin* or amoxycillin* or penicillin* or cefdinir* or cefpodoxime* or cefuroxime* or azithromycin* or clarithromycin* or erythromycin*) or AB (ampicillin* or cephalosporin* or macrolide* or amoxicillin* or amoxycillin* or penicillin* or cefdinir* or cefpodoxime* or cefuroxime* or azithromycin* or clarithromycin* or erythromycin*)
S16 (MH "Penicillins+")
S15 (MH "Antibiotics, Macrolide+")
S14 (MH "Cephalosporins+")
S13 (MH "Ampicillin+")
S12 TI antibacterial* or AB antibacterial*
S11 TI antibiotic* or AB antibiotic*
S10 (MH "Antiinfective Agents")
S9 (MH "Drug Therapy")
S8 (MH "Antibiotics+")
S7 S1 or S2 or S3 or S4 or S5 or S6
S6 TI (aom or ome) or AB (aom or ome)
S5 TI middle ear inflam* or AB middle ear inflam*
S4 TI middle ear infect* or AB middle ear infect*
S3 AB glue ear* or TI glue ear*
S2 TI otitis media or AB otitis media
S1 (MH "Otitis Media+")

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Appendix 6. LILACS search strategy

> Search > (MH:"otitis media" OR "otitis media" OR "Otite Média" OR MH:C09.218.705.633$) AND (MH:"Anti-Bacterial Agents" OR antibiotic$ OR Antibacterianos OR Antibióticos OR MH:"Drug Therapy" OR Quimioterapia OR "Terapia por Drogas" OR Farmacoterapia OR MH:"Anti-Infective Agents" OR Antiinfecciosos OR MH:ampicillin OR Ampicilina OR ampicillin$ OR MH:D02.065.589.099.750.750.050$ OR MH:D02.886.108.750.750.050$ OR MH:D03.438.460.825.750.050$ OR MH:D03.605.084.737.750.050$ OR D04.075.080.875.099.221.750.750.050$ OR MH:cephalosporins OR cephalosporin$ OR Cefalosporinas OR MH:D02.065.589.099.249$ OR D02.886.665.074$ OR D04.075.080.875.099.221.249$ OR MH:macrolides OR macrolide$ OR Macrólidos OR Macrolídeos OR D02.540.505$ OR D02.540.576.500$ OR D04.345.674.500$ OR MH:penicillins OR penicillin$ OR Penicilinas OR MH:D02.065.589.099.750$ OR D02.886.108.750$ OR D03.438.260.825$ OR D03.605.084.737$ OR D04.075.080.875.099.221.750$ OR amoxicillin$ OR Amoxicilina OR cefdinir OR cefpodoxim$ OR cefuroxim$ OR azithromycin$ OR Azitromicina OR clarithromycin$ OR Claritromicina OR erythromycin OR Eritromicina) > clinical_trials

F E E D B A C K

Antibiotics for AOM, 22 November 2000

Summary

1. Types of interventions includes surgical procedures versus placebo which are not dealt with in this review and should therefore be deleted.

2. The authors included only six studies in the analysis but in 1994 another meta-analysis by Rosenfeld and colleagues to which the authors refer was published which included 33 randomized trials with 5400 children. Were any studies with a no-treatment control excluded and if so why?

3. The meta-analysis by Rosenfeld is only mentioned in the text; there is no reference to it. How many patients were included in the meta-analysis?

4. It is stated that trials analysed on an intention to treat basis were preferred. This indicates that other trials were excluded which does not seem reasonable?

5. The description of the factorial trial is unclear; I suppose the authors excluded all patients who were randomised to myringotomy?

6. In the trial by Laxdal the control group was more closely monitored. The trial therefore violates the principle that all other Traitement etc. should be the same in the two randomised groups and it should therefore be excluded.

7. The strategy described by Dickersin lacks a publication year and it is not cited in the references.

8. The search was done in August 1994 and the Cochrane review was published in April 1997. The search should therefore have been updated before publication since Cochrane reviews are meant to be up-to-date.

9. There is no information whether the original authors and the pharmaceutical industry were contacted about additional data including unpublished trials and trials not registered in Medline. Useful trial data might be expected to be available in books published in connection with symposia arranged by the drug industry for example.

10. What is quality methodology?

11. The term blinded randomisation should be avoided since it may be confused with blinded treatments; the term concealed allocation should be used.

12. The elaborated quality assessment scale for the trials does not appear under Results and should therefore be deleted.

13. The authors refer to Rosenfeld's meta-analysis when they state that 80% of the children have recovered spontaneously after 24 hours. Since such a percentage refers to untreated patients it raises the question why the authors did not use their own data? If these data are used in a meta-analysis of the risk difference the NNTB will be 23 not 12 as stated in the Cochrane review.

14. For several of the excluded studies the authors gave no reason for the exclusion.

15. There should be a cross-reference to the authors' nearly identical review in the BMJ (24 May 1997).

Reply

The changes made were:

1. We updated the search. (see Johansen criticism 7 & 8). No recent trials were found but we recognised that the Appelman trial qualifies (originally we had thought this was only prevention of recurrent otitis, rather than treatment of acute otitis in children with a recurrent episode).

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

2. We have corrected and updated the Relative Risk Reduction and consequent Number-Needed-to-Treat (see Johansen criticism 13).

3. We have separate the four arms of the Van Buchem factorial trial, and treated this as "two" trials (i.e., two separate strata): (a) without myringotomy - antibiotics versus placebo (b) with myringotomy - antibiotics versus placebo. (see Johansen criticism 5)

4. As suggested by Andrew Herxheimer, we have added several references including (a) Chris Cates BMJ, and (b) Kozrskyj's meta-analysis of short versus long duration of antibiotics (rather than just the de Saintonge paper).

5. We have made small text changes in response to Johansen's criticisms 5 (description added), 7 (dropped), 10 (- methodological quality), 11 (- allocation concealment), 13 (corrected in text), 14 (exclusions explained), and 15 (reference added).

6. As we have pointed out to Johansen in the BMJ correspondence, and point out in the discussion here, the Rosenfeld meta-analysis is largely concerned with comparison between antibiotics. (see Johansen criticism 2 & 3).

Contributors

Helle Krogh Johansen
Peter C. Gøtzsche

Antibiotic versus placebo for acute otitis media, 22 November 2010

Summary

This excellent and important review was completed in 1996, and I hope it will soon be updated. It is especially worth noting and discussing the new study by Christopher Cates (BMJ 13 March 1999, p715-6), who has successfully tried a method in his general practice of substantially reducing the use of antibiotic in children with acute otitis media. This would considerably strengthen the 'implications for practice' in the conclusion.

I would like to suggest that in updating this review the objectives be amended and the trial by Chaput de Saintonge et al be added, because it contributes an important piece of evidence about the duration of amoxicillin therapy. The review concludes that some children will benefit from antibiotic treatment, and it would be valuable to say (as a result of the Chaput trial) that the evidence indicates that a 3-day course is no less effective than a 10-day course.

Reply

Chris and I have revised the acute otitis media review. We have made a number of modest changes, though none of these change the conclusions. However, because a new trial is included we've called it a "substantive update".

Contributors

Andrew Herxheimer

Antibiotic versus placebo for acute otitis media, 22 November 2000

Summary

1. I am glad to see this has been updated but the text does not explain what was updated, forcing the reader who wants to know to compare the previous version with the new one. Is it the sentence referring to Cates 99 [in implics for practice] or other points as well?

2. There are embarrassingly many typos in the refs to excluded and additional studies: Chaput de SaintoNGE, amoxyciillin, author not in bold in the first few additional refs, below that several authors' names begin in lower case when they should all begin with a capital.

3. It is implied that no comcrit was received before the final submission date for CL99 issue 3. Is this true? I think I sent one early this year.

CONFLICT OF INTEREST: None.

Reply

Excluded and additional references have been corrected and completed.

Contributors

Andrew Herxheimer

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Antibiotic versus placebo for acute otitis media, 22 June 2000

Summary

1. The new study also reported diarrhoea and rashes. Shouldn't it be included in this outcome (side effects) also?
2. I think the methods used for calculating the NNTB should be made explicit.
3. The new trial is important because it looks at a sub-group who were believed to be a greater risk of poor outcomes. In EBM OM Rosenfeld and Bluestone review the study inclusion criteria and state that the meta-analysis 'most likely can be applied to children 2 years of age or older with non severe AOM, and most likely cannot be applied to infants with severe symptoms'. This study provides the best evidence that the conclusions of the meta-analysis do appear to apply to this group. Perhaps this point needs to be emphasised (the peak incidence of AOM is 9 months).
4. I think the comment that 80% resolve spontaneously within 2 to 7 days is now slightly misleading as about 70% of the control children were clinical failures in this new study.
5. The entry in the table 'characteristics of included studies' should be consistent with previous entries.
6. Some typographical errors and inconsistent spelling.

Reply

Thank you for your comments and suggestions.
The Absolute risk difference was used to calculate the NNTB in this systematic review. This has now been stated in the Results section of the review. A comment regarding the application of the conclusions to infants with severe symptoms has been added to the discussion section. The 70% incidence of clinical failure in the Damoiseaux, 2000 study have been included and typographical errors and inconsistencies have been corrected.

Contributors

Peter Morris

Antibiotics for acute otitis media, 19 February 2002

Summary

The second graph (comparison of outcome Abnormal Tympanometry) has wrong labels on the X-axis.

It says 'antibiotics better' (left) and 'placebo worse' (right). The second should probably be 'placebo better'.

The other graphs are correctly labelled.

I certify that I have no affiliations with or involvement in any organisation or entity with a direct financial interest in the subject matter of my criticisms.

Reply

The label on the x-axis has been corrected and now reads 'Placebo better'.

Contributors

Johannes C van der Wouden

Antibiotics reduce the risk of mastoiditis?, 26 August 2002

Summary

I agree with other commentators that this is a very good and important review. However, I would like some more clarity concerning one statement in your conclusions: Antibiotic treatment may play an important role in reducing the risk of mastoiditis in populations where it is more common.

What is the basis for this statement? In the included studies with more than 2000 children only one mastoiditis case occurred in a patient in a penicillin treated group. In the review you mention two articles concerning the mastoiditis. Firstly, the study of Rudberg (1954), which was excluded since it was not properly randomised. Even if it were, the rate of 17 % of mastoiditis cases is in these times highly unlikely, as is shown in the included studies. The second article by Berman (1995) is a literature review, where only the available literature concerning developing countries were reviewed. The goal of this review was to determine the extent to which otitis media impacts mortality and morbidity in developing countries, not to study the effect of antibiotics on (acute) otitis media or mastoiditis.

In neither of these studies evidence is shown that antibiotic treatment reduces the risk of mastoiditis, certainly not in developed countries. Since I think the rest of the review is excellent, I wonder if you could explain to me the reasons for including this statement in the conclusions.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

I certify that I have no affiliations with or involvement in any organisation or entity with a direct financial interest in the subject matter of my criticisms.

Reply

Dear Markus,

We included the caveat about mastoiditis because we, and the reviewers, were concerned about misinterpretation of the results in situations with high rates of mastoiditis. We were mindful that "an absence of evidence is not equal to evidence of absence". Since the trials we analysed did not include high rates of mastoiditis, we can use them as the sole basis. Given that we have two weaker pieces of evidence:

1. The trials do show a modest reduction in other infective complications

2. The excluded Rudberg trial did show dramatic effects that we don't think explicable from the potential biases of that study.

Prudence would then suggest that antibiotics are advisable if there is a substantial risk of mastoiditis,

Regards,

Paul Glasziou

I certify that I have no affiliations with or involvement in any organisation or entity with a direct financial interest in the subject matter of my criticisms.

Contributors

Markus Oei (ENT surgeon)

Incorrect NNTB, 19 June 2005

Summary

I am a bit troubled by the way the conclusions of this review are written. By combining results of treatment at Days 2 to 7 in arriving at a NNTB of 15 one is going to underestimate treatment benefit after 2 days. In your abstract though you say the ARR is 7% and NNTB 15 for some pain after two days. This is simply not correct. If one carefully looks at trials that record pain at the end of day 2 the ARR is in fact 20% giving a NNTB of 5. Clearly acute otitis media is an acute condition and the main benefit of antibiotics is pain control and symptom relief. If this is measured at the end of 2 days the benefits are greater than one would surmise just from reading the review. It would be absurd to do a review of pain relief for biliary colic treated with pethidine and measuring the outcome 7 days later. For acute conditions symptom control in the first few days should be the outcome of interest. NNTB are meaningless unless giving a time period at which they apply. I think the review needs correcting. This is not just of academic interest but of direct relevance to parents and doctors faced with a child with AOM in pain. Unfortunately your review gets quoted uncritically and invariably the NNTB of 15 is given for symptom control after 2 days. I am currently trying to correct a brochure produced here in New Zealand for GPs to give to parents of children with AOM and it uncritically repeats this misleading information. If you want to comment on symptom control after Day 2 DO NOT pool it with data from Day 7 or later!

I certify that I have no affiliations with or involvement in any organisation or entity with a direct financial interest in the subject matter of my criticisms.

Reply

Thank you for your comment. We agree that we should be clearer about the time frame to which the ARR 7% and NNTB 15 applies. With the availability of results of the individual patient data meta-analysis (Rovers 2006) we are able to obtain a clearer indication of the recovery pattern over time. We have reported this in the text and included an extra figure.

Contributors

Paul Corwin

Comment on two of the meta-analyses, 9 June 2007

Summary

Summary
Feedback: This is a comment on two of the meta-analyses in the Cochrane Review, Glasziou et al. (2004). These analyses are for the outcomes "Vomiting, Diarrhea or Rash" and "Contralateral AOM."

1) Vomiting, Diarrhea or Rash

First we consider the meta-analysis relating possible adverse effects of treatment. In Glasziou et al. (2004), this is done using the composite outcome "Vomiting, Diarrhea or Rash." The data used for this meta-analysis are reproduced in the table below.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Outcome: Vomiting, Diarrhea or Rash
Study Treatment Control
Thalin et al. (1985) 1/159 1/158
Burke et al. (1991) 53/114 36/116
Mygind et al. (1981) 3/72 1/77
Damoiseaux et al. (2000) 20 12

We noted five major problems with this meta-analysis. The first relates to clinical heterogeneity. This was manifested in variations in terms of the types of adverse effects recorded, who recorded them (parent or physician) and the time period over which they were recorded (from 3-4 days to 21 days). In Thalin et al. (1985), the effects were recorded by an ENT physician on days 3-4 or days 8-10. In Burke et al. (1991), they were recorded by a parent in a 21-day diary. In Mygind et al. (1981), it was done with 7 day parental score card. And in Damoiseaux et al. (2000), this was done by a physician on day 4 and day 11.

Another related problem is the use of the outcome "Vomiting, Diarrhea or Rash" as an entity. Vomiting is only reported in Burke et al. (1991). It is not clear whether it was not observed, or observed but not reported in the other studies. Also, in Burkeat al. (1991), as noted, such effects were recorded over a 21-day period while the maximum recording period for the other studies was 11 days. The totals then gave a much higher weight to Burke et al. (1991) than may be appropriate.

A third problem is possible double or triple counting with the use of the composite outcome. For Burke et al. (1991), the group numerator is the sum of the cases for each effect. A number of children may well have had two or three of these effects at the same time.

A fourth problem is also with the numbers used. Damoiseaux et al. (2000) gives two sets of numbers for "de novo diarrhoea," for day 4 and for day 11. Glasziou et al. (2004) uses the day 4 numbers only. The reason for this choice is not clear. It may be better to use the sums of the numbers for the two days (provided this does not involve double counting.)

Further, the group denominators used for Burke et al. (1991) are perhaps not what they should be. In this study, the adverse effects were recorded by parents. Only 220 (treatment = 107, control = 113) out of a total of 232 (treatment = 114, control = 118) diaries were collected. Using the total group size in the numerator (also done in Burke at al. (1991)) is thus not appropriate.

Finally, it is not clear if the numbers for adverse effects in Burke et al. (1991) and Damoiseaux et al. (2000) included the cases known or suspected to have dropped out of the study due to an adverse effect.

In our view, this meta-analysis should be modified as follows: First, do not use the data on vomiting until it is reported in at least one other study. Second, do not use a composite adverse effect outcome. Instead, perform separate meta-analyses for diarrhoea and rash. Third, for Damoiseaux et al. (2000), use the total numbers for day 4 and day 11, with the above noted qualification in mind. Fourth, for Burke et al. (1991) change the denominators as noted above. Finally, include drop outs due to side effects in the meta-analyses. The table below gives the possible numerators to be used for these meta-analysis.

Separated Data on Side Effects
Vomiting Diarrhea Rash
Study T C T C T C
Thalin et al. (1985) ? ? 0 0 1 1
Burke et al. (1991)+ 20 14 24 16 16 9
Mygind et al. (1981) ? ? 2 1 1/2? 0
Damoiseaux et al. (2000)*,+ ? ? 20 12 0 3
Damoiseaux et al. (2000)? ? ? 34 22 0 3
Note: ? Unclear if vomiting not observed or not reported.
Note: ? = 2 if a dropout was not counted; else = 1.
* Day 4; ? Day 4 and Day 11; + unclear if dropouts counted.

2) Contralateral AOM

The occurrence of contralateral AOM, as is made clear in Glasziou et al. (2004), is relevant for only the cases with unilateral AOM at the outset. This numbers in the table below are used for the meta-analysis of this outcome in Glasziou et al. (2004).

Outcome: Contralateral AOM
Study Treatment Control
Thalin et al. (1985) 4/159 17/158
Burke et al. (1991) 29/98 33/102
Mygind et al. (1981) 2/72 6/77
Overall 35/329 56/337

The first problem is clinical heterogeneity, as noted in the table below. The issues in that respect are similar to those stated for the meta-analysis of adverse effect.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Clinical Heterogeneity: Contralateral AOM
Study Time Period Evaluator(s)
Thalin et al. (1985) day 8-10 or day 30 ENT Physician
Burke et al. (1991) 21 days Parent
Mygind et al. (1981) 1 week Physician

A further problem with this meta-analysis is the denominators used. Consider this issue for each study.

Thalin et al. (1985): The denominators in Glasziou et al. (2004) include unilateral and bilateral cases. Only 82% of the episodes were unilateral at the start but the breakdown by group is not given in the paper. We obtained adjusted denominators as follows. Treatment: 0.82?159 = 130; Control: 0.82?158 = 130. The bias now remains the same but the precision level is now corrected.

Burke et al. (1991): The denominators represent the total unilateral cases for each group. The study authors used these denominators. Completed 21-day diaries, the source of data on contralateral otitis, were, however, available only for 107 (of 114) in the treatment group and 113 (of 118) in the control group. So either one assumes that only the bilateral cases had missing diaries (which is unlikely) or that the rate of missingness in each group was not affected by laterality. In the latter case, the adjusted denominators are: Treatment: (98?107)/114 = 92; Control: (102?113)/118 = 98. The level of bias remains unknown but the precision level is possibly better.

Mygind et al. (1991): The denominators used include unilateral and bilateral cases. But there were 8 bilateral cases in the placebo group and 14 in the control group. So the appropriate denominators are Treatment: 72 - 8 = 64; Control: 77 - 14 = 65. The bias and precision levels are now corrected.

The appropriately adjusted data for this meta analysis are given below.

Contralateral AOM: Adjusted Data
Study Treatment Control
Thalin et al. (1985) 4/130 17/130
Burke et al. (1991) 29/92 33/98
Mygind et al. (1981) 2/64 6/65
Overall 35/286 56/294

References

1. Burke P, Bain J, Robinson D and Dunleavey J (1991) Acute red ear in children: Controlled trial of non-antibiotic treatment in general practice, British Medical Journal, 303, 558?562.

2. Damoiseaux RAMJ, van Balen FAM, Hoes AW, Verheij TJM and de Melker RA (2000) Primary care based randomised, double blind trial of amoxicillin versus placebo for acute otitis media in children aged under 2 years, British Medical Journal, 320: 330?334.

3. Glasziou PP, Del Mar CB, HayemMand Sanders SL (2004) Antibiotics for acute otitis media in children, Cochrane Database of Systematic Reviews, 2004; (1): CD000219. Art. No: CD000219, DOI: 10.1002/14651858.CD000219.pub2 (21pages)

4. Mygind N, Meistrup-Larsen K-I, Thomsen J, Thomsen VF, Josefsson K and Sorenson H (1981) Penicillin in acute otitis media: a double-blind placebo-controlled trial, Clinical Otolaryngology, 6: 5?13.

5. Thalin A, Densert O, Larsson A, Lyden E and Ripa T (1986) Is penicillin necessary in the treatment of acute otitis media? In: Proceedings of the International Conference on Acute and Secretory Otitis Media, Amsterdam, The Netherlands, Kegler Publications, pages 441?446.

Submitter agrees with default conflict of interest statement:
I certify that I have no affiliations with or involvement in any organization or entity with a financial interest in the subject matter of my feedback.

Reply

1) We acknowledge the variation in methods of collecting and recording information on adverse events and in the types of adverse events reported in the included trials. We contend however, that considering vomiting, diarrhoea or rash as an entity is justified by the easier interpretation it provides. Though the events are biologically very different, they are of similar seriousness; irritating and difficult to manage but minor in nature. Also, as pointed out in the above comments, dividing the adverse events into each type would not be helpful as they are infrequently reported (i.e. vomiting is only reported in one study). We recognise that 'lumping' the adverse events together is a crude approach but believe the benefits in continuing to do so outweigh the drawbacks. In the discussion section of this update we have made reference to the results of the individual patient data meta analysis (Rovers 2006) (which included a subset [n = 6] of the trials included in this review [n = 10]) which reports separately on the frequency of diarrhoea and rash in the treatment and control groups. We appreciate your consideration and suggestions related to the inclusion of drop outs due to side effects in the Burke and Damoiseaux studies. Corrections to the data have been incorporated.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

2) Thank you for pointing out the numerical errors in the meta analysis of contralateral AOM. We have corrected the analysis as suggested. This results in a minor changed to the pooled random effects OR (OR 0.44 95% CI 0.16, 1.26 versus 0.45 95% CI 0.16, 1.23) with antibiotics appearing to reduce contralateral AOM though the effect was not significant with the random effects model.

Contributors

Karim F. Hirji, D.Sc
Peter C. Gøtzsche

Antibiotics for acute otitis media in children, 8 March 2011

Summary

The title and conclusion of the review need revising as it is just reviewing the effect of penicillin family antibiotic on the AOM and not other antibiotics. It is suggesting to changed the title to "Usage of penicillin family Antibiotics for acute otitis media in children".

Warm regards.

PS: The only included trials were too old and they just used the publish data:

Halsted 1968 ampicillin 100 mg/kg/day or phenethicillin 30 mg/kg/day plus sulphisoxazole 150 mg/kg/day
Howie 1973 one of erythromycin, ampicillin, or triple sulphonamide plus erythromycin

Submitter agrees with default conflict of interest statement: I certify that I have no affiliations with or involvement in any organization or entity with a financial interest in the subject matter of my feedback.

Reply

The title is our intention. However, as you point out, it just so happens that most (but not all) antibiotics trialled against placebo for acute otitis media were from the penicillin group. Moreover more trials might be undertaken using non-penicillin antibiotics. So it is appropriate to retain the original title.

Chris Del Mar, 19 June, 2012

Contributors

Amirkambiz Hamedanizadeh, Medical Doctor

WHAT'S NEW

Date	Event	Description
6 December 2023	Amended	Plain language summary edited to remove one redundant sentence.

HISTORY

Protocol first published: Issue 1, 1995
Review first published: Issue 3, 1996

Date	Event	Description
15 November 2023	New search has been performed	We updated the searches in February 2023. One new trial was eligible for the review of immediate antibiotics against expectant observation (Shahbaznejad 2021). This study included children aged six months to six years and provided data on pain at 24 to 72 hours, adverse events, tympanic membrane perforation, AOM recurrences at one month and serious complications related to AOM. We excluded one new trial (Hay 2021).

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Date	Event	Description
		No new trials were identified for the review of antibiotics against placebo. Furthermore, we did not identify any further ongoing trials apart from the one identified previously (the WATCH trial). For each outcome, we have now rated the certainty of evidence using the GRADE approach.
15 November 2023	New citation required but conclusions have not changed	The overall conclusions of the review have not changed.
26 April 2015	New citation required but conclusions have not changed	The conclusions regarding the effectiveness and safety of antibiotics have essentially not changed, except for some new outcomes (e.g. long-term effects on AOM recurrences) and minor changes to the risk of bias.
26 April 2015	New search has been performed	We updated the searches in April 2015. In this updated review, we now provide outcome data on: • pain at 24 hours, two to three days, four to seven days and 10 to 14 days (in earlier versions outcome data on pain were presented at 24 hours, two to three days and four to seven days); • abnormal tympanometry findings at two to four weeks, six to eight weeks and three months (in earlier versions outcome data on abnormal tympanometry findings were presented at four to six weeks and three months); • long-term effects including number of parent-reported AOM symptom episodes, antibiotic prescriptions, health care utilisation as assessed at least one year after randomisation (in earlier versions no data on long-term effects were presented). The outcome 'Adverse effects likely to be related to the use of antibiotics such as vomiting, diarrhoea or rash' has been added to primary outcomes (in earlier versions this outcome was listed as a secondary outcome) according to the recommendations described in Chapter 5.4.2 of the *Cochrane Handbook for Systematic Reviews of Interventions* ("the primary outcomes should include at least one desirable and at least one undesirable outcome") (Higgins 2011). One new trial was identified for the review of antibiotics against placebo (Tapiainen 2014). This study included children aged between six months and 15 years and provided data on pain at days four to seven, adverse effects likely to be related to the use of antibiotics, abnormal tympanometry findings at two to four weeks and six to eight weeks, tympanic membrane perforation and serious complications. New data were added to the review from previously included trials. For the review of antibiotics against placebo: • data on pain at 10 to 12 days (Hoberman 2011); • data on abnormal tympanometry findings at two to four weeks (Kaleida 1991); • data on long-term effects (Burke 1991; Damoiseaux 2000). For the review of immediate antibiotics against expectant observation:

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Date	Event	Description
		• data on pain at 11 to 14 days (Spiro 2006); • data on abnormal tympanometry findings at two to four weeks (McCormick 2005); • data on long-term effects (Little 2001). We identified one ongoing trial (the WATCH trial). The objective of this double-blind, placebo-controlled randomised clinical trial is to assess the effectiveness of azithromycin for seven days in aboriginal children with asymptomatic AOM, defined as a bulging tympanic membrane without associated symptoms at the time of diagnosis. The primary outcome is the proportion of children with a bulging tympanic membrane or ear discharge, or withdrawn due to complications or side effects at 14 days. Quality of evidence is now described based on the GRADE framework.
8 November 2012	New search has been performed	The general conclusions and recommendations regarding the effectiveness of antibiotics on pain and adverse events remained unchanged. Antibiotic treatment led to a statistically significant reduction of children with AOM experiencing pain at two to seven days compared with placebo, but since most children (82%) settle spontaneously, about 20 children must be treated to prevent one suffering from ear pain at two to three and four to seven days. (In the previous version the number needed to treat to benefit (NNTB) was 16). However, in this updated review antibiotic treatment appeared to have a statistically significant beneficial effect on the number of tympanic membrane perforations (risk ratio (RR) 0.37, 95% confidence interval (CI) 0.18 to 0.76; NNTB 33) and contralateral acute otitis media (AOM) episodes (RR 0.49, 95% CI 0.25 to 0.95; NNTB 11) compared with placebo. For every 14 children treated with antibiotics one child experienced an adverse event (such as vomiting, diarrhoea or rash) that would not have occurred if antibiotics were withheld. (In the previous version the number needed to treat to harm (NNTH) was 24). Antibiotics are most useful in children under two years of age with bilateral AOM, or with both AOM and otorrhoea. For most other children with mild disease, an expectant observational approach seems justified. We have no data on populations with higher risks of complications.
8 November 2012	New search has been performed	A new review author joined the team to update this review. We updated the searches in November 2012. Two new trials were identified for the review of antibiotics against placebo (Hoberman 2011; Tähtinen 2011). These studies included children < 35 months of age and provided data on pain (Tähtinen 2011), contralateral otitis, late recurrences (Hoberman 2011), perforation and adverse events (Hoberman 2011; Tähtinen 2011). The Laxdal 1970 trial has been removed from the review of antibiotics against placebo and added to the review of immediate antibiotics versus expectant observation.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Date	Event	Description
		No new trials were identified for the review of immediate antibiotics compared with expectant observation. Furthermore, we did not identify ongoing trials. In this updated review, we now provide outcome data for pain at 24 hours, two to three days and four to seven days (in earlier versions outcome data for pain were presented at 24 hours and two to seven days).
19 June 2012	Feedback has been incorporated	Feedback added to review.
2 September 2009	Amended	95% confidence intervals corrected for the outcome pain at two to seven days, and adverse events are stated in the abstract and body of the review.
2 July 2008	New search has been performed	The search was updated in July 2008. Four new trials were identified and included in the review (Le Saux 2005; McCormick 2005; Neumark 2007; Spiro 2006). One of these trials compared antibiotics with placebo (Le Saux 2005). For the outcome pain at 24 hours and two to seven days, inclusion of this trial did not alter the overall conclusions of the primary analysis. The three other new trials compared immediate antibiotics with various observational approaches (McCormick 2005; Neumark 2007; Spiro 2006). One of the new trials compared immediate antibiotics with delayed prescribing (Spiro 2006). The other trials compared immediate antibiotics with 'watchful waiting', in which no prescription was supplied but advice on when to seek treatment was provided (McCormick 2005; Neumark 2007). Outcome data on pain at three to seven days from these trials were analysed with data from another trial of immediate versus delayed prescription (Little 2001). In earlier versions of the review, data from the Little trial had been included in a sensitivity analysis (Little 2001). In this update, data from the four trials comparing immediate versus observational management strategies have been included in the main analysis. Information on subgroups of children who are most likely to benefit from treatment with antibiotics, obtained from a meta-analysis of individual patient data, has been included in this review (Rovers 2006). Methods of the IPD meta-analysis, conducted by two authors on this review (and others) are also included. Survival curves from the IPD meta-analysis showing the pattern of recovery from acute otitis media over time have been included as an extra figure. Two ongoing trials comparing antibiotics with placebo in children < 35 months have been identified.
17 January 2008	Amended	Converted to new review format.
4 September 2007	Feedback has been incorporated	Feedback added.
18 February 2005	Feedback has been incorporated	Feedback and reply added.
24 March 2003	New search has been performed	Searches conducted.
24 August 2002	Feedback has been incorporated	Feedback added.
17 February 2002	Feedback has been incorporated	Feedback added.
20 November 2000	Feedback has been incorporated	Feedback comments and replies added.

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Date	Event	Description
3 February 2000	New search has been performed	Searches conducted.
3 February 2000	New citation required and conclusions have changed	Conclusions changed.
30 December 1998	New search has been performed	Searches conducted.
30 July 1994	New search has been performed	Searches conducted.

CONTRIBUTIONS OF AUTHORS

Roderick P Venekamp reviewed search results, identified studies, extracted data and prepared the manuscript for the updated review in 2023.
Paul P Glasziou, Maroeska M Rovers and Sharon L Sanders have reviewed and provided comments on the 2023 updated version of the review.

DECLARATIONS OF INTEREST

Maroeska M Rovers has participated in workshops and educational activities on otitis media organised by GlaxoSmithKline and received a grant from GlaxoSmithKline for a study on the microbiology of otitis media in 2009.
Roderick P Venekamp is an Editor for Cochrane Acute Respiratory Infections and Cochrane ENT.
Sharon L Sanders declares no conflicts of interests in the current work.
Paul P Glasziou is a co-investigator on a NHMRC-funded grant Antibiotic Resistance.

SOURCES OF SUPPORT

Internal sources

- No sources of support provided

External sources

- No sources of support provided

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

In this 2023 updated review, we updated the methodology according to the recommendations described in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2021). This included the use of random-effects models in all meta-analyses, which slightly changed some of the effect estimates compared to the 2015 version of our review (Venekamp 2015). Overall, the conclusions of the review did not change.

In the 2015 updated review, we provided outcome data on:

- pain at 24 hours, two to three days, four to seven days and 10 to 14 days (in earlier versions outcome data on pain were presented at 24 hours, two to three days and four to seven days);
- abnormal tympanometry findings at two to four weeks, six to eight weeks and three months (in earlier versions outcome data on abnormal tympanometry findings were presented at four to six weeks and three months);
- long-term effects including number of parent-reported AOM symptom episodes, antibiotic prescriptions and health care utilisation as assessed at least one year after randomisation (in earlier versions no data on long-term effects were presented).

The outcome 'Adverse effects likely to be related to the use of antibiotics such as vomiting, diarrhoea or rash' was added to primary outcomes (in earlier versions this outcome was listed as a secondary outcome) according to the recommendations described in Chapter 5.4.2 of the *Cochrane Handbook for Systematic Reviews of Interventions* ("the primary outcomes should include at least one desirable and at least one undesirable outcome") (Higgins 2011).

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

INDEX TERMS

Medical Subject Headings (MeSH)

Acute Disease; Anti-Bacterial Agents [adverse effects]; Diarrhea [chemically induced]; *Exanthema; *Otitis Media [drug therapy] [epidemiology]; Pain [drug therapy]; *Tympanic Membrane Perforation [chemically induced] [drug therapy]; Vomiting [chemically induced]

MeSH check words

Child; Humans

Copyright © 2023 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.